Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()



    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)

    #df = df[valid_features]
    # Removing of last np.nan
    mask = np.isfinite(df[valid_features]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(valid_features_list + #target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    new_feats_X = [x for x in features_X if x in valid_features_list]
    new_feats_add = [x for x in feats_eng if x in valid_features_list]

    data["df"]=df
    data["features_X"] = new_feats_X
    data["feats_added"] = new_feats_add

    column_removed = [x for x in cols_to_check if x not in valid_features_list]
    print(f'Column removed because np.isnan > 15%: {column_removed}')

    return data
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
```

```
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r"
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as 
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
```

```
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Column removed because np.isnan > 15%: ['r_NH3_wl_2h_D2_energy', 'r_NH3_wl_2h_D2_energy_rel', 'r_temperature_wl_2h_D2_energy', 'r_temperature_wl_2h_D2_energy_rel', 'r_umidity_wl_2h_D2_energy', 'r_umidity_wl_2h_D2_energy_rel']

1.2 Metadata

Nombre de lignes df	7793
Nombre de lignes df clean	7793
Nombre de features	47
% de NaN	0.0
Taille dataset entraînement	5455
Taille dataset test	2337
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_D1_energy_rel, r_NH3_wl_2h_A_energy_rel, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_NH3_wl_8h_A_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D2_energy, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D2_energy_rel, r_temperature_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy_rel, r_umidity_wl_8h_D1_energy, r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_energy, r_umidity_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

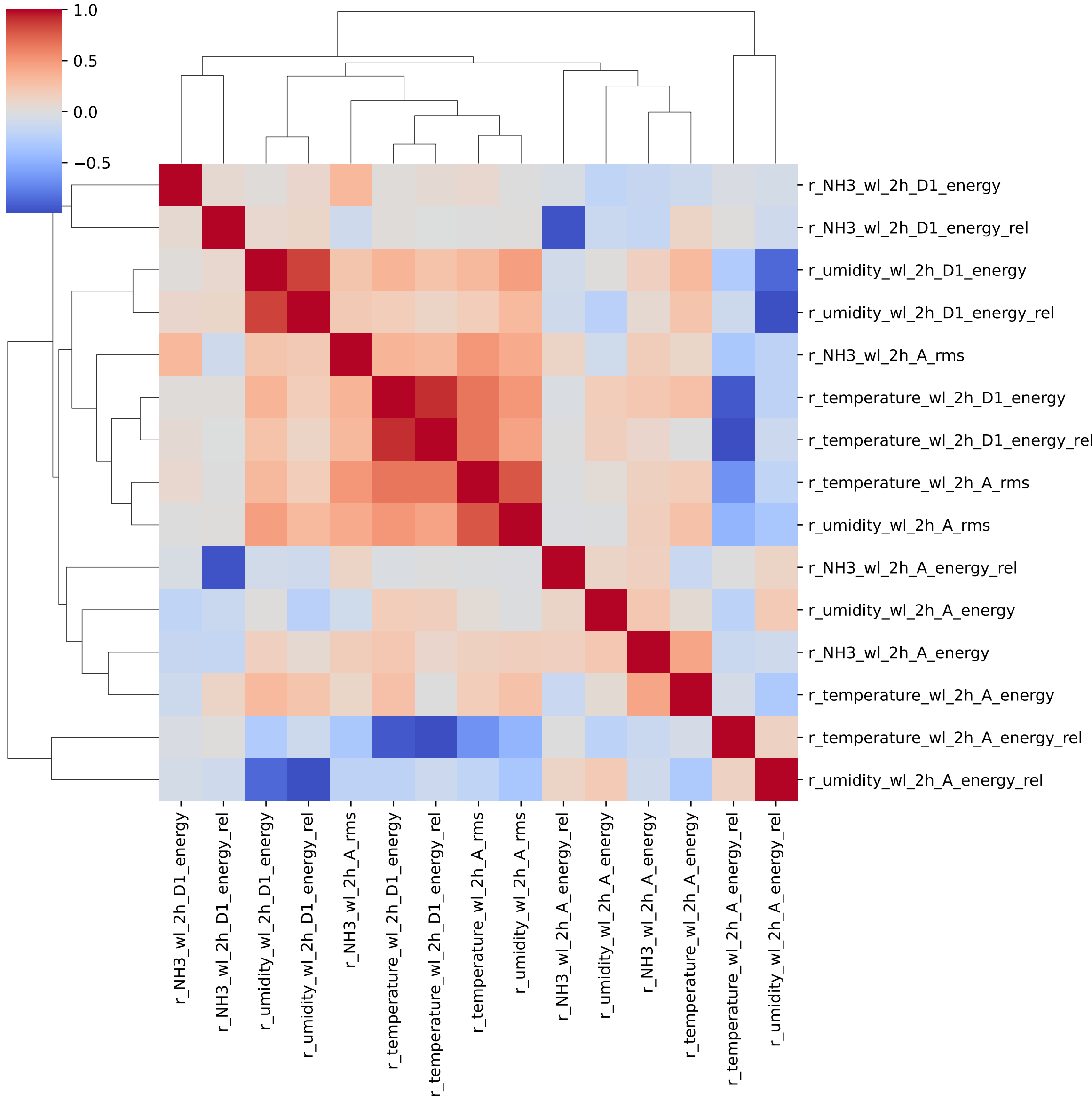
cluster 1: r, *, N, H, 3 ;* ***cluster 2:*** *r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ;* ***cluster 4:*** *r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_wl_2h_D1_energy ; **cluster 7:** r_NH3_wl_2h_A_energy ; **cluster 8:** r_NH3_wl_2h_A_rms ; **cluster 9:** r_NH3_wl_2h_D1_energy_rel, r_NH3_wl_2h_A_energy_rel ; **cluster 10:** r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_A_energy_rel ; **cluster 11:** r_temperature_wl_2h_A_energy ; **cluster 12:** r_temperature_wl_2h_A_rms ; **cluster 13:** r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy_rel ; **cluster 14:** r_umidity_wl_2h_A_energy ; **cluster 15:** r_umidity_wl_2h_A_rms ; **cluster 16:** r_NH3_wl_8h_D1_energy ; **cluster 17:** r_NH3_wl_8h_D2_energy ; **cluster 18:** r_NH3_wl_8h_D3_energy ; **cluster 19:** r_NH3_wl_8h_A_energy ; **cluster 20:** r_NH3_wl_8h_A_rms ; **cluster 21:** r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D1_energy_rel ; **cluster 22:** r_NH3_wl_8h_D3_energy_rel ; **cluster 23:** r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D2_energy_rel, r_temperature_wl_8h_D2_energy ; **cluster 24:** r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_D3_energy_rel ; **cluster 25:** r_temperature_wl_8h_A_energy, r_umidity_wl_8h_A_energy ; **cluster 26:** r_temperature_wl_8h_A_rms ; **cluster 27:** r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D1_energy, r_umidity_wl_8h_A_energy_rel ; **cluster 28:** r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D2_energy_rel ; **cluster 29:**

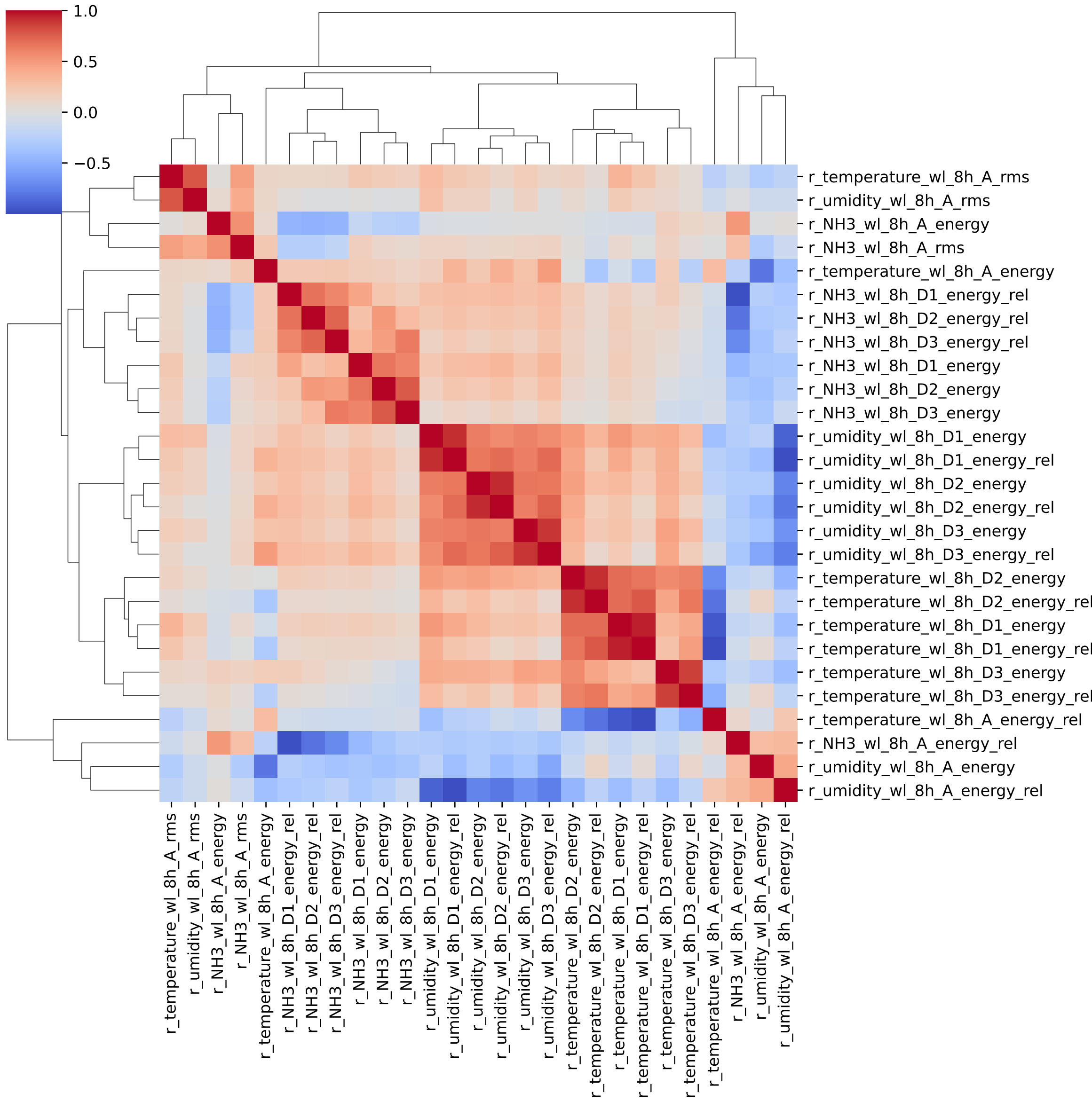
r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_D3_energy_rel ; **cluster 30:** r_umidity_wl_8h_A_rms ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_D1_energy_rel, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_rms


1.0
0.5
0.0
−0.5
r_NH3_wl_2h_D1_energy
r_NH3_wl_2h_D1_energy_rel
r_umidity_wl_2h_D1_energy
r_umidity_wl_2h_D1_energy_rel
r_NH3_wl_2h_A_rms
r_temperature_wl_2h_D1_energy
r_temperature_wl_2h_D1_energy_rel
r_temperature_wl_2h_A_rms
r_umidity_wl_2h_A_rms
r_NH3_wl_2h_A_energy_rel
r_umidity_wl_2h_A_energy
r_NH3_wl_2h_A_energy
r_temperature_wl_2h_A_energy
r_temperature_wl_2h_A_energy_rel
r_umidity_wl_2h_A_energy_rel



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_D1_energy_rel, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_rms

Len(Features_X): 30

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

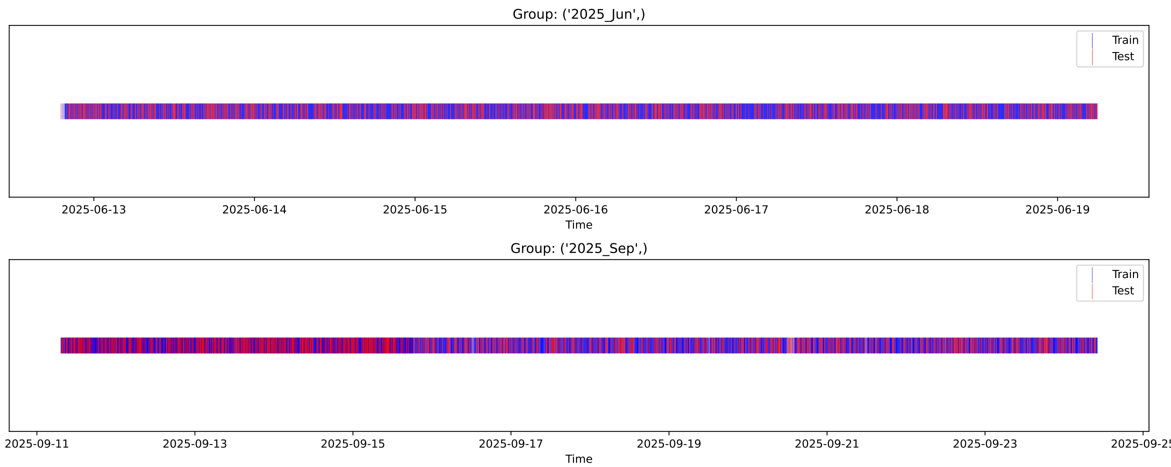
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5455 samples Test: 2338 samples



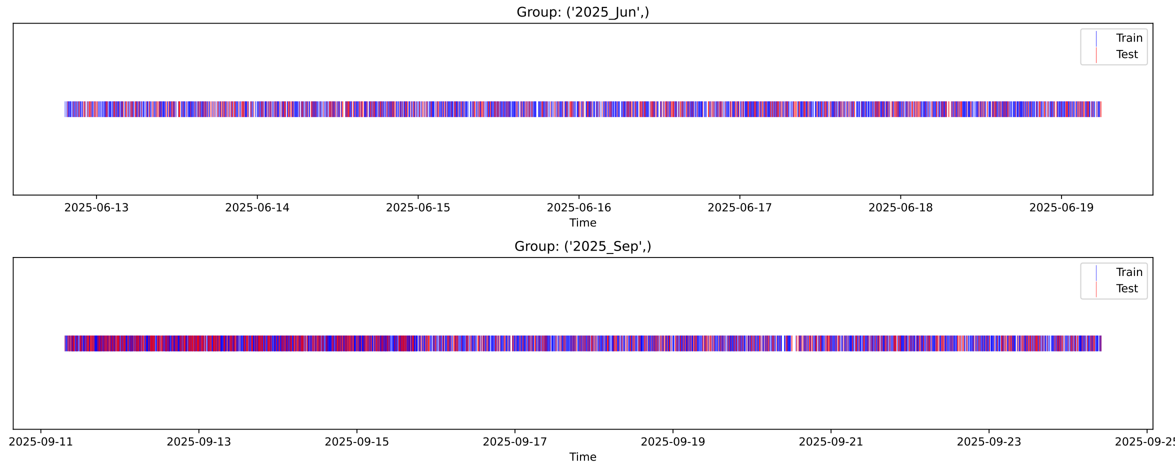
3.3.2 Internal CV Folds

Fold 1/5

Train: 3818 samples Test: 1637 samples Excluded: 0 samples Window exclusion: 0h

Fold 2/5

Train: 3818 samples Test: 1637 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3818 samples Test: 1637 samples Excluded: 0 samples Window exclusion: 0h

Fold 4/5

Train: 3818 samples Test: 1637 samples Excluded: 0 samples Window exclusion: 0h

Fold 5/5

Train: 3818 samples Test: 1637 samples Excluded: 0 samples Window exclusion: 0h

3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.343, std=0.800, min=0.959, max=13.845 y_test: mean=2.367, std=0.801, min=0.973, max=7.822

Fold 0 — y_test: mean=2.33, std=0.78, y_train: mean=2.35, std=0.81

Fold 1 — y_test: mean=2.34, std=0.79, y_train: mean=2.34, std=0.80

Fold 2 — y_test: mean=2.32, std=0.77, y_train: mean=2.35, std=0.81

Fold 3 — y_test: mean=2.33, std=0.78, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.34, std=0.76, y_train: mean=2.34, std=0.82

train scores CV: [0.56639743 0.55583491 0.56100634 0.55118532 0.56172342]

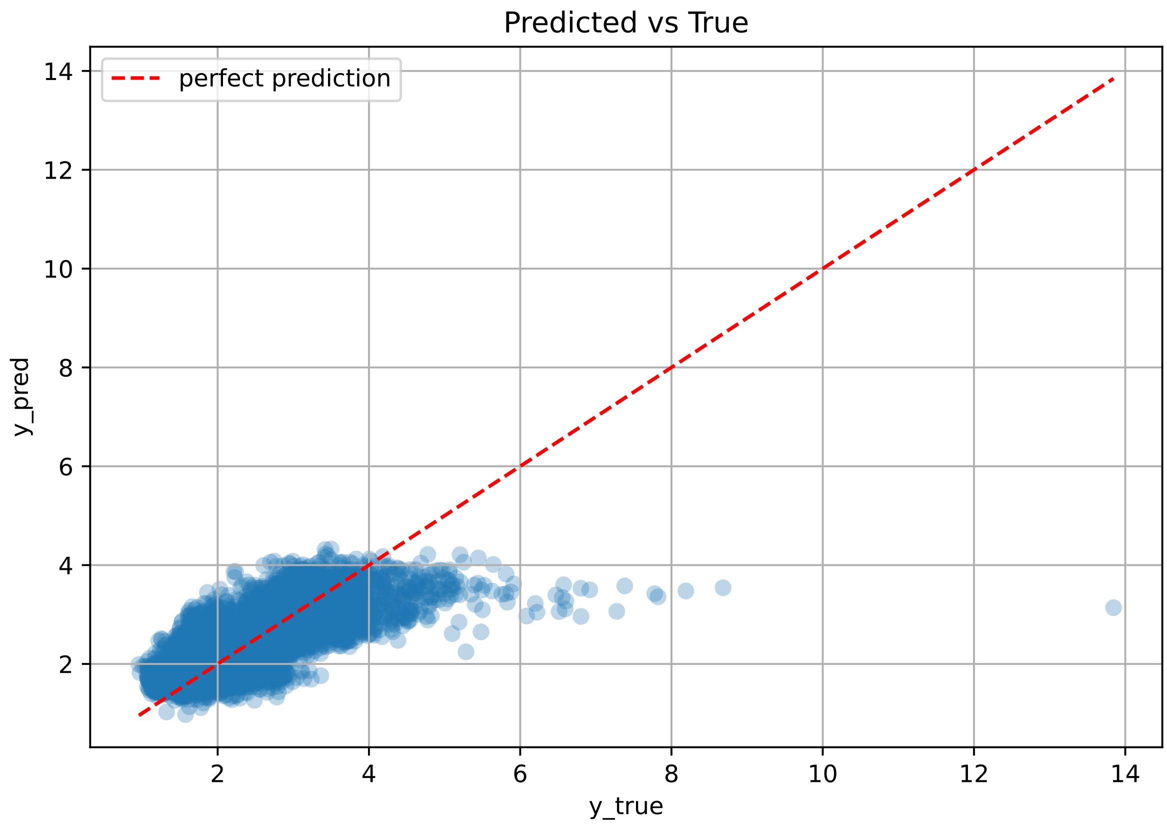
test scores CV: [0.571965 0.59795672 0.58628477 0.60638137 0.58079305]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5592	0.5887	0.0122	0.5711	0.5717	-0.017	0.9712	0.95

Coefficients:

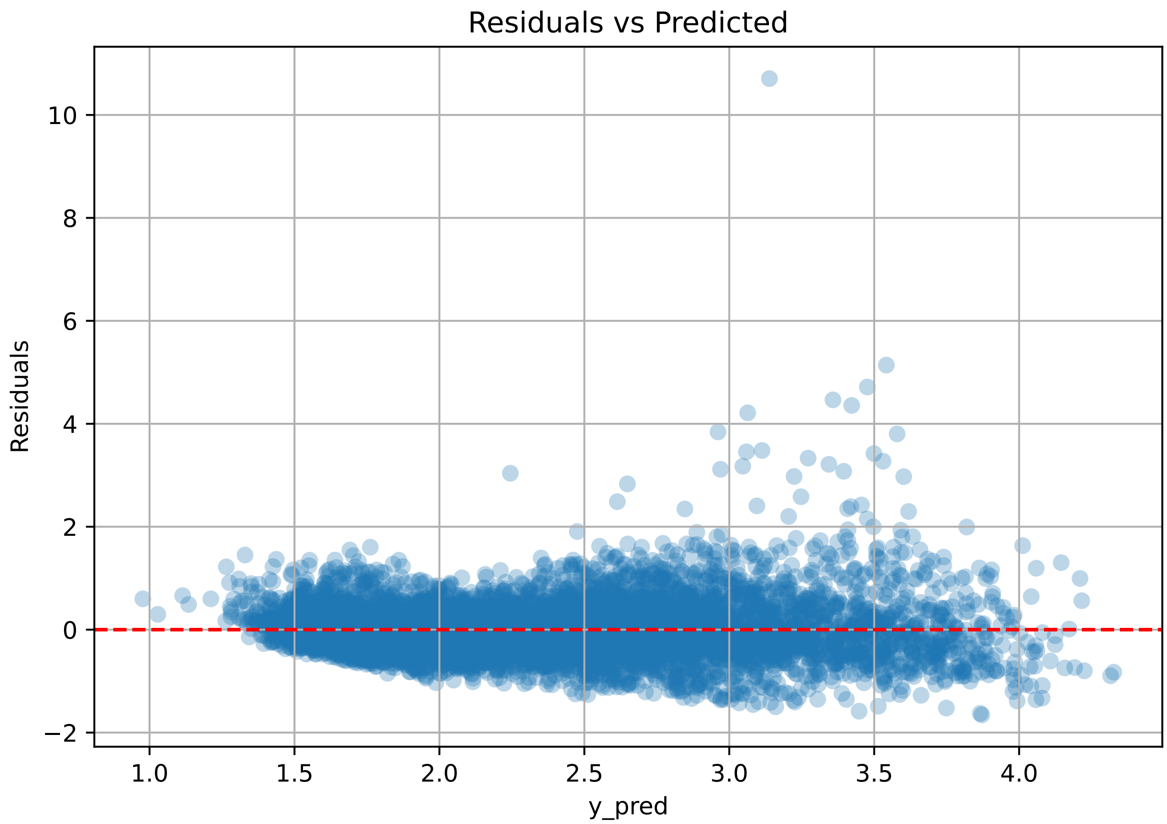
coef_r_CO2 = -0.0611, coef_r_NH3 = 0.2051, coef_r_NH3_wl_2h_A_energy = -0.1810, coef_r_NH3_wl_2h_A_rms = -0.0038, coef_r_NH3_wl_2h_D1_energy = -0.0025, coef_r_NH3_wl_2h_D1_energy_rel = -0.0227, coef_r_NH3_wl_8h_A_energy = -0.1426, coef_r_NH3_wl_8h_A_rms = 0.0224, coef_r_NH3_wl_8h_D1_energy = 0.0551, coef_r_NH3_wl_8h_D2_energy = 0.1340, coef_r_NH3_wl_8h_D2_energy_rel = -0.0615, coef_r_NH3_wl_8h_D3_energy = -0.0586, coef_r_NH3_wl_8h_D3_energy_rel = -0.0287, coef_r_THI = 0.6246, coef_r_temperature = -0.4561, coef_r_temperature_wl_2h_A_energy = -0.0359, coef_r_temperature_wl_2h_A_rms = 0.1530, coef_r_temperature_wl_2h_D1_energy_rel = 0.0232, coef_r_temperature_wl_8h_A_energy = -0.1777, coef_r_temperature_wl_8h_A_rms = 0.0053, coef_r_temperature_wl_8h_D1_energy = -0.0168, coef_r_temperature_wl_8h_D3_energy = -0.0714, coef_r_umidity = -0.5538, coef_r_umidity_wl_2h_A_energy = 0.0990, coef_r_umidity_wl_2h_A_rms = 0.0531, coef_r_umidity_wl_2h_D1_energy_rel = 0.0239, coef_r_umidity_wl_8h_A_rms = -0.1159, coef_r_umidity_wl_8h_D1_energy_rel = -0.0356, coef_r_umidity_wl_8h_D2_energy = -0.0292, coef_r_umidity_wl_8h_D3_energy = 0.0380, intercept = 2.3427,

3.4.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

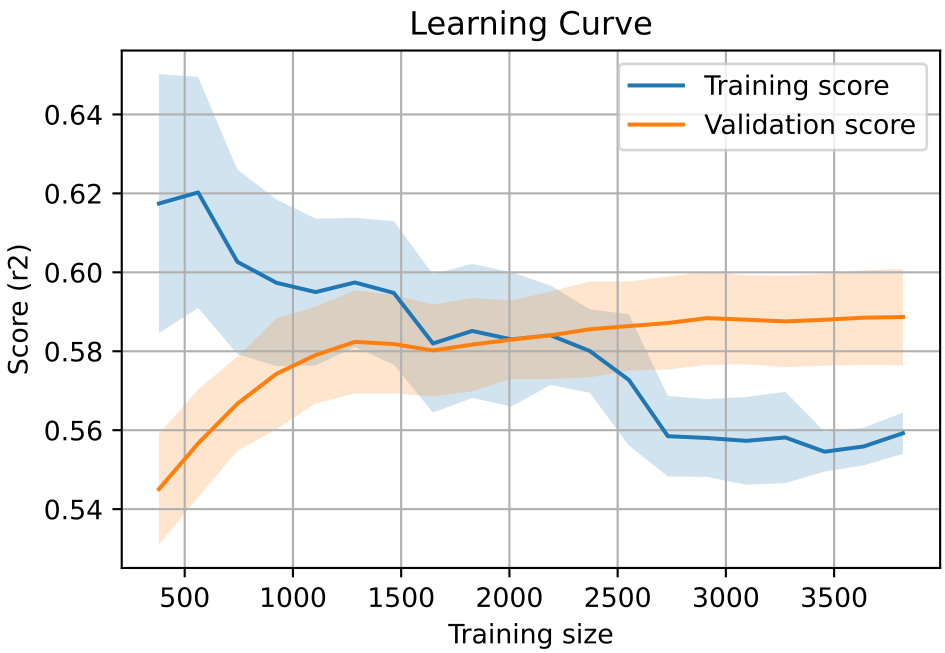
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 7793, **Len(training)** : 5456,

Len(training_real) : 3820, **Len(test_of_training)** : 1636, **Len(test)** : 2337

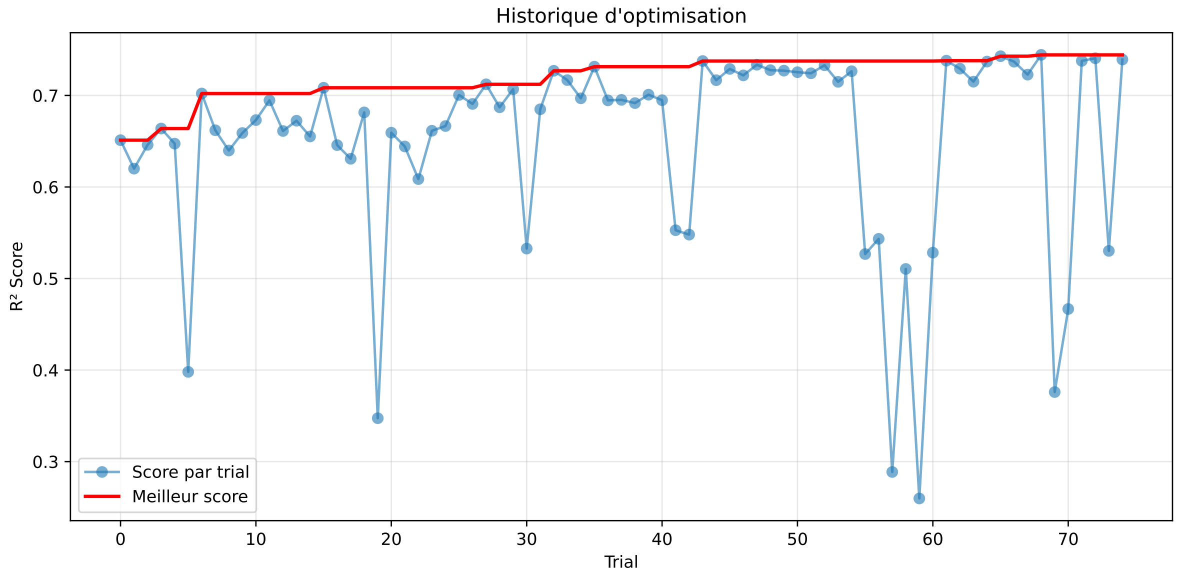


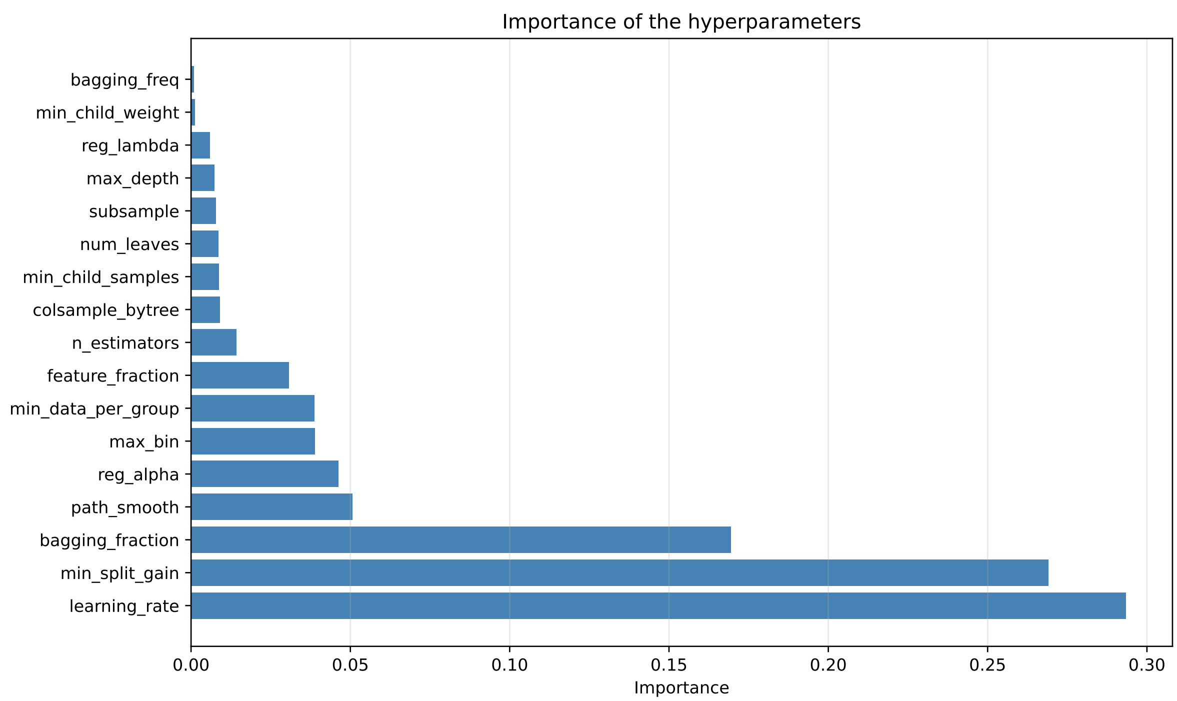
3.5 Model : LGBM

3.5.1 Gridsearch

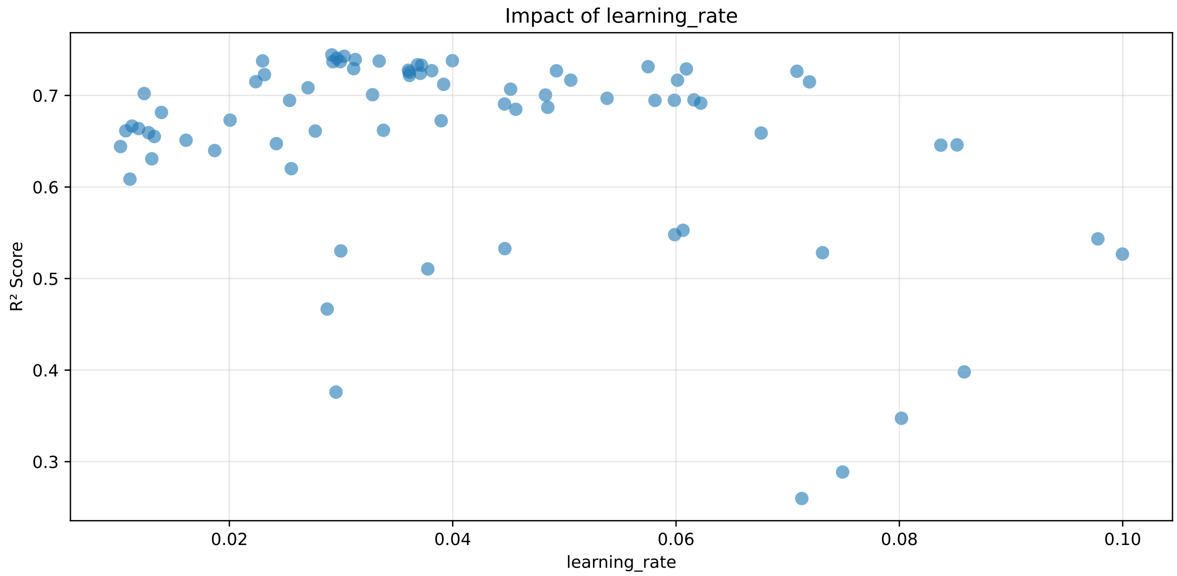
Distribution y_train vs y_test: y_train: mean=2.343, std=0.800, min=0.959, max=13.845 y_test: mean=2.367, std=0.801, min=0.973, max=7.822

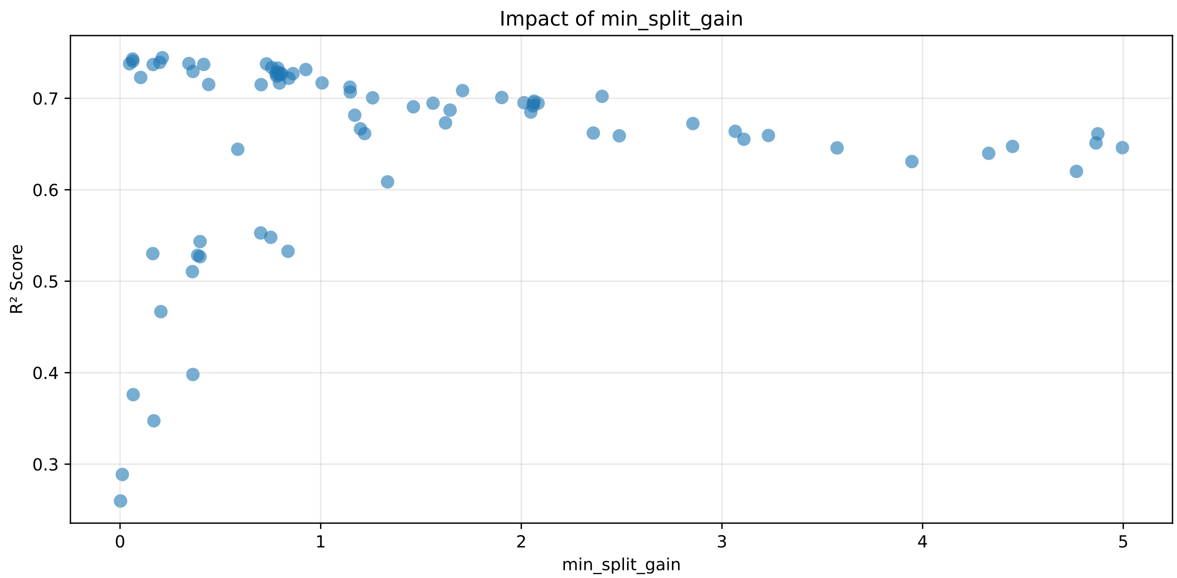
==================== TOP Importance FEATURES ==================== feature importance r_CO2 497 r_temperature_wl_8h_A_energy 354 r_NH3_wl_8h_A_energy 349 r_NH3 309 r_umidity 307 r_NH3_wl_8h_D3_energy_rel 254 r_NH3_wl_2h_A_energy 230 r_THI 222 r_temperature 215 r_temperature_wl_8h_A_rms 211 r_temperature_wl_2h_A_rms 206 r_umidity_wl_2h_A_energy 202 r_NH3_wl_8h_D3_energy 197 r_temperature_wl_2h_A_energy 188 r_umidity_wl_8h_A_rms 183 r_NH3_wl_8h_D1_energy 169 r_NH3_wl_8h_A_rms 152 r_temperature_wl_8h_D1_energy 143 r_umidity_wl_2h_A_rms 138 r_NH3_wl_8h_D2_energy 138 r_umidity_wl_8h_D3_energy 131 r_temperature_wl_8h_D3_energy 123 r_NH3_wl_2h_A_rms 114 r_NH3_wl_2h_D1_energy 110 r_umidity_wl_2h_D1_energy_rel 107 r_NH3_wl_8h_D2_energy_rel 104 r_umidity_wl_8h_D2_energy 101 r_temperature_wl_2h_D1_energy_rel 94 r_NH3_wl_2h_D1_energy_rel 80 r_umidity_wl_8h_D1_energy_rel 65

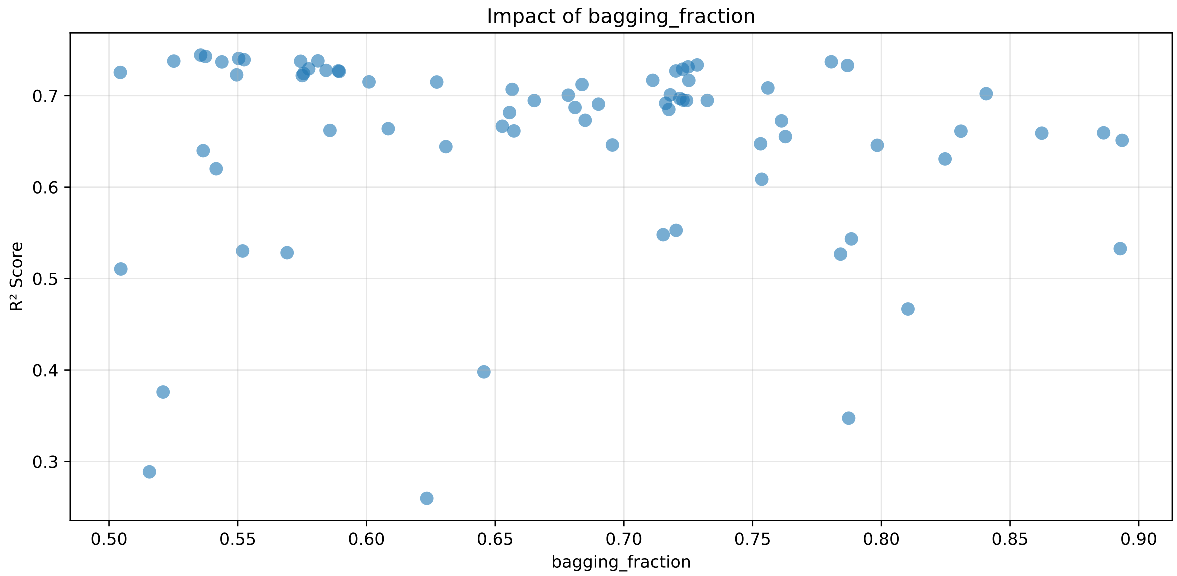


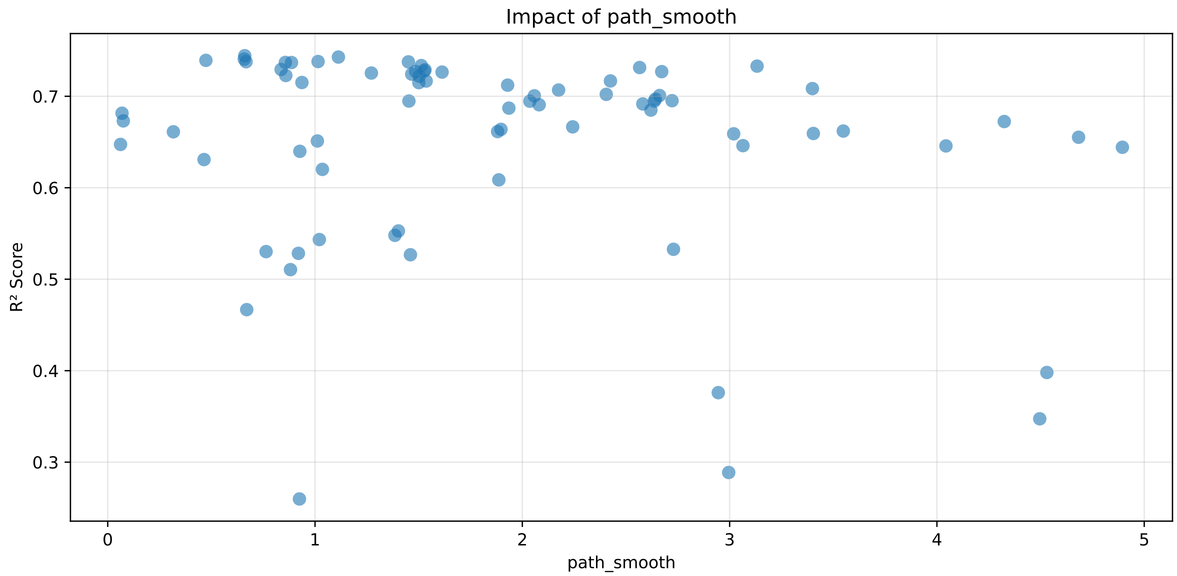


Paramètres avec >5% d'importance : ['learning_rate', 'min_split_gain', 'bagging_fraction', 'path_smooth']


Impact of learning_rate
R² Score
0.3
0.4
0.5
0.6
0.7
0.02
0.04
0.06
0.08
0.10
learning_rate

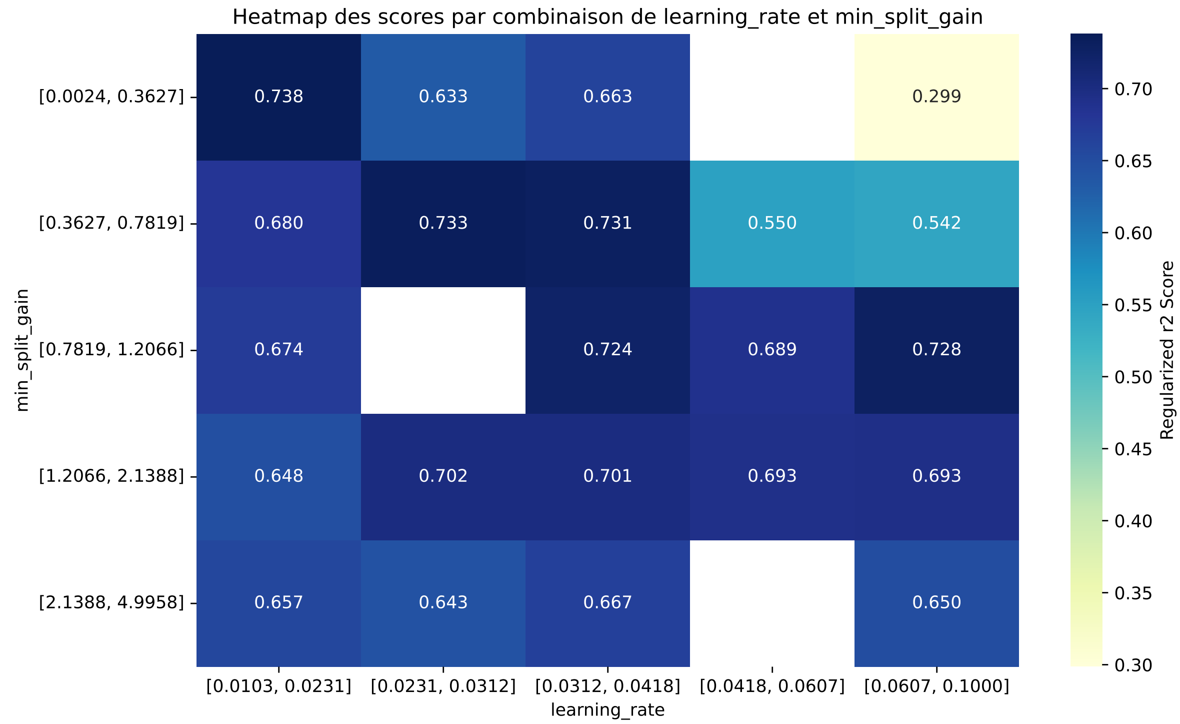

Impact of min_split_gain
R² Score
0.3
0.4
0.5
0.6
0.7
0
1
2
3
4
5
min_split_gain






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7781	0.7443	0.7404	0.0073	0.7443	0.7595	0.0151	1.0204	1.0453
2	0.7789	0.7429	0.7396	0.0088	0.7429	0.7535	0.0107	1.0144	1.0484
3	0.773	0.7406	0.7365	0.0088	0.7406	0.7526	0.012	1.0162	1.0437
4	0.774	0.7393	0.7368	0.0075	0.7393	0.7545	0.0153	1.0206	1.047
5	0.7683	0.738	0.7338	0.0077	0.738	0.752	0.014	1.019	1.041
6	0.7703	0.7378	0.7335	0.0089	0.7378	0.7473	0.0095	1.0129	1.0441
7	0.7727	0.7376	0.7315	0.0083	0.7376	0.7515	0.0139	1.0188	1.0475
8	0.7715	0.737	0.7358	0.0089	0.737	0.7508	0.0138	1.0187	1.0468
9	0.7673	0.7369	0.7333	0.0082	0.7369	0.75	0.0131	1.0178	1.0413
10	0.7586	0.7336	0.7285	0.0081	0.7336	0.745	0.0114	1.0155	1.0342
---	---	---	---	---	---	---	---	---	---
75	0.892	0.7867	0.7833	0.0061	0.2597	0.7947	0.0081	1.0102	1.134

Hyperparameters:

Rank 1: {'reg___n_estimators': 200, 'reg___max_depth': 7, 'reg___learning_rate': 0.029175433510489703, 'reg___num_leaves': 32, 'reg___subsample': 0.5276467704143013, 'reg___colsample_bytree': 0.7443098551242944, 'reg___min_child_samples': 22, 'reg___reg_alpha': 2.0522488014468294, 'reg___reg_lambda': 29.277305995892462, 'reg___min_split_gain': 0.21069797715021266, 'reg___path_smooth': 0.6610225905405192, 'reg___min_child_weight':

0.0018826078263896474, ’reg__feature_fraction’: 0.7745019850480404, ’reg__bagging_fraction’: 0.535590925021902, ’reg__bagging_freq’: 1, ’reg__max_bin’: 80, ’reg__min_data_per_group’: 184}

Rank 2: {’reg__n_estimators’: 200, ’reg__max_depth’: 6, ’reg__learning_rate’: 0.030272084400266977, ’reg__num_leaves’: 43, ’reg__subsample’: 0.7829160577143272, ’reg__colsample_bytree’: 0.7588885121145779, ’reg__min_child_samples’: 12, ’reg__reg_alpha’: 4.830870512751267, ’reg__reg_lambda’: 23.376433399547803, ’reg__min_split_gain’: 0.0625587224839661, ’reg__path_smooth’: 1.112796868410542, ’reg__min_child_weight’: 0.002551925042811203, ’reg__feature_fraction’: 0.7684330173851326, ’reg__bagging_fraction’: 0.5374559627651283, ’reg__bagging_freq’: 1, ’reg__max_bin’: 83, ’reg__min_data_per_group’: 186}

Rank 3: {’reg__n_estimators’: 200, ’reg__max_depth’: 6, ’reg__learning_rate’: 0.029613691549491346, ’reg__num_leaves’: 41, ’reg__subsample’: 0.5248037762581725, ’reg__colsample_bytree’: 0.7479184319880272, ’reg__min_child_samples’: 12, ’reg__reg_alpha’: 4.629603242573304, ’reg__reg_lambda’: 29.829814455407156, ’reg__min_split_gain’: 0.06387113883653905, ’reg__path_smooth’: 0.6586716007829659, ’reg__min_child_weight’: 0.0023399217897504014, ’reg__feature_fraction’: 0.7624806026555826, ’reg__bagging_fraction’: 0.5503769494278119, ’reg__bagging_freq’: 1, ’reg__max_bin’: 79, ’reg__min_data_per_group’: 186}

Rank 4: {’reg__n_estimators’: 200, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.03127557223936297, ’reg__num_leaves’: 41, ’reg__subsample’: 0.5145580523150526, ’reg__colsample_bytree’: 0.725365531831792, ’reg__min_child_samples’: 14, ’reg__reg_alpha’: 4.574588408688166, ’reg__reg_lambda’: 27.02407233711927, ’reg__min_split_gain’: 0.19814274508379115, ’reg__path_smooth’: 0.47357804799987324, ’reg__min_child_weight’: 0.002122790200095867, ’reg__feature_fraction’: 0.7926929349946349, ’reg__bagging_fraction’: 0.5524758393978347, ’reg__bagging_freq’: 1, ’reg__max_bin’: 81, ’reg__min_data_per_group’: 186}

Rank 5: {’reg__n_estimators’: 200, ’reg__max_depth’: 6, ’reg__learning_rate’: 0.03997069244880404, ’reg__num_leaves’: 41, ’reg__subsample’: 0.7819228612782761, ’reg__colsample_bytree’: 0.7446403264945213, ’reg__min_child_samples’: 12, ’reg__reg_alpha’: 4.958353114615016, ’reg__reg_lambda’: 23.645191520714356, ’reg__min_split_gain’: 0.34311674239181267, ’reg__path_smooth’: 1.0146835376229024, ’reg__min_child_weight’: 0.002567173681436995, ’reg__feature_fraction’: 0.7807656205222608, ’reg__bagging_fraction’: 0.5811333253575008, ’reg__bagging_freq’: 2, ’reg__max_bin’: 81, ’reg__min_data_per_group’: 186}

Rank 6: {’reg__n_estimators’: 200, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.022974759358492247, ’reg__num_leaves’: 40, ’reg__subsample’: 0.7432819187857975, ’reg__colsample_bytree’: 0.7441341449549754, ’reg__min_child_samples’: 13, ’reg__reg_alpha’: 2.292467921257349, ’reg__reg_lambda’: 28.481257587189248, ’reg__min_split_gain’: 0.04783341576104827, ’reg__path_smooth’: 0.6673459321937609, ’reg__min_child_weight’:

0.002452720971121236, 'reg___feature_fraction': 0.6804672174463259, 'reg___bagging_fraction': 0.525161735226821, 'reg___bagging_freq': 1, 'reg___max_bin': 83, 'reg___min_data_per_group': 186}

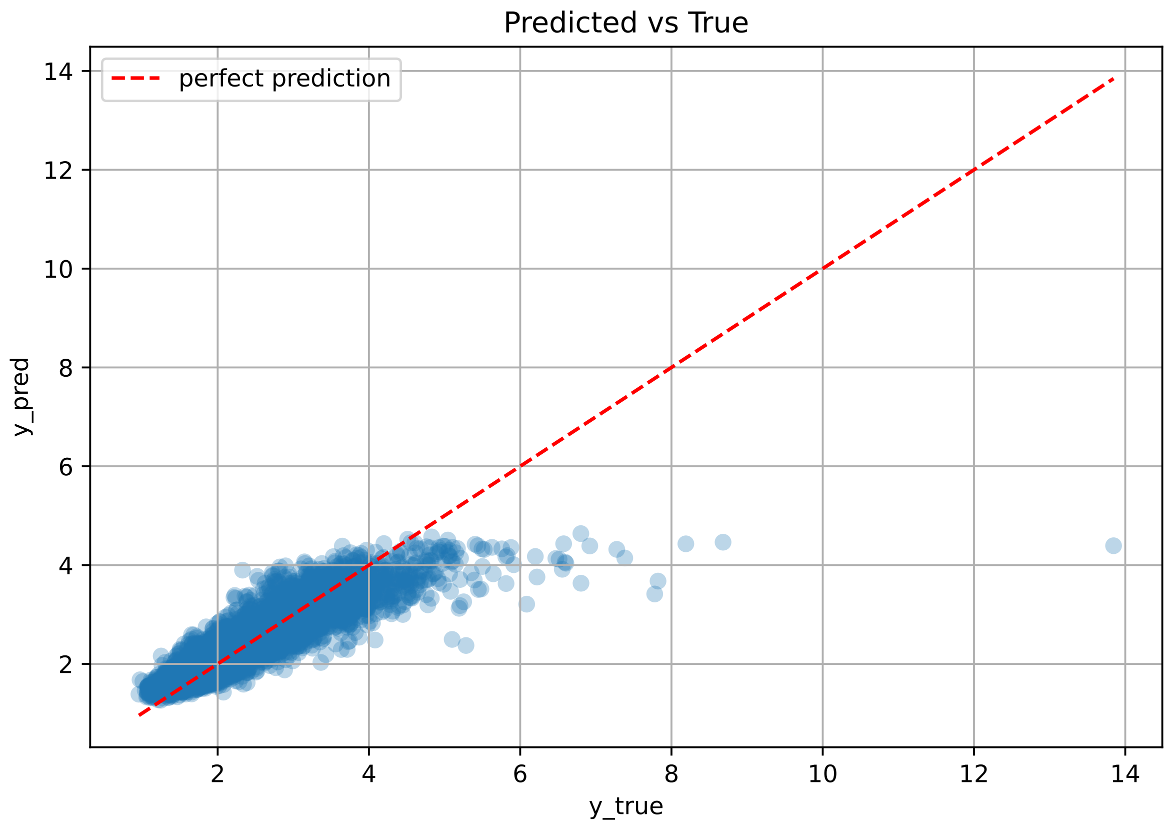
Rank 7: {'reg___n_estimators': 700, 'reg___max_depth': 8, 'reg___learning_rate': 0.03341407876494182, 'reg___num_leaves': 34, 'reg___subsample': 0.6241083696302151, 'reg___colsample_bytree': 0.8177934413790569, 'reg___min_child_samples': 31, 'reg___reg_alpha': 1.4348261946954024, 'reg___reg_lambda': 19.23211902177032, 'reg___min_split_gain': 0.7301073543479554, 'reg___path_smooth': 1.4504073660501755, 'reg___min_child_weight': 0.06001794836798195, 'reg___feature_fraction': 0.6883793653669353, 'reg___bagging_fraction': 0.5744386570951125, 'reg___bagging_freq': 4, 'reg___max_bin': 215, 'reg___min_data_per_group': 151}

Rank 8: {'reg___n_estimators': 200, 'reg___max_depth': 6, 'reg___learning_rate': 0.029929724321552133, 'reg___num_leaves': 31, 'reg___subsample': 0.7757854950602404, 'reg___colsample_bytree': 0.7551851407095295, 'reg___min_child_samples': 13, 'reg___reg_alpha': 4.727881769796012, 'reg___reg_lambda': 24.71273725206512, 'reg___min_split_gain': 0.4170012669142886, 'reg___path_smooth': 0.8568463787212514, 'reg___min_child_weight': 0.0027582708676406023, 'reg___feature_fraction': 0.7719091290233886, 'reg___bagging_fraction': 0.7805670927416478, 'reg___bagging_freq': 1, 'reg___max_bin': 82, 'reg___min_data_per_group': 187}

Rank 9: {'reg___n_estimators': 200, 'reg___max_depth': 6, 'reg___learning_rate': 0.02927154759917635, 'reg___num_leaves': 32, 'reg___subsample': 0.5326611383834472, 'reg___colsample_bytree': 0.7471038505844755, 'reg___min_child_samples': 12, 'reg___reg_alpha': 4.821395829877199, 'reg___reg_lambda': 23.49657042028577, 'reg___min_split_gain': 0.16518754657721058, 'reg___path_smooth': 0.8864860270949011, 'reg___min_child_weight': 0.002575913558391351, 'reg___feature_fraction': 0.7694799427635985, 'reg___bagging_fraction': 0.5438368400318739, 'reg___bagging_freq': 1, 'reg___max_bin': 80, 'reg___min_data_per_group': 187}

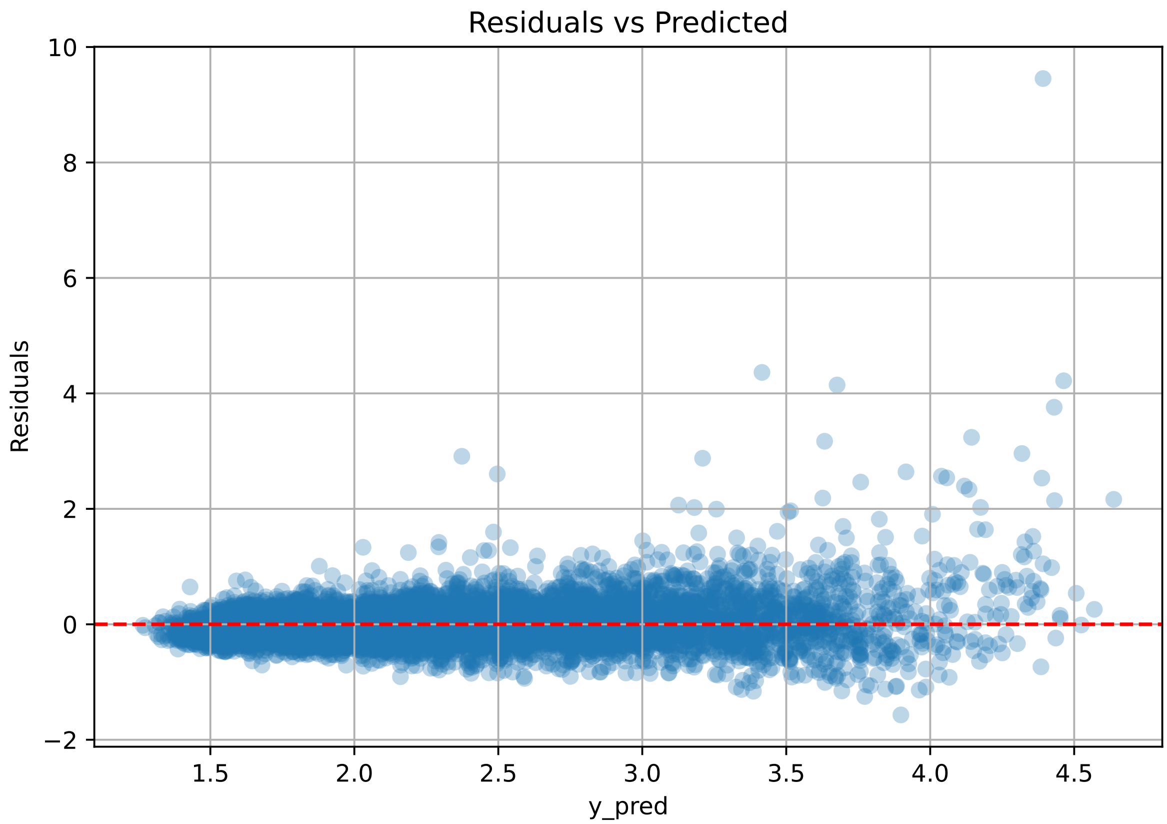
Rank 10: {'reg___n_estimators': 300, 'reg___max_depth': 5, 'reg___learning_rate': 0.03684317516880486, 'reg___num_leaves': 21, 'reg___subsample': 0.8417718483928641, 'reg___colsample_bytree': 0.8020874802444473, 'reg___min_child_samples': 12, 'reg___reg_alpha': 3.0102074355098054, 'reg___reg_lambda': 18.703329862533586, 'reg___min_split_gain': 0.7563459877844805, 'reg___path_smooth': 1.51267235807677, 'reg___min_child_weight': 0.004478180192996574, 'reg___feature_fraction': 0.7226379414775057, 'reg___bagging_fraction': 0.7284753478746115, 'reg___bagging_freq': 2, 'reg___max_bin': 79, 'reg___min_data_per_group': 197}

3.5.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

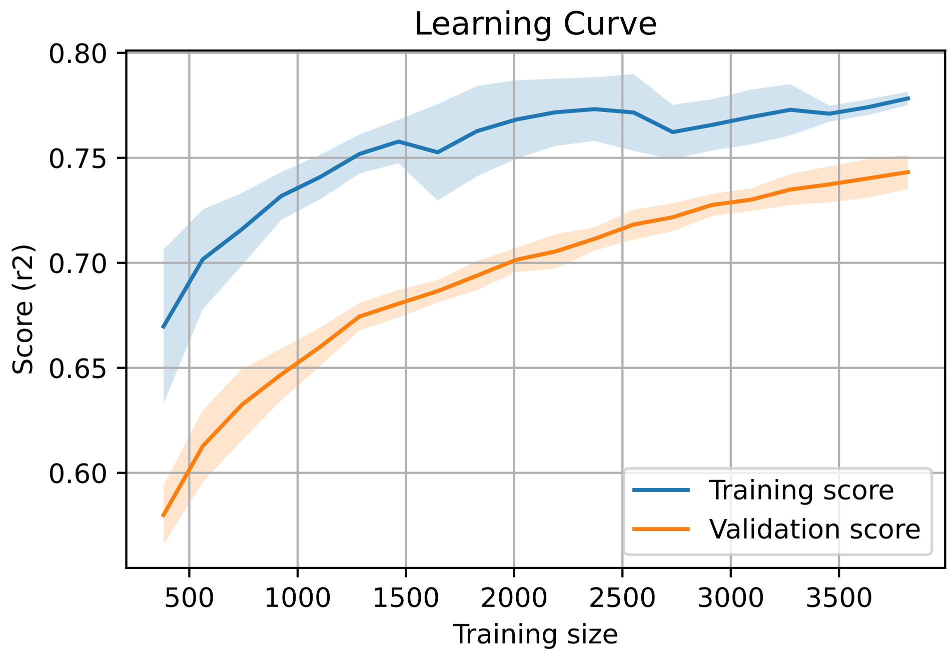
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 7793, **Len(training)** : 5456,

Len(training_real) : 3820, **Len(test_of_training)** : 1636, **Len(test)** : 2337

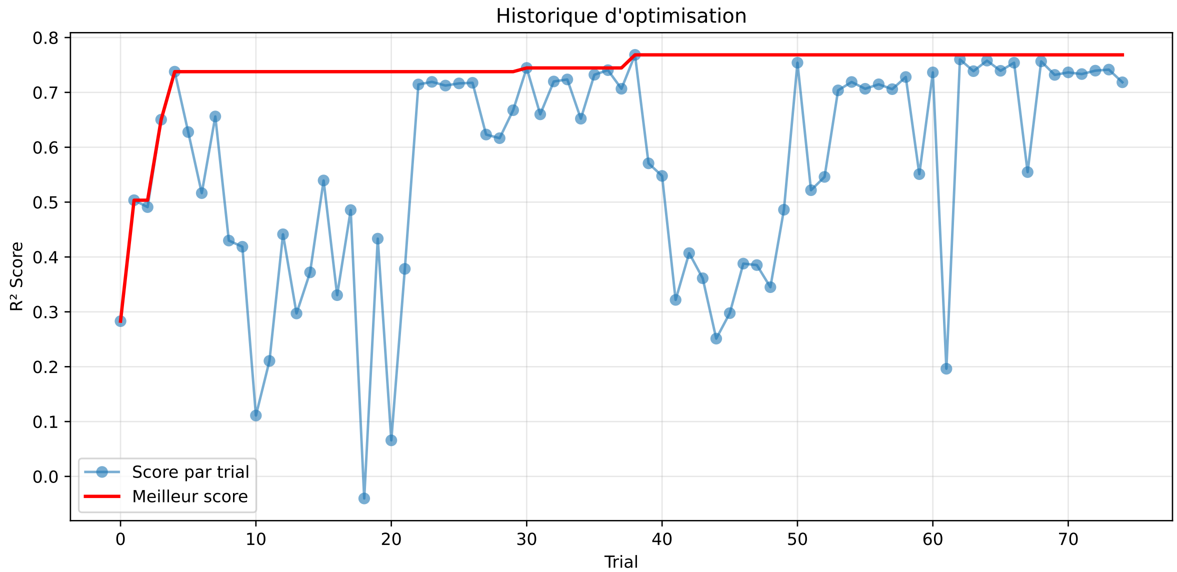


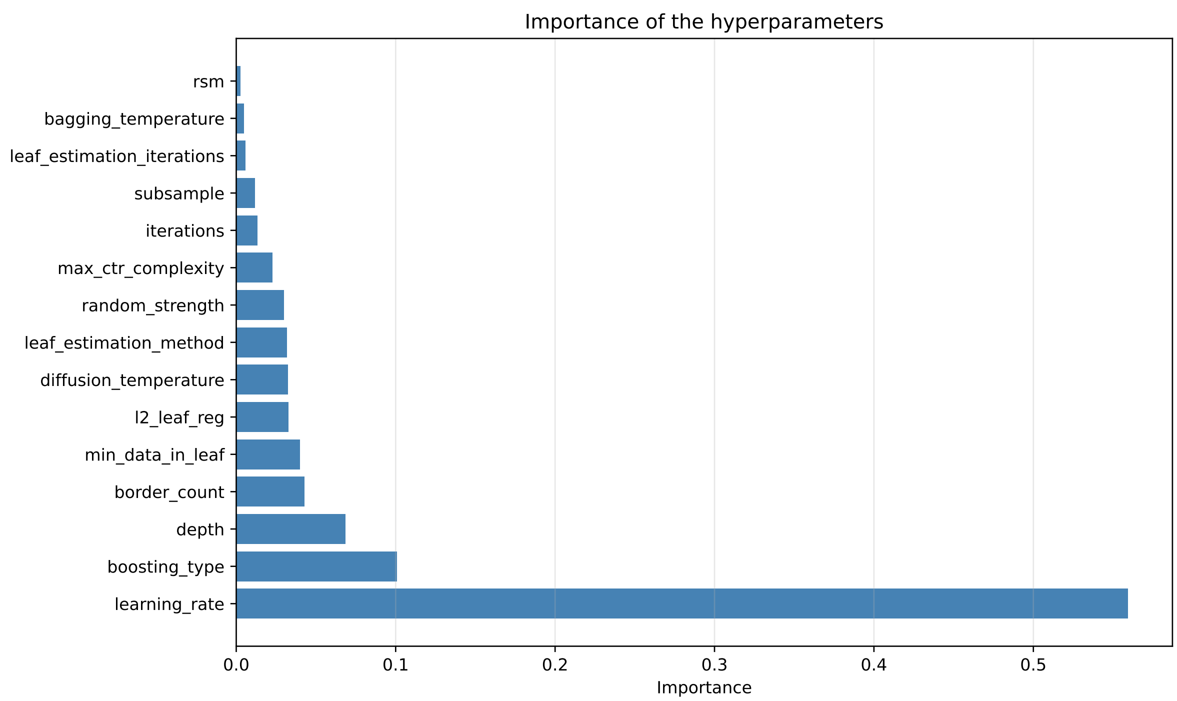
3.6 Model : CatBoost

3.6.1 Gridsearch

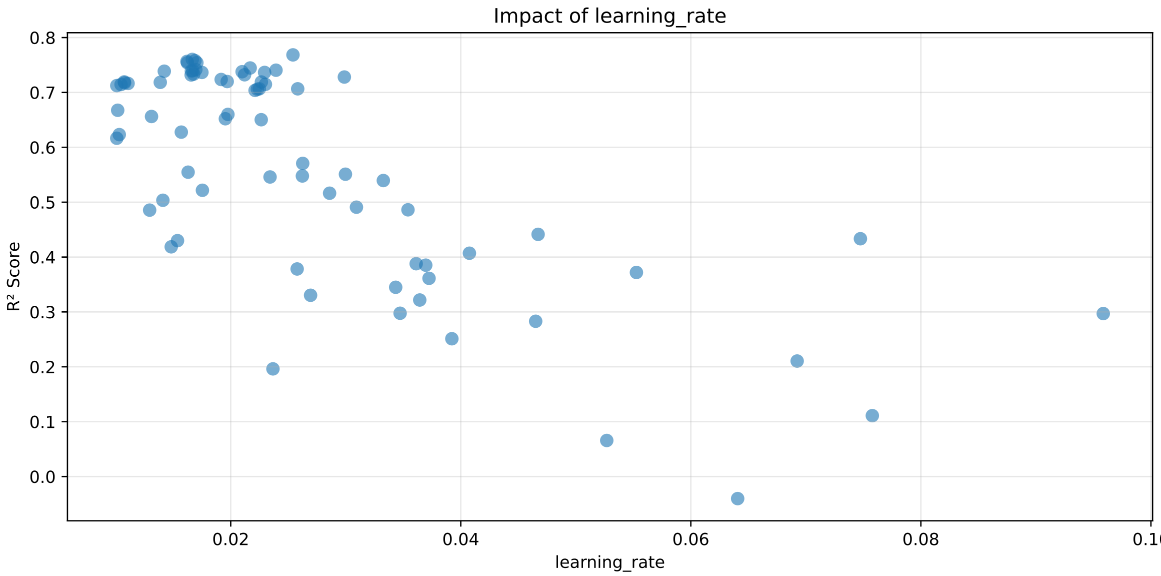
Distribution y_train vs y_test: y_train: mean=2.343, std=0.800, min=0.959, max=13.845 y_test: mean=2.367, std=0.801, min=0.973, max=7.822

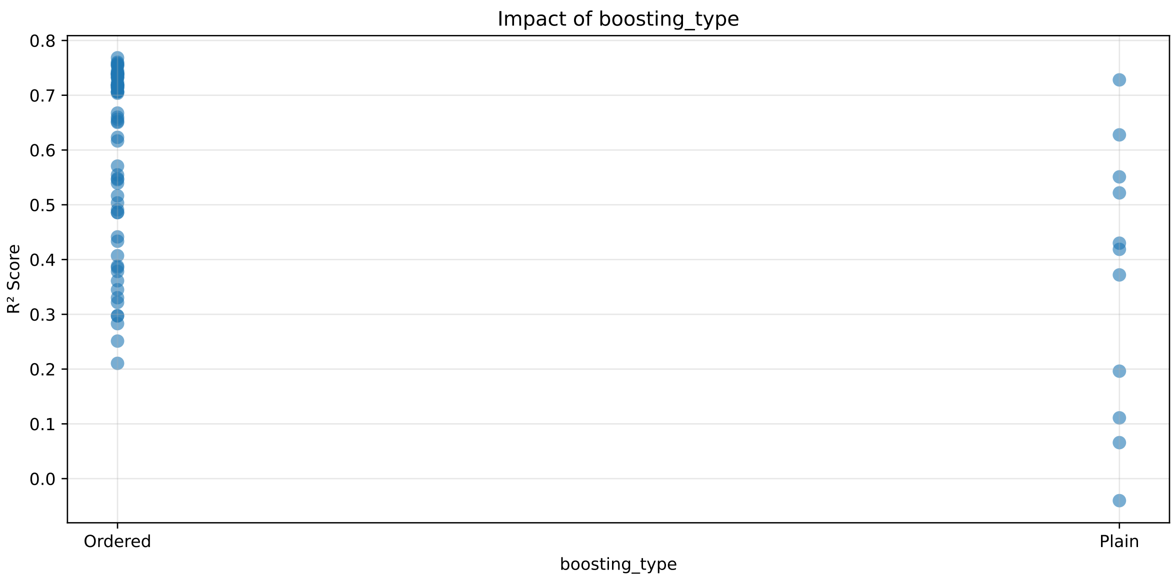
==================== TOP Importance FEATURES ==================== feature importance r_umidity 15.654697 r_temperature_wl_8h_A_energy 12.656480 r_temperature_wl_2h_A_rms 8.477805 r_CO2 6.399321 r_temperature 5.926442 r_NH3 5.126047 r_NH3_wl_8h_D1_energy 3.694218 r_temperature_wl_2h_D1_energy_rel 3.482798 r_THI 3.290036 r_temperature_wl_8h_D1_energy 3.277090 r_NH3_wl_8h_A_energy 3.161339 r_NH3_wl_8h_D3_energy 2.501864 r_umidity_wl_2h_A_energy 2.454859 r_NH3_wl_2h_A_energy 2.407787 r_NH3_wl_8h_D2_energy 2.091083 r_temperature_wl_2h_A_energy 2.076338 r_NH3_wl_8h_D3_energy_rel 1.928850 r_umidity_wl_2h_D1_energy_rel 1.850531 r_umidity_wl_8h_A_rms 1.779715 r_NH3_wl_2h_A_rms 1.555328 r_NH3_wl_8h_A_rms 1.384256 r_NH3_wl_2h_D1_energy_rel 1.267607 r_NH3_wl_2h_D1_energy 1.259944 r_umidity_wl_8h_D3_energy 1.169510 r_temperature_wl_8h_A_rms 1.122324 r_umidity_wl_2h_A_rms 1.084003 r_temperature_wl_8h_D3_energy 1.066382 r_NH3_wl_8h_D2_energy_rel 0.742095 r_umidity_wl_8h_D2_energy 0.640261 r_umidity_wl_8h_D1_energy_rel 0.470992

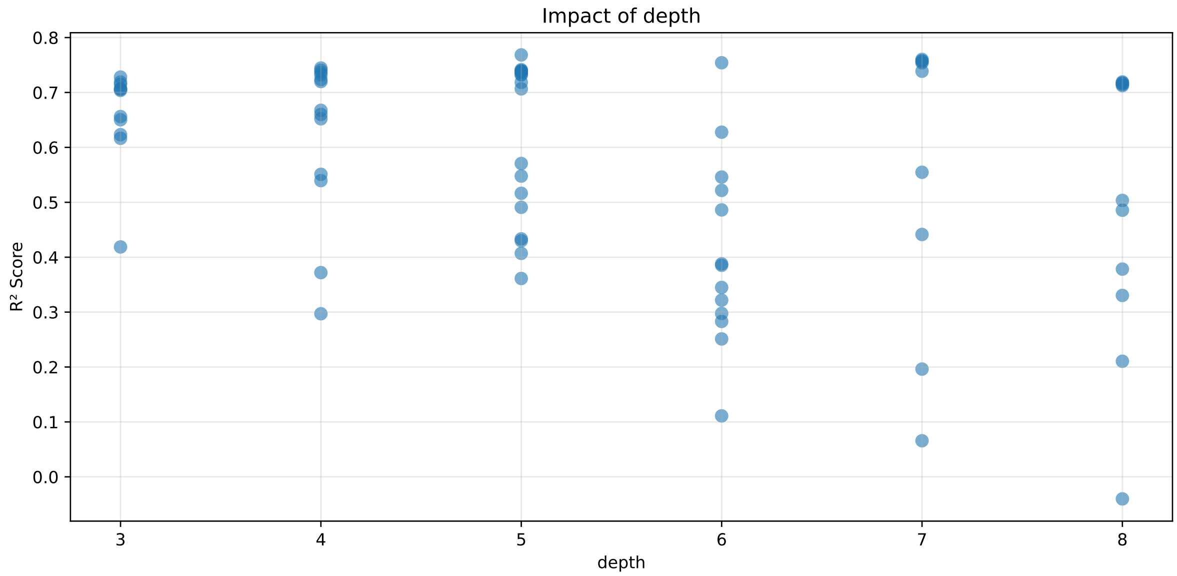




Paramètres avec >5% d'importance : ['learning_rate', 'boosting_type', 'depth']


Impact of learning_rate
R² Score
learning_rate
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.02
0.04
0.06
0.08
0.10


Impact of boosting_type
R² Score
boosting_type
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Ordered
Plain



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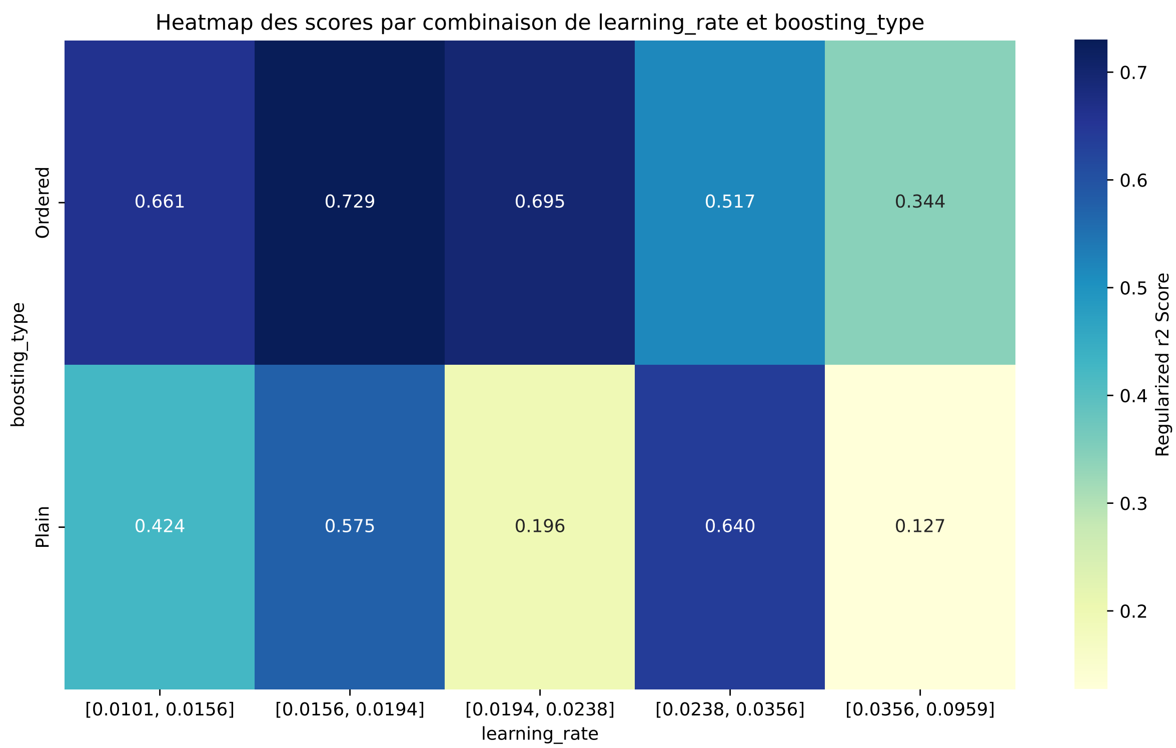
`The default value of observed=False is deprecated and will change to observed=True in a futu`



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.806	0.7684	0.7613	0.0066	0.7684	0.7657	-0.0027	0.9964	1.0489
2	0.7958	0.7602	0.7556	0.0069	0.7602	0.7613	0.0011	1.0014	1.0468
3	0.7864	0.7579	0.7513	0.0078	0.7579	0.7605	0.0026	1.0034	1.0376
4	0.7932	0.7565	0.754	0.008	0.7565	0.7601	0.0035	1.0047	1.0484
5	0.7801	0.7541	0.7487	0.007	0.7541	0.7554	0.0012	1.0016	1.0344
6	0.7911	0.7541	0.7525	0.0086	0.7541	0.7594	0.0054	1.0071	1.0491
7	0.7661	0.7446	0.7392	0.0073	0.7446	0.7435	-0.001	0.9986	1.0289
8	0.7635	0.7415	0.7392	0.0076	0.7415	0.7414	-0.0002	0.9997	1.0296
9	0.7475	0.7405	0.7298	0.0089	0.7405	0.7332	-0.0073	0.9901	1.0094
10	0.7623	0.7396	0.7367	0.0071	0.7396	0.7403	0.0007	1.0009	1.0307
---	---	---	---	---	---	---	---	---	---
75	0.9867	0.8156	0.8066	0.008	-0.0402	0.8227	0.0071	1.0088	1.2099

Hyperparameters:

Rank 1: {'reg___iterations': 700, 'reg___depth': 5, 'reg___learning_rate': 0.025410915873423163, 'reg___l2_leaf_reg': 9.475553247228575, 'reg___bagging_temperature': 4.541035484889177, 'reg___random_strength': 2.4995866428691778, 'reg___subsample': 0.720317013975402, 'reg___min_data_in_leaf': 48, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.8209452534318471, 'reg___border_count': 71, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 5039.130027285362}

Rank 2: {'reg___iterations': 500, 'reg___depth': 7, 'reg___learning_rate': 0.016656707892830077, 'reg___l2_leaf_reg': 10.936773293641428, 'reg___bagging_temperature': 4.183812717567081, 'reg___random_strength': 0.6911076961045061, 'reg___subsample': 0.6718192501025233, 'reg___min_data_in_leaf': 10, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.8364581442434892, 'reg___border_count': 47, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 8619.301895516395}

Rank 3: {'reg___iterations': 500, 'reg___depth': 7, 'reg___learning_rate': 0.016891860146872946, 'reg___l2_leaf_reg': 18.728030193906886, 'reg___bagging_temperature': 4.285000064742104, 'reg___random_strength': 0.6873390207508813, 'reg___subsample': 0.8519491496323758, 'reg___min_data_in_leaf': 48, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.7861024310815352, 'reg___border_count': 108, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 6518.675459382472}

Rank 4: {'reg___iterations': 500, 'reg___depth': 7, 'reg___learning_rate': 0.01619558375596506, 'reg___l2_leaf_reg': 32.84866152887886, 'reg___bagging_temperature': 4.253683107493004, 'reg___random_strength': 0.045760757185013046, 'reg___subsample': 0.753547298186511, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 3, 'reg___rsm':

0.8317925363636849, 'reg___border_count': 46, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 3630.7828533113193}

Rank 5: {'reg___iterations': 600, 'reg___depth': 6, 'reg___learning_rate': 0.01705745264578418, 'reg___l2_leaf_reg': 23.865474728570305, 'reg___bagging_temperature': 3.9639134279260086, 'reg___random_strength': 0.9526550032270874, 'reg___subsample': 0.8410538653836868, 'reg___min_data_in_leaf': 46, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.862920799870267, 'reg___border_count': 47, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 7334.3572194768385}

Rank 6: {'reg___iterations': 500, 'reg___depth': 7, 'reg___learning_rate': 0.016234125743842392, 'reg___l2_leaf_reg': 32.612595832428724, 'reg___bagging_temperature': 4.271898383477527, 'reg___random_strength': 0.05913808361027306, 'reg___subsample': 0.8924946938090805, 'reg___min_data_in_leaf': 36, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.8314393028481026, 'reg___border_count': 49, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 6701.61781669394}

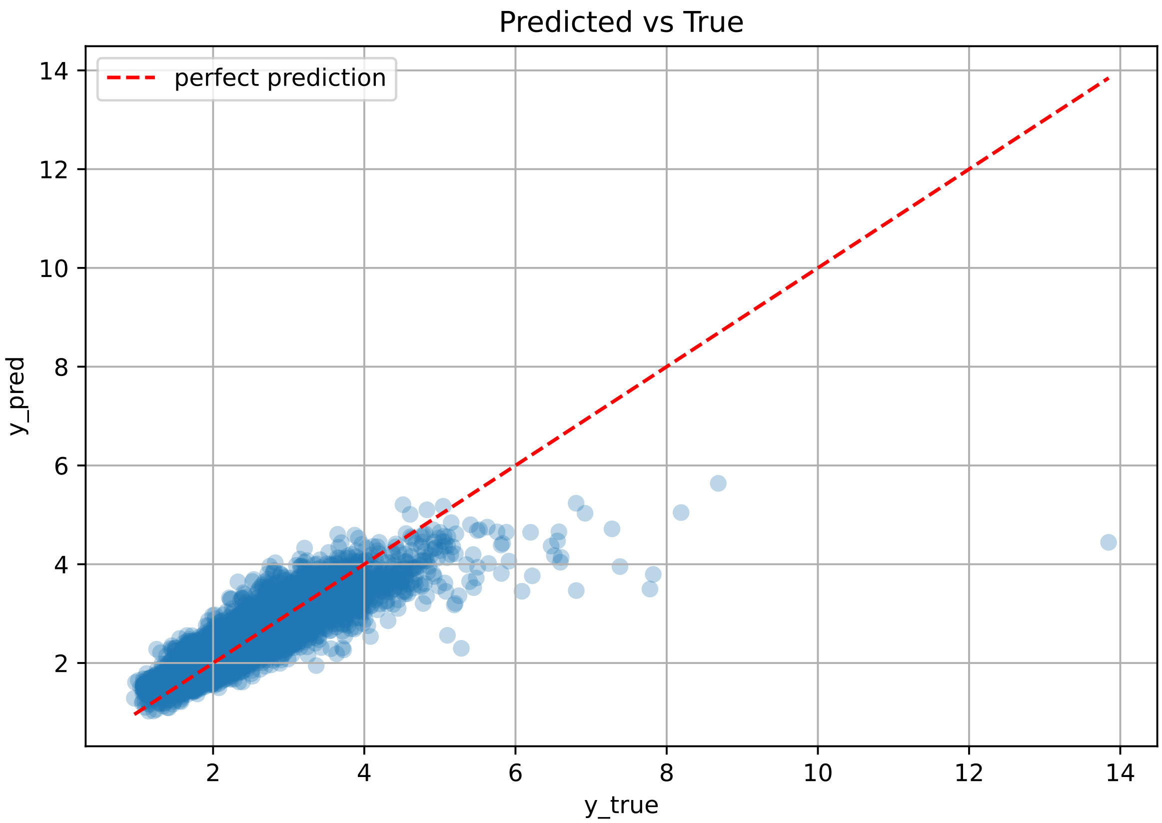
Rank 7: {'reg___iterations': 600, 'reg___depth': 4, 'reg___learning_rate': 0.021689863226557304, 'reg___l2_leaf_reg': 14.910828228127702, 'reg___bagging_temperature': 3.5934147104016505, 'reg___random_strength': 0.041671081478908994, 'reg___subsample': 0.7408559772773025, 'reg___min_data_in_leaf': 46, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.8140116699568187, 'reg___border_count': 128, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 2203.1529565422466}

Rank 8: {'reg___iterations': 500, 'reg___depth': 5, 'reg___learning_rate': 0.016965211478751713, 'reg___l2_leaf_reg': 11.692355553627491, 'reg___bagging_temperature': 4.259324572956628, 'reg___random_strength': 0.02721791239966178, 'reg___subsample': 0.6702573641179076, 'reg___min_data_in_leaf': 48, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.8280013298136547, 'reg___border_count': 48, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 3574.281038124375}

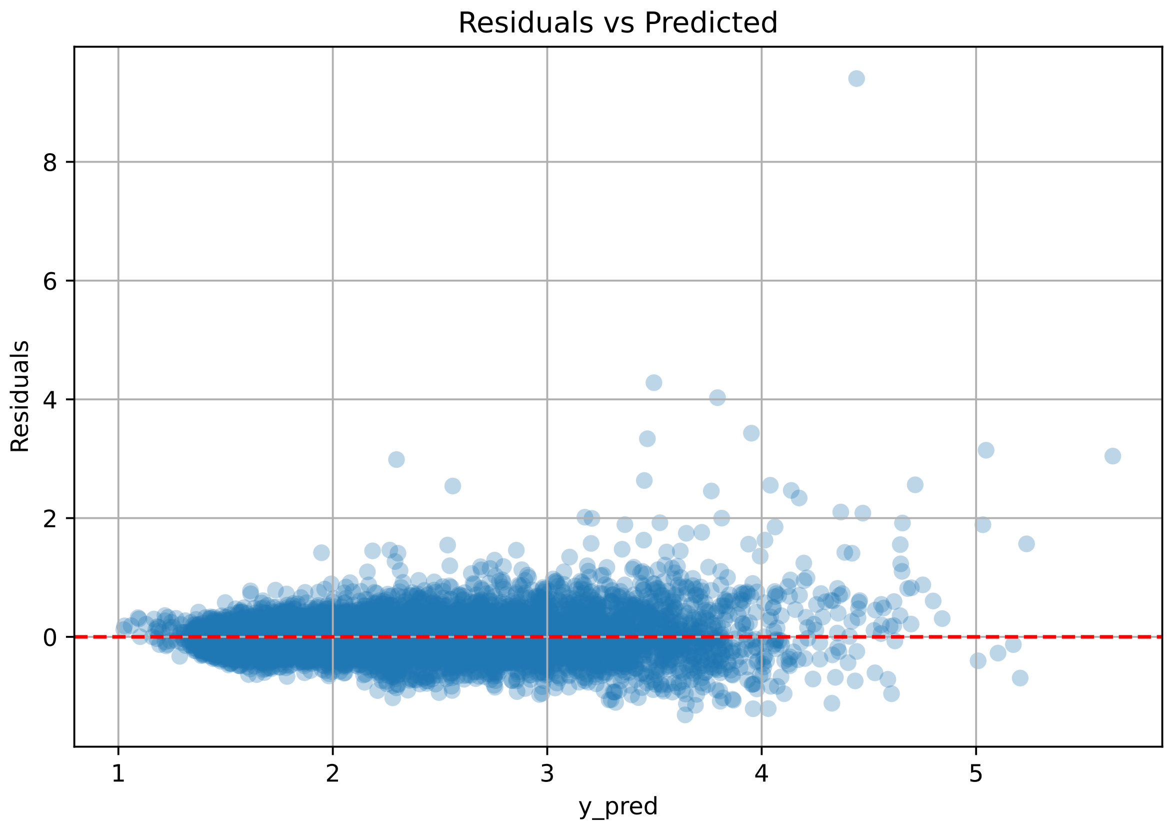
Rank 9: {'reg___iterations': 600, 'reg___depth': 4, 'reg___learning_rate': 0.02393526504152566, 'reg___l2_leaf_reg': 10.495039847602827, 'reg___bagging_temperature': 4.501606033174903, 'reg___random_strength': 1.8055807008607951, 'reg___subsample': 0.7248185857017988, 'reg___min_data_in_leaf': 50, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.8081023525899037, 'reg___border_count': 124, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 252.94265085271152}

Rank 10: {'reg___iterations': 500, 'reg___depth': 5, 'reg___learning_rate': 0.01669963975187979, 'reg___l2_leaf_reg': 11.306624429028755, 'reg___bagging_temperature': 4.265328971782635, 'reg___random_strength': 0.000344168080042756, 'reg___subsample': 0.7535697263237877, 'reg___min_data_in_leaf': 11, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.8283312512708929, 'reg___border_count': 47, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 3866.9395182342887}

3.6.2 Scatter



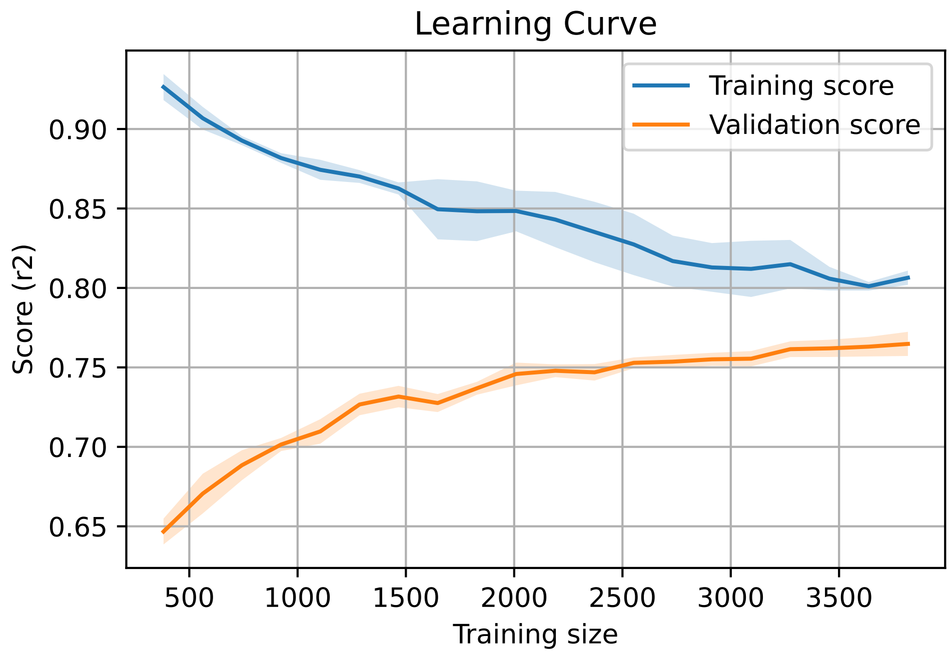
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 7793, **Len(training)** : 5456,

Len(training_real) : 3820, **Len(test_of_training)** : 1636, **Len(test)** : 2337



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.559	0.589	0.012	0.571	0.572	-0.017	0.971	0.95	nan
LGBM	0.778	0.744	0.007	0.74	0.76	0.015	1.02	1.045	0.7443
CatBoost	0.806	0.768	0.007	0.761	0.766	-0.003	0.996	1.049	0.7684

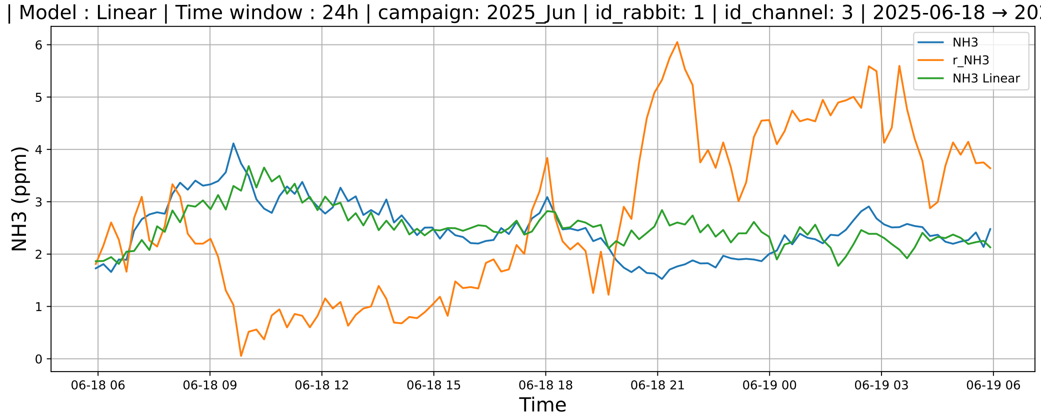
4 Plot

4.1 Standalone plots

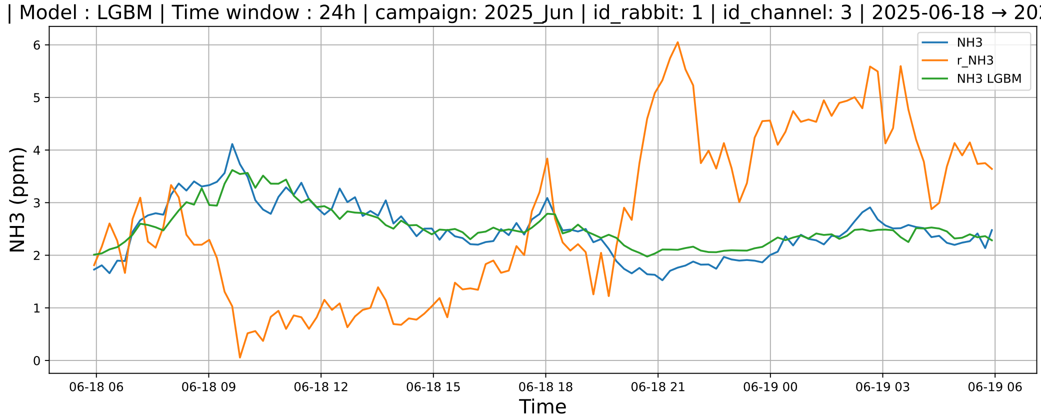
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

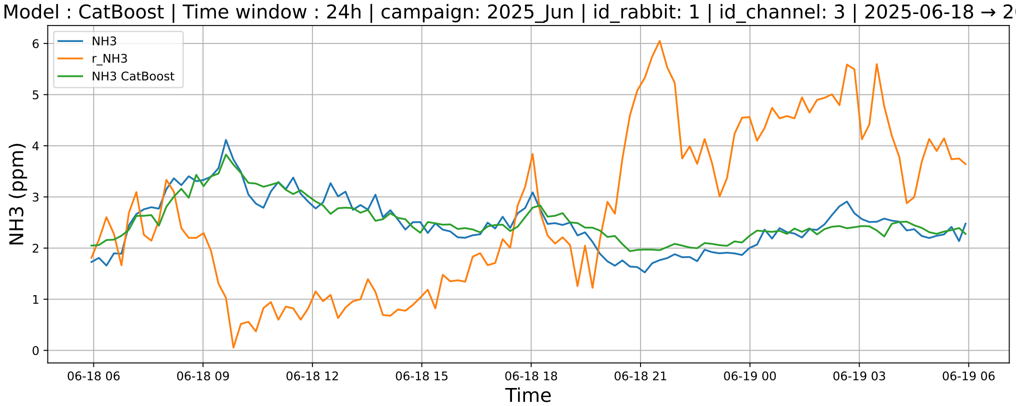
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

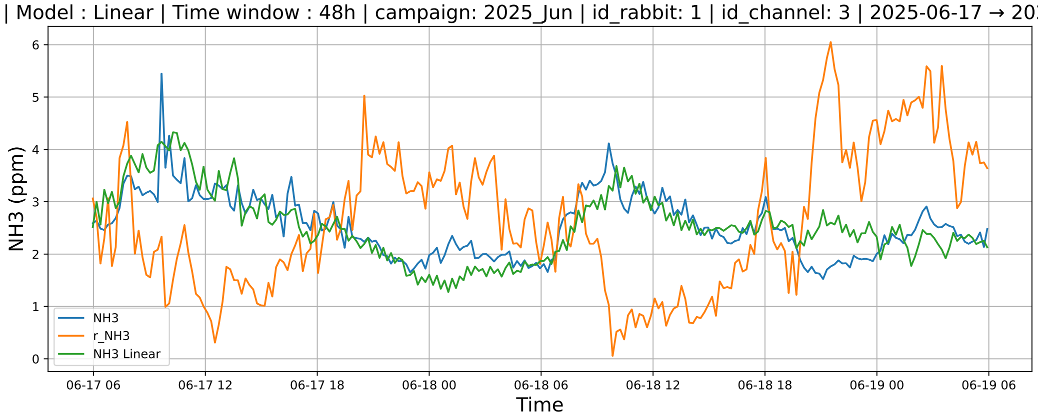


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

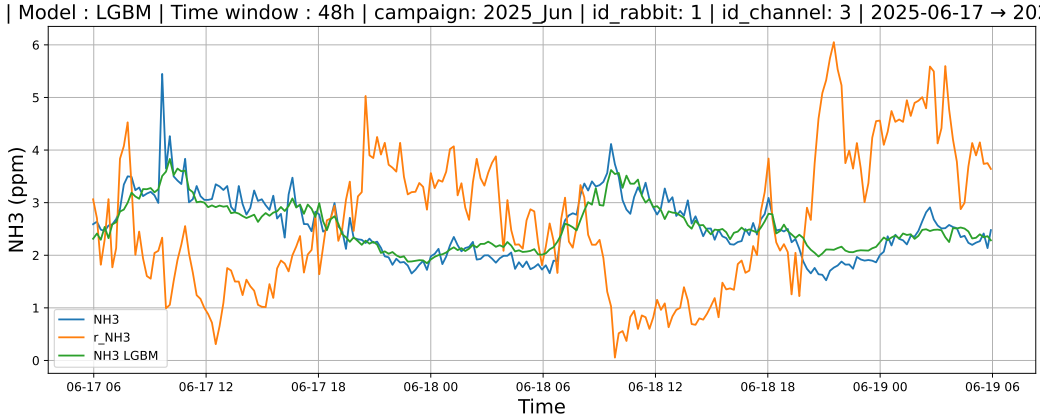


4.1.1.2 Time window : 48

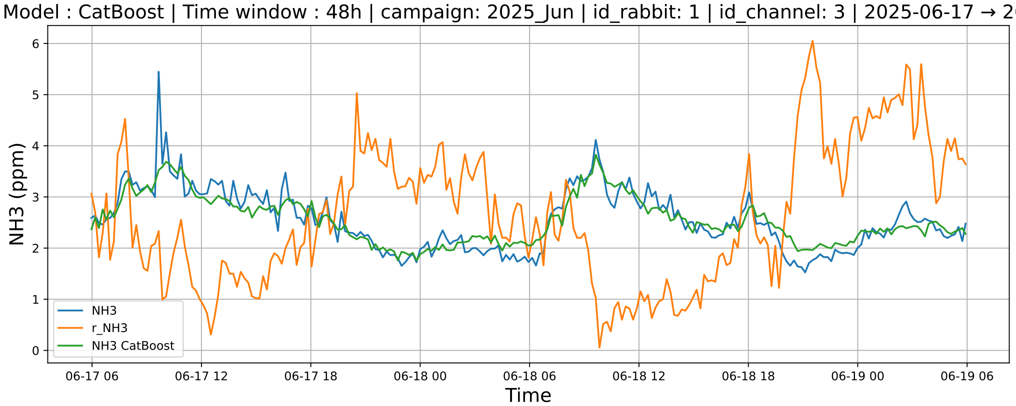
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



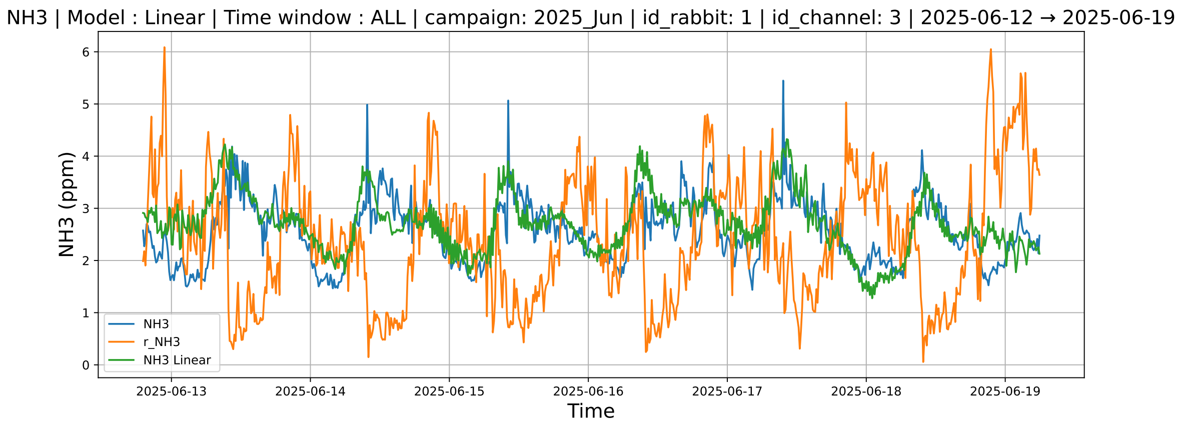
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

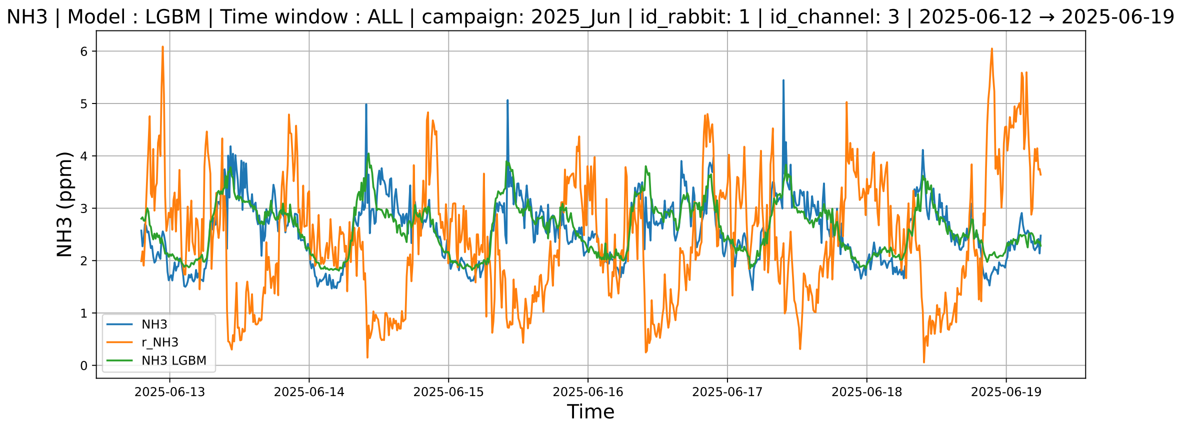


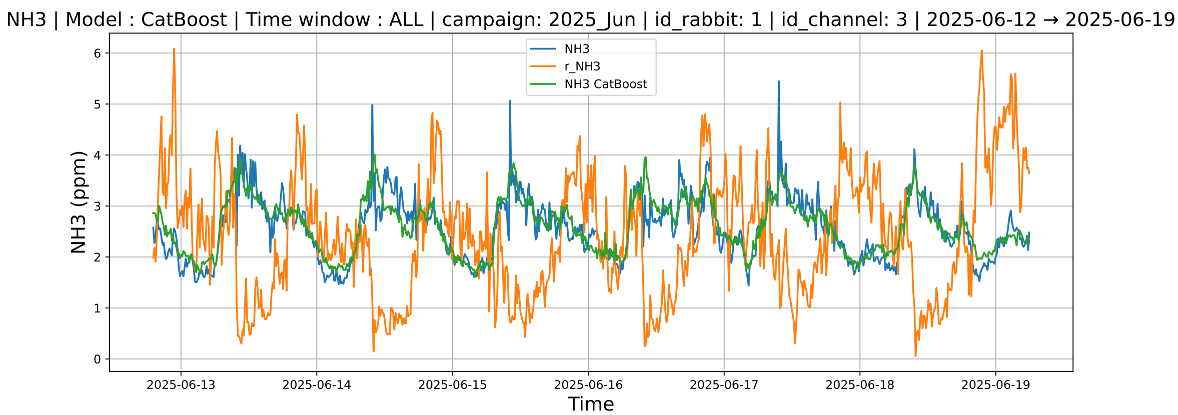
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.1.1.3 Time window : ALL



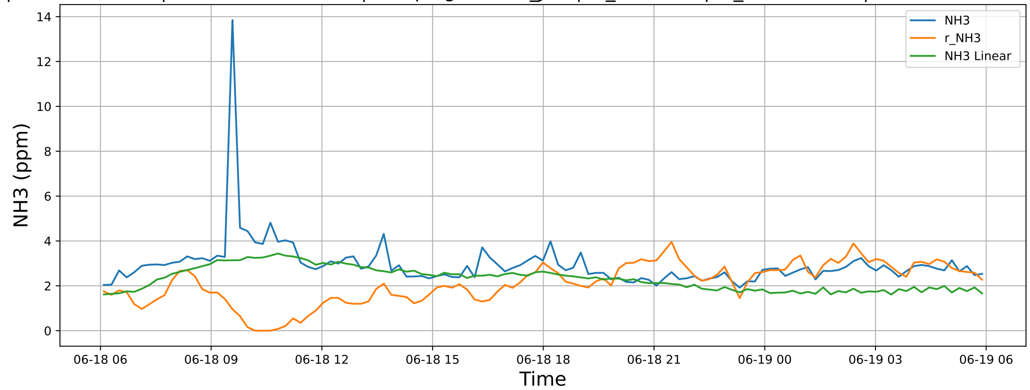




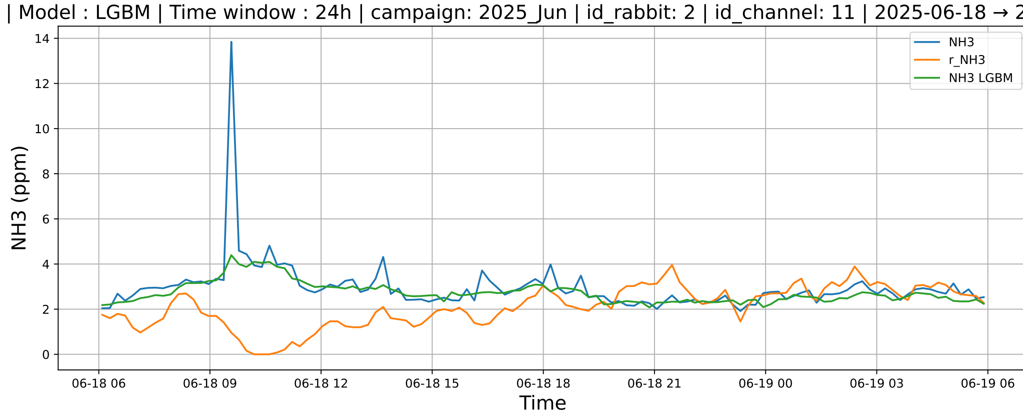
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

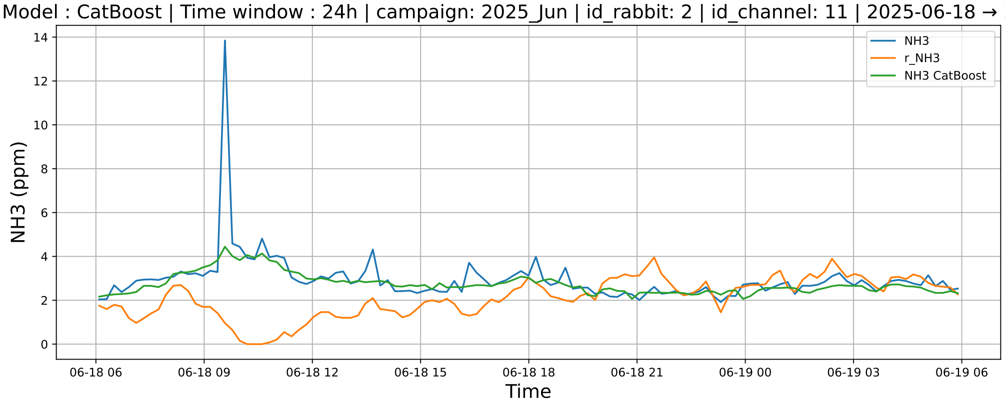
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

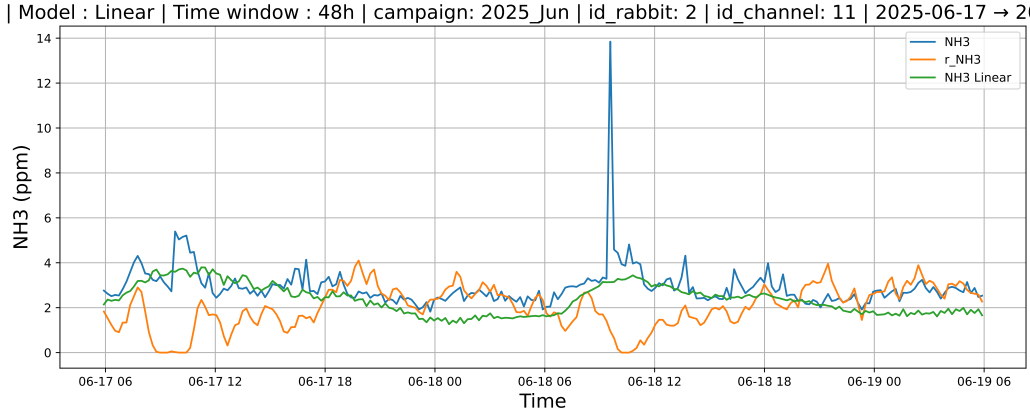


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

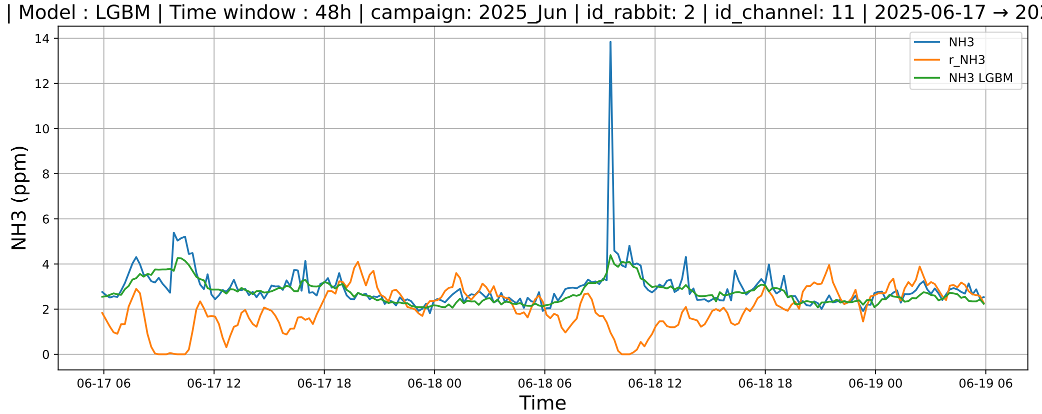


4.1.2.2 Time window : 48

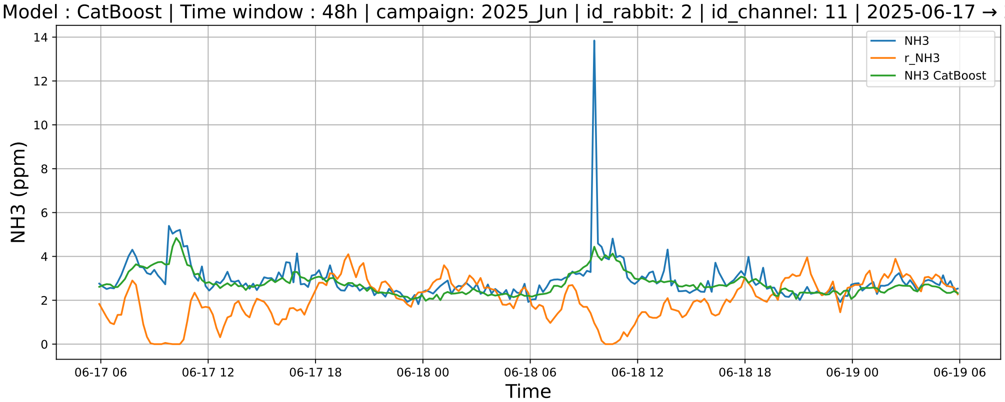
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

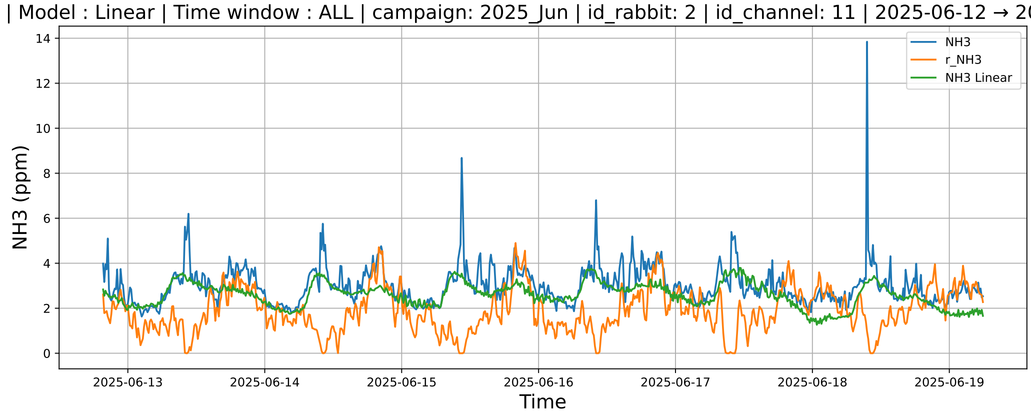


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

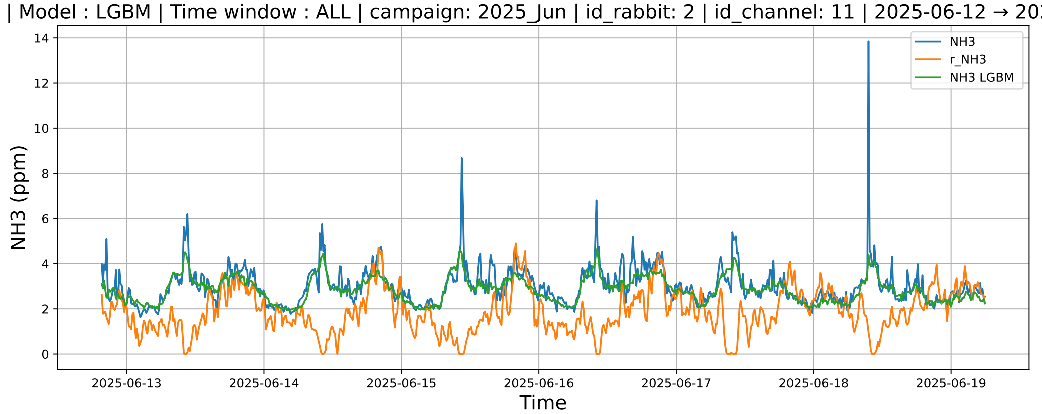


4.1.2.3 Time window : ALL

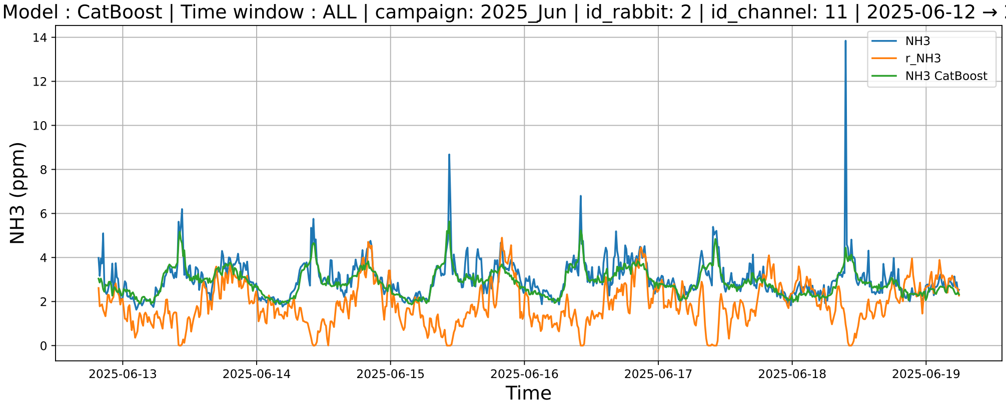
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

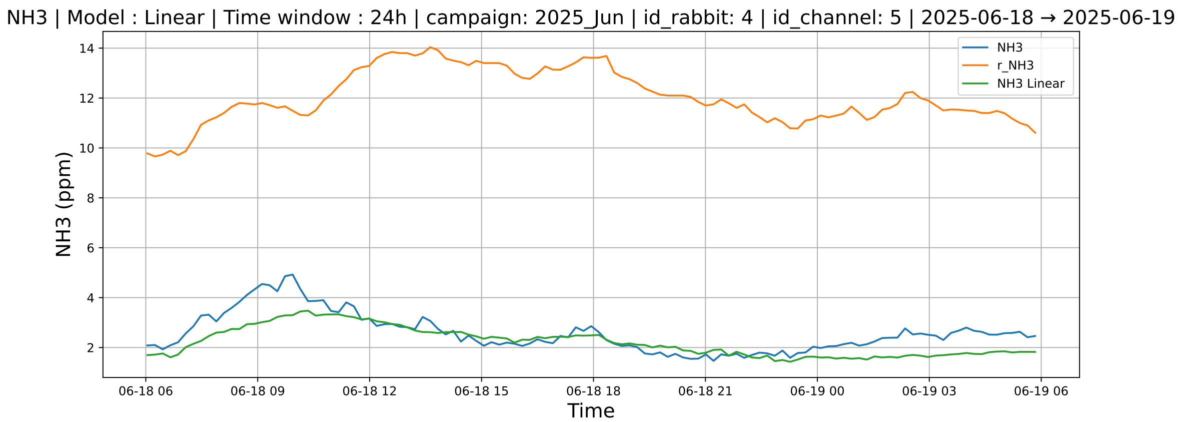


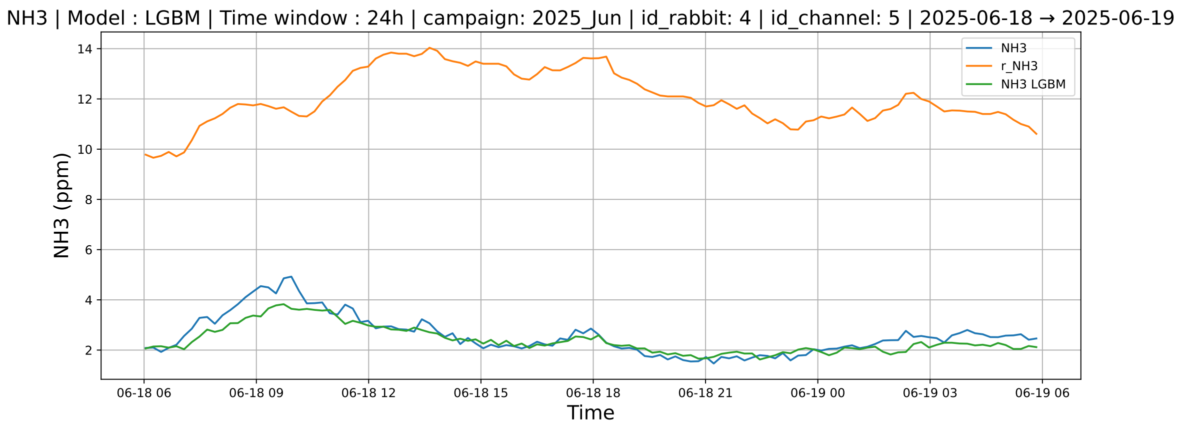
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

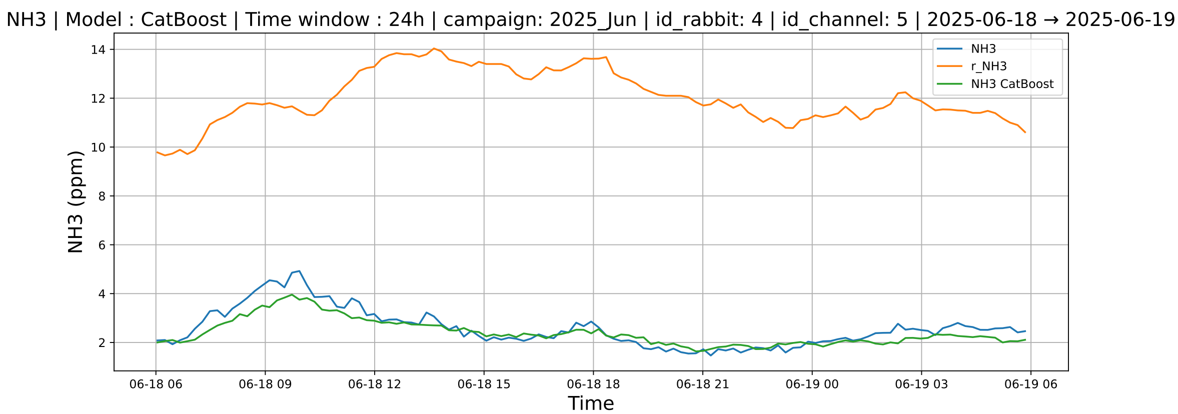


4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

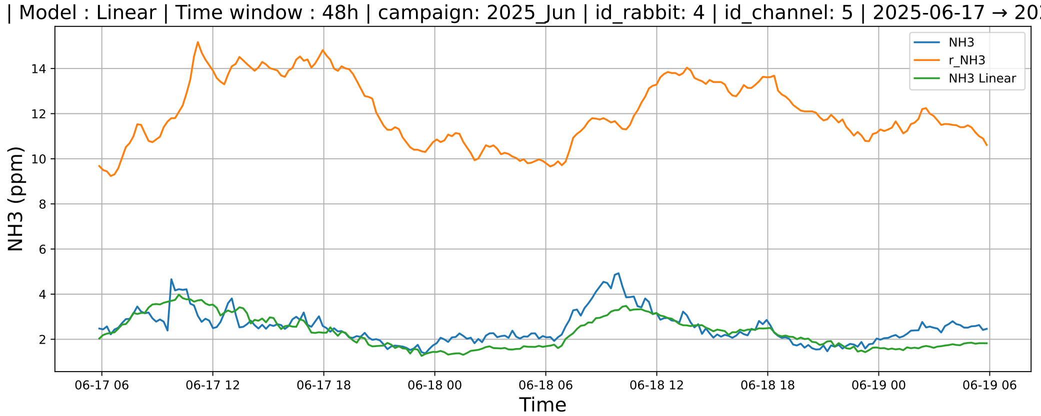




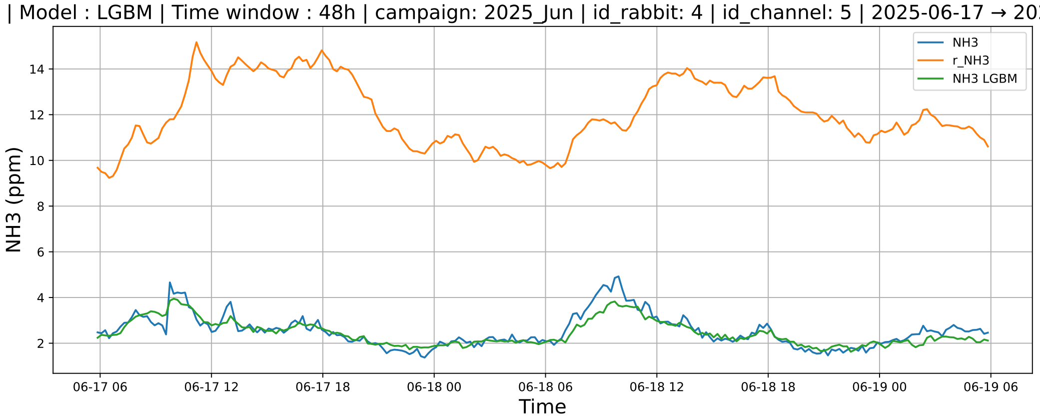


4.1.3.2 Time window : 48

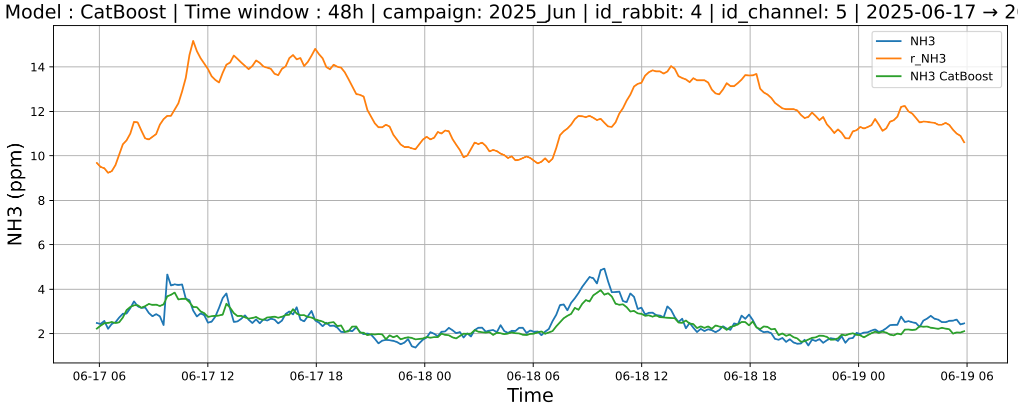
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



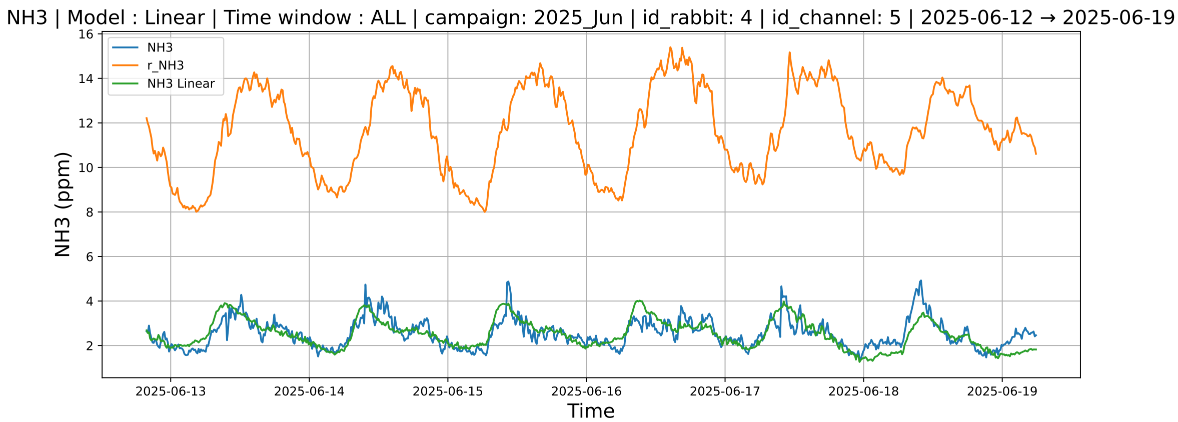
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

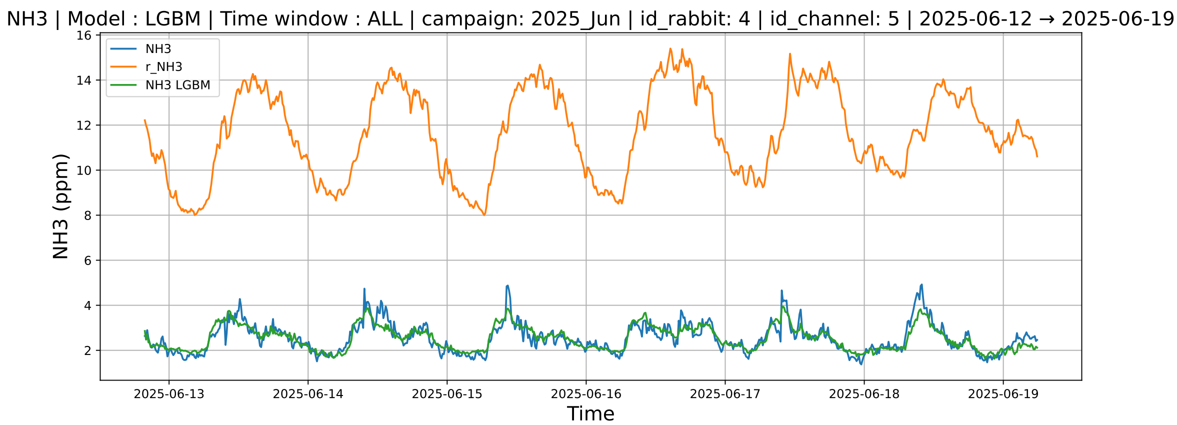


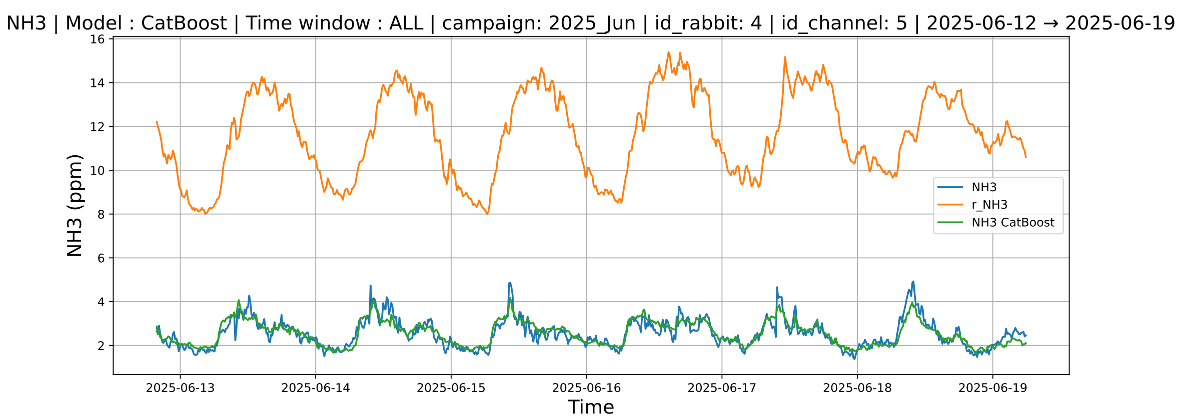
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.1.3.3 Time window : ALL



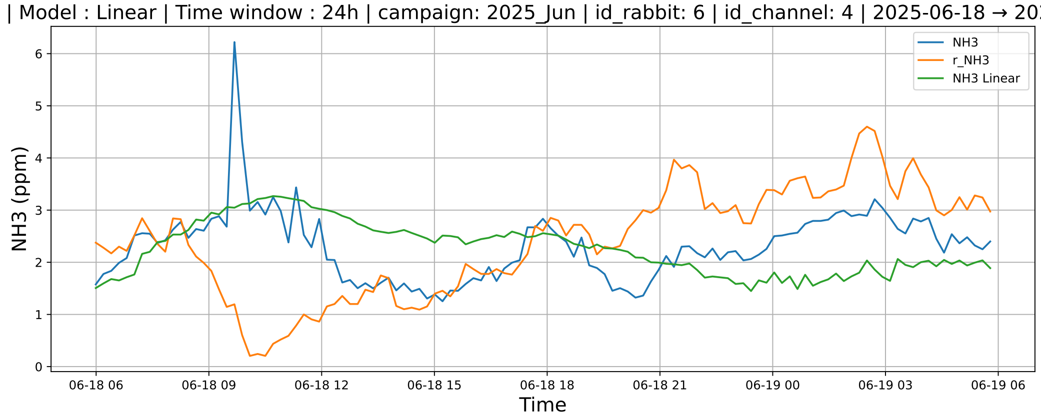




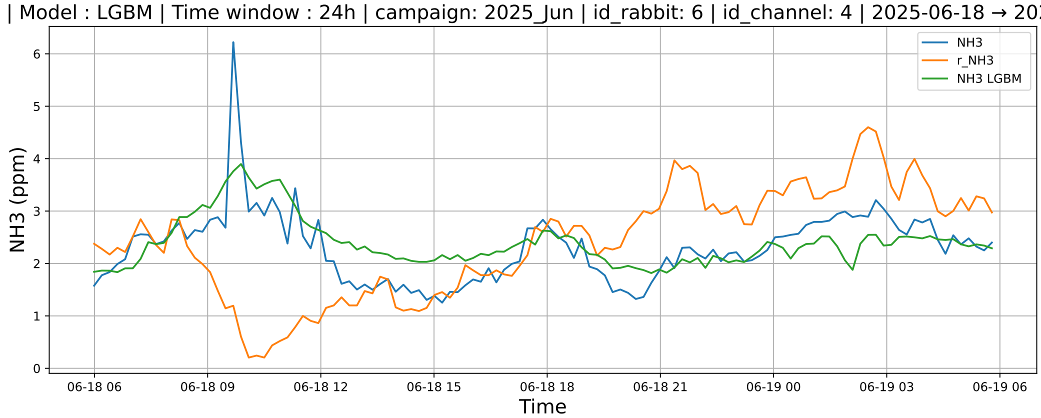
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

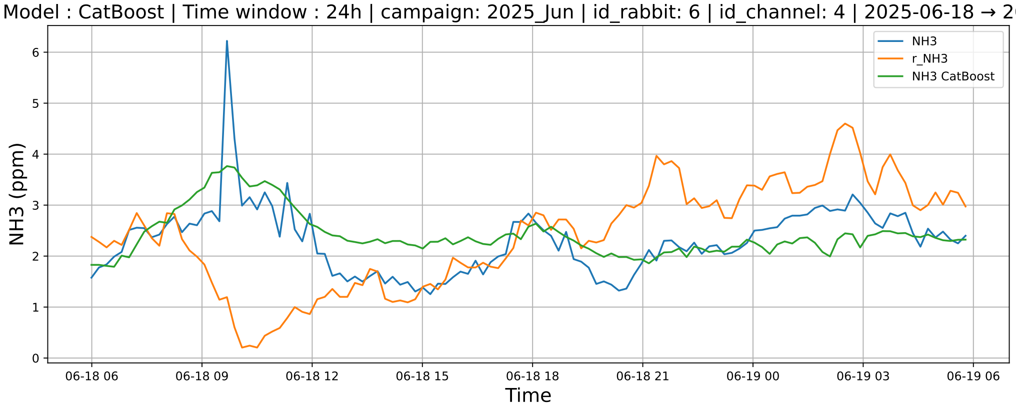
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

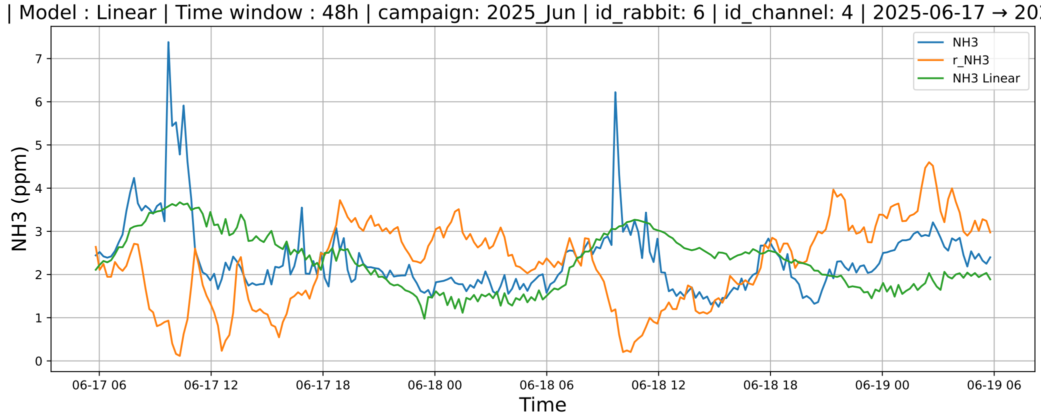


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

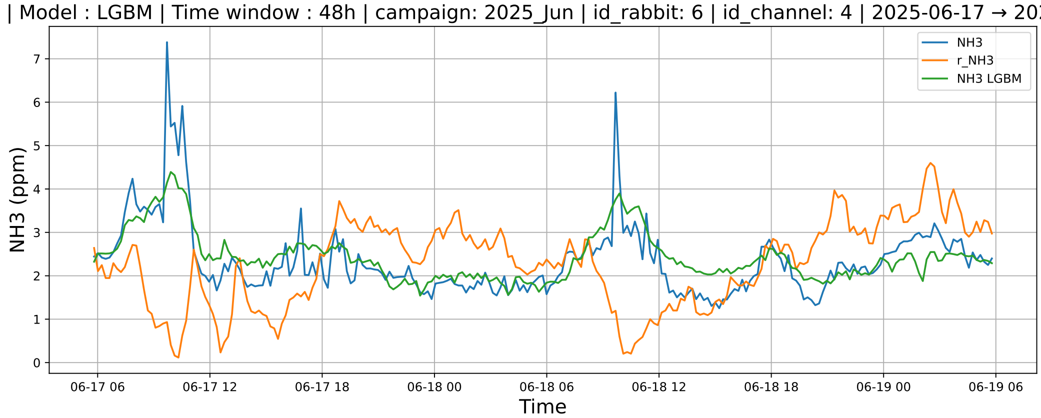


4.1.4.2 Time window : 48

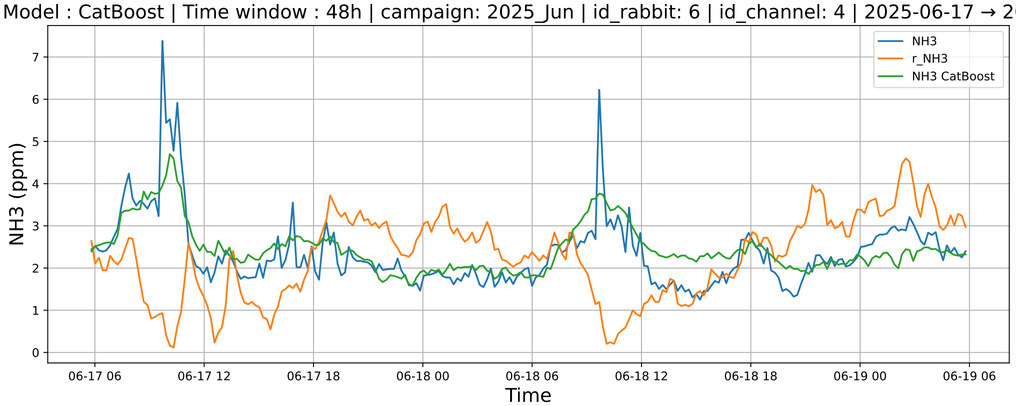
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

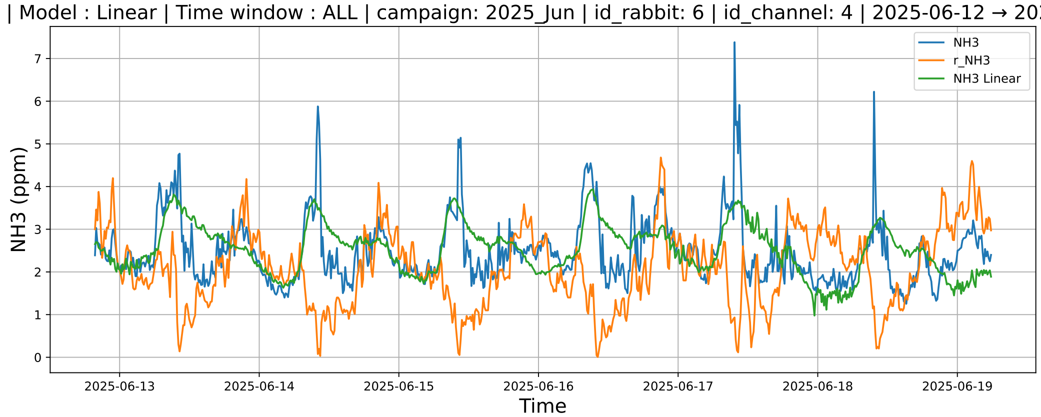


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

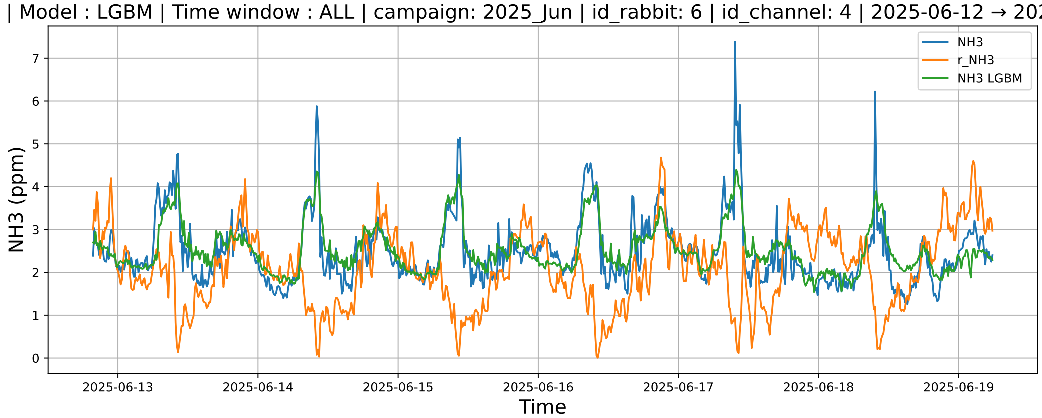


4.1.4.3 Time window : ALL

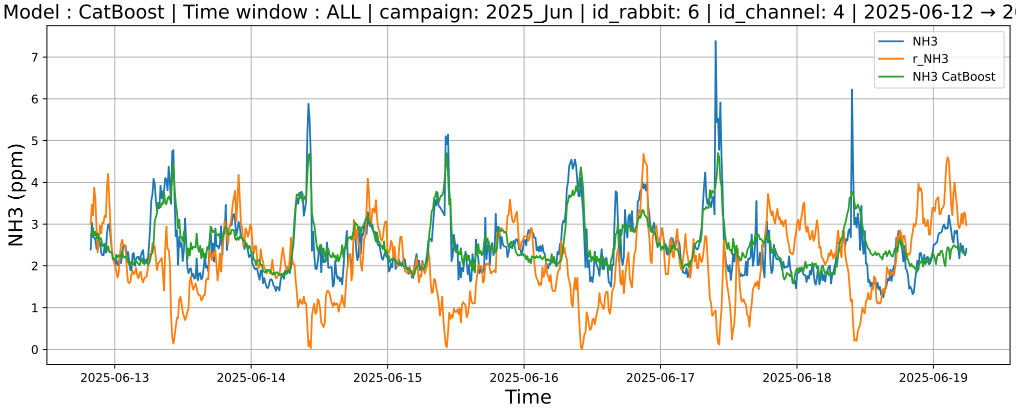
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



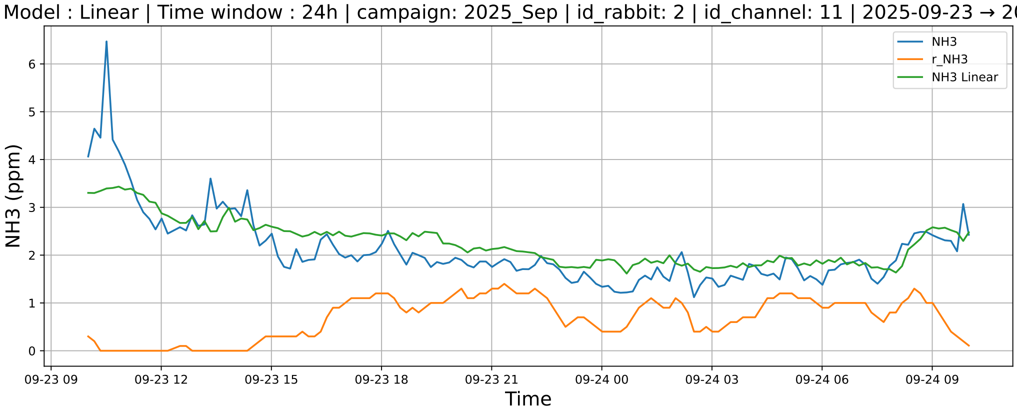
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



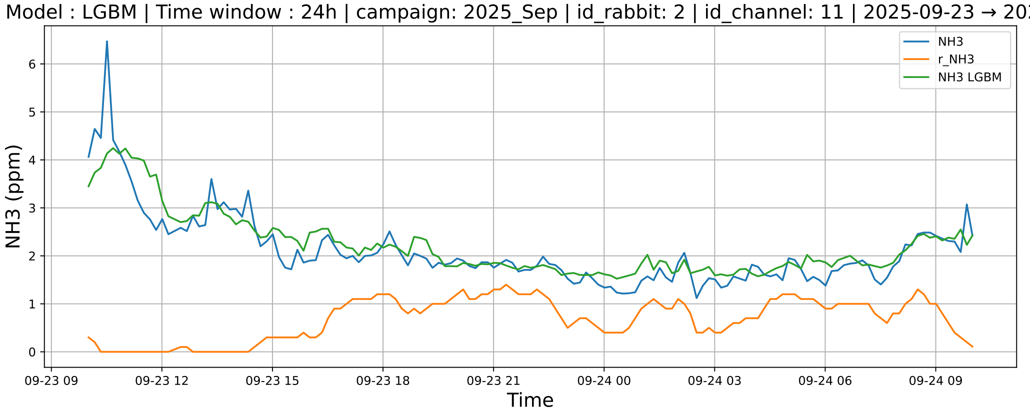
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

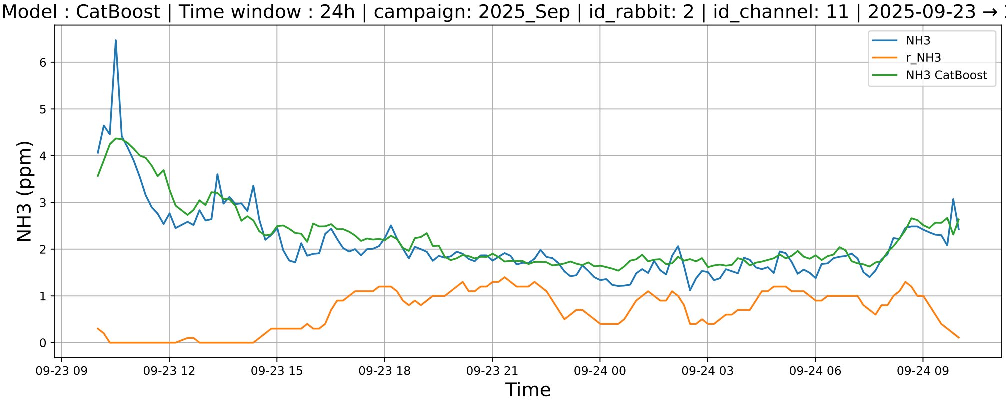
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

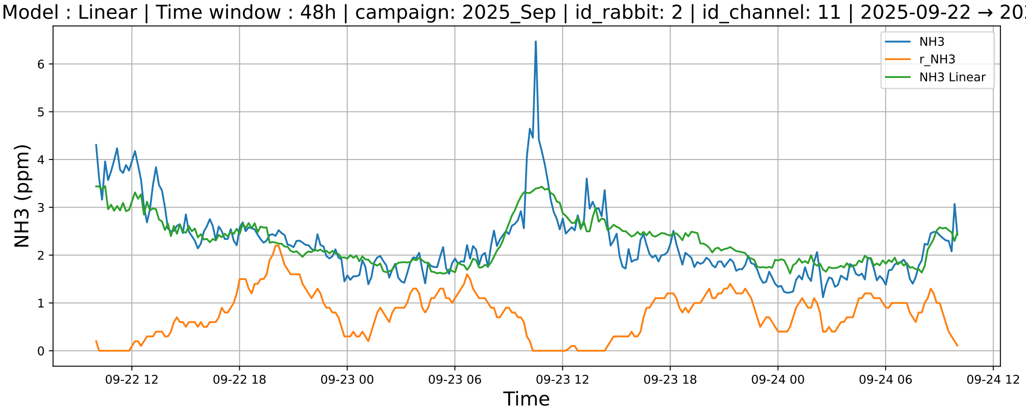


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

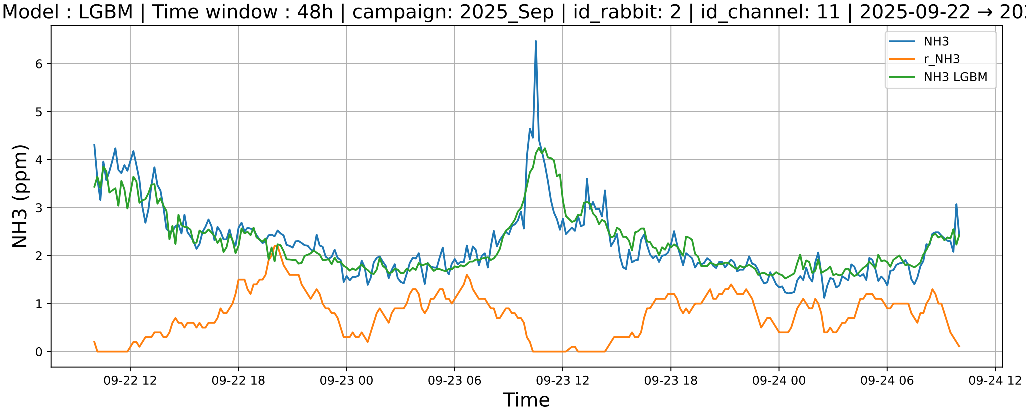


4.1.5.2 Time window : 48

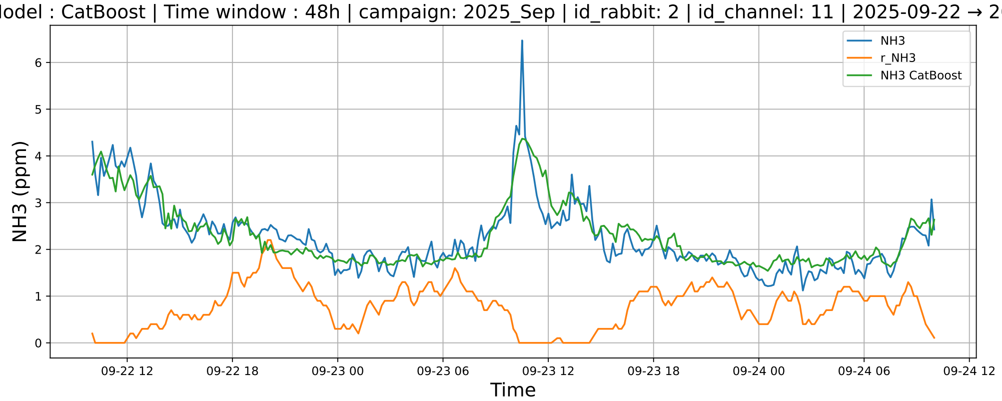
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

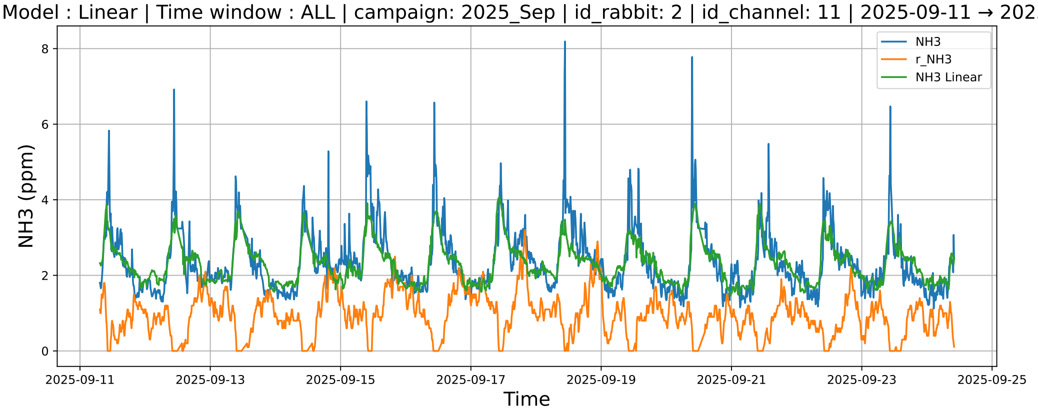


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

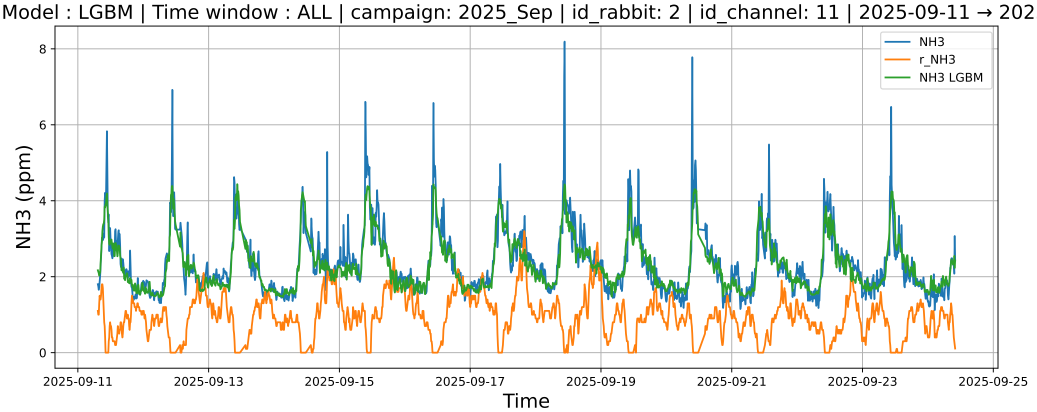


4.1.5.3 Time window : ALL

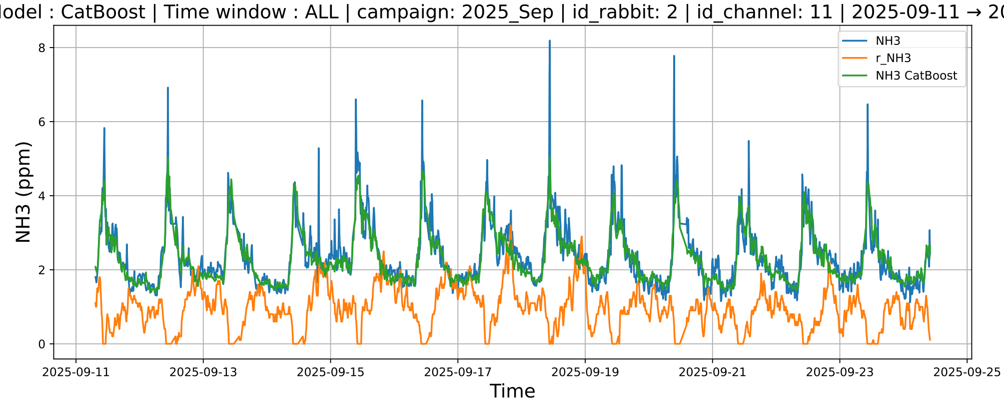
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

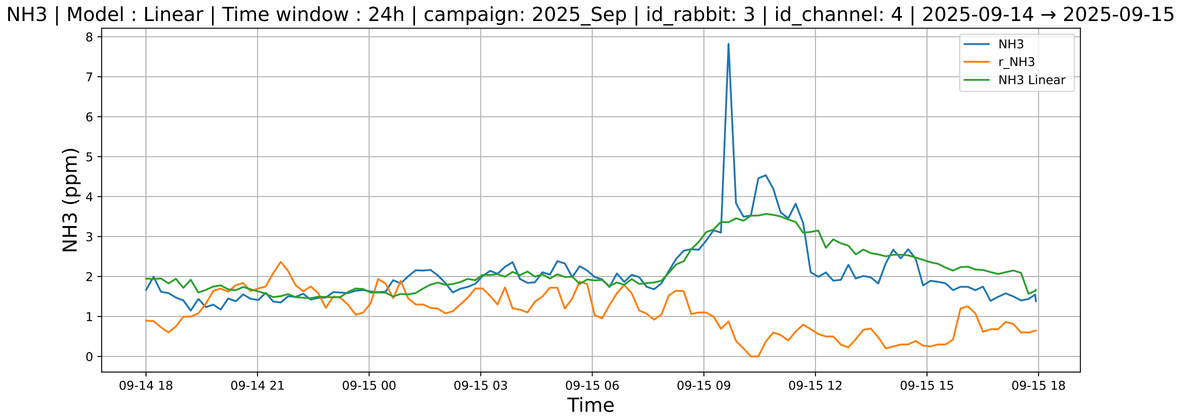


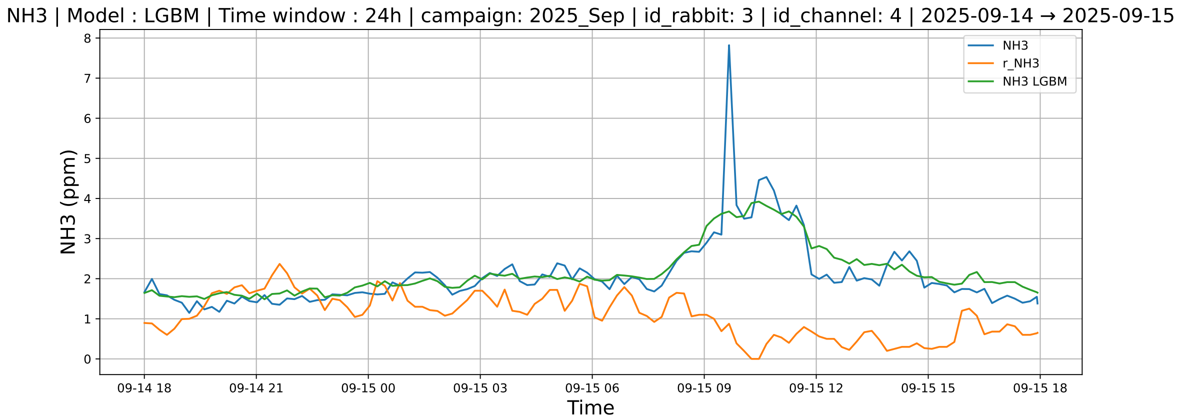
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

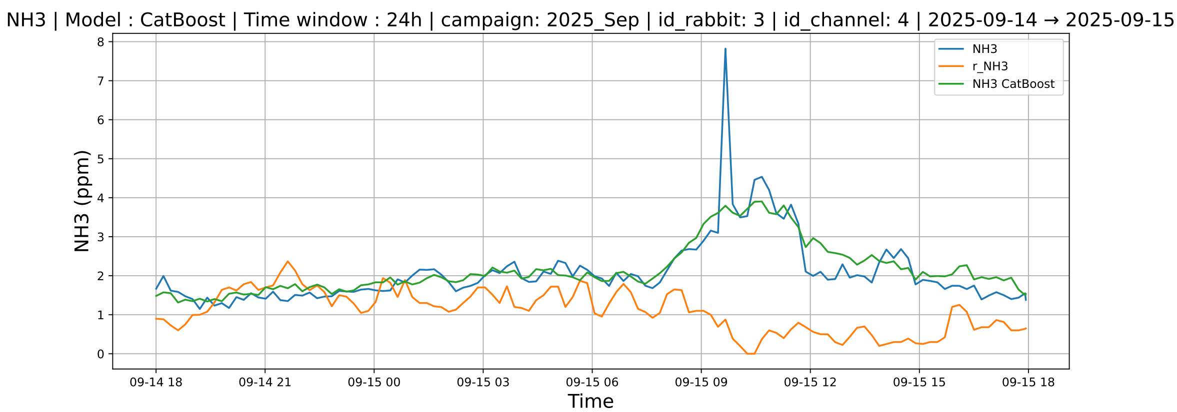


4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

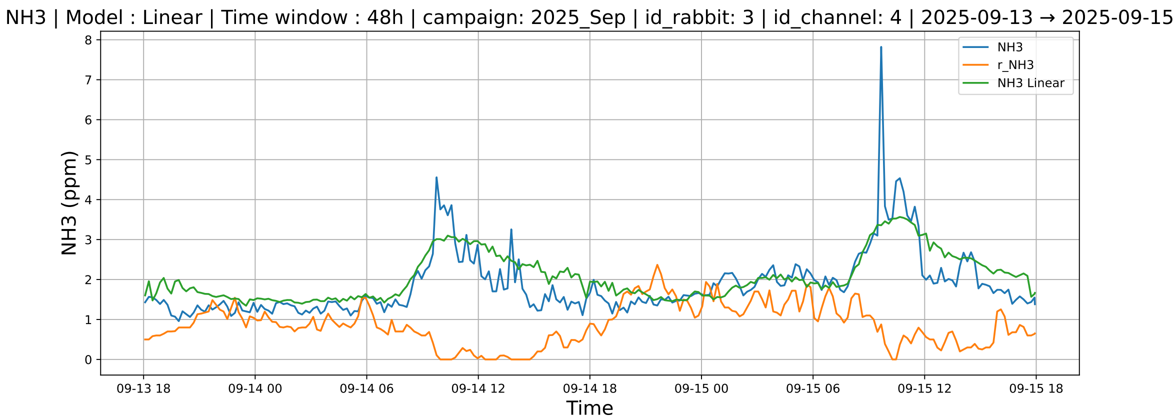
4.1.6.1 Time window : 24

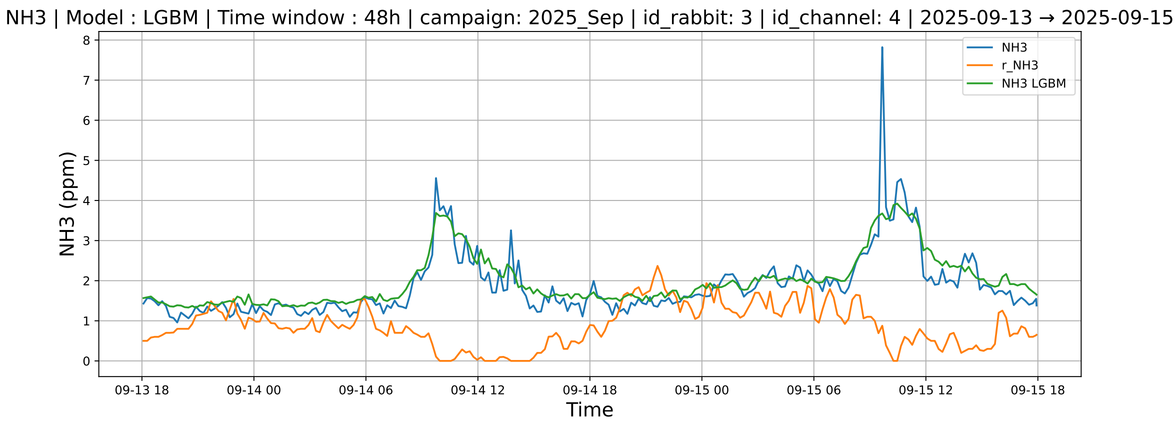


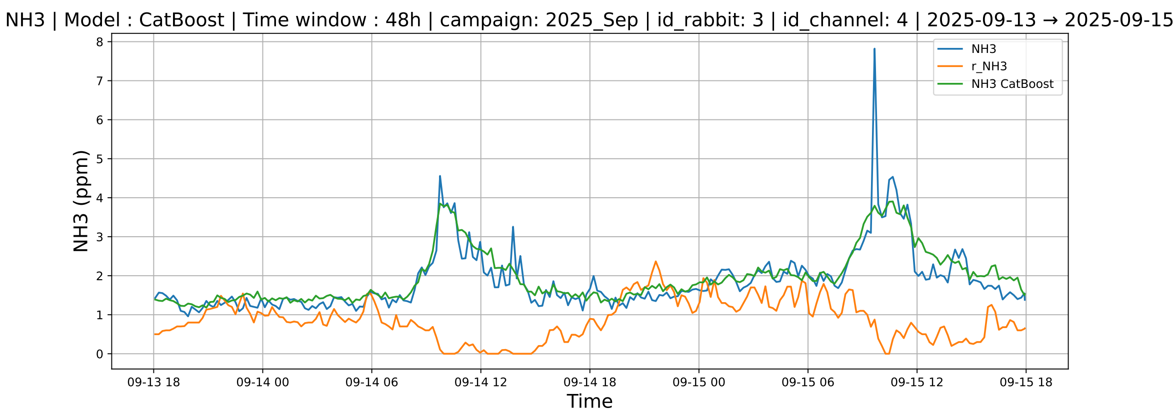




4.1.6.2 Time window : 48

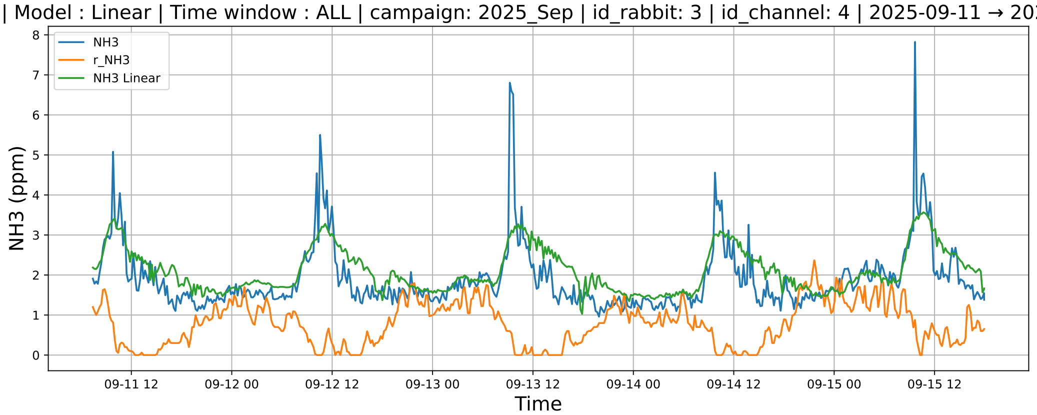




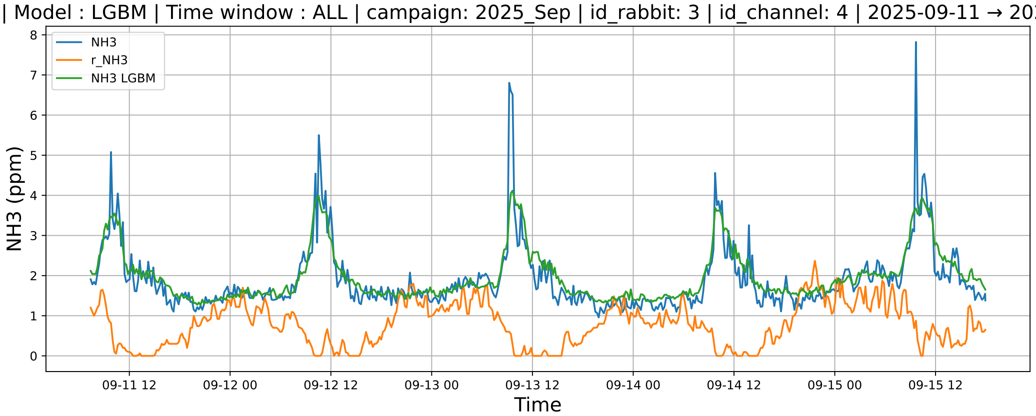


4.1.6.3 Time window : ALL

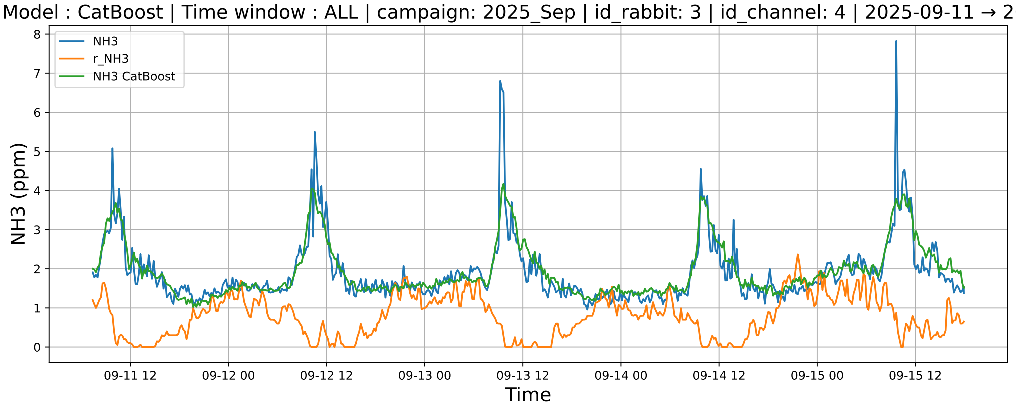
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

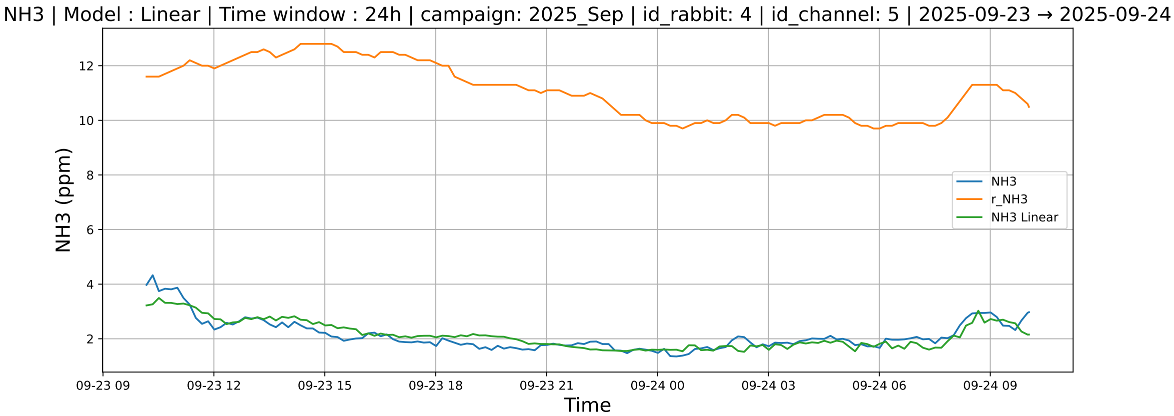


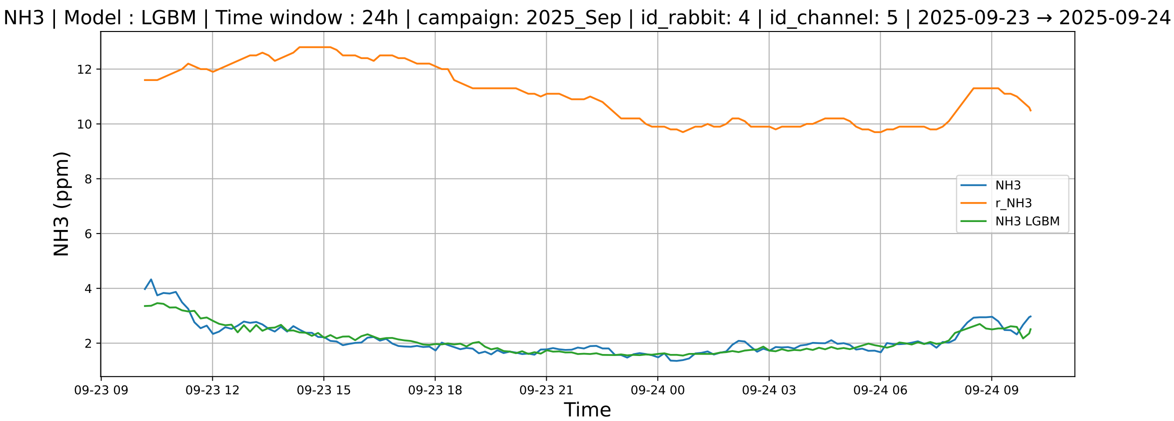
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

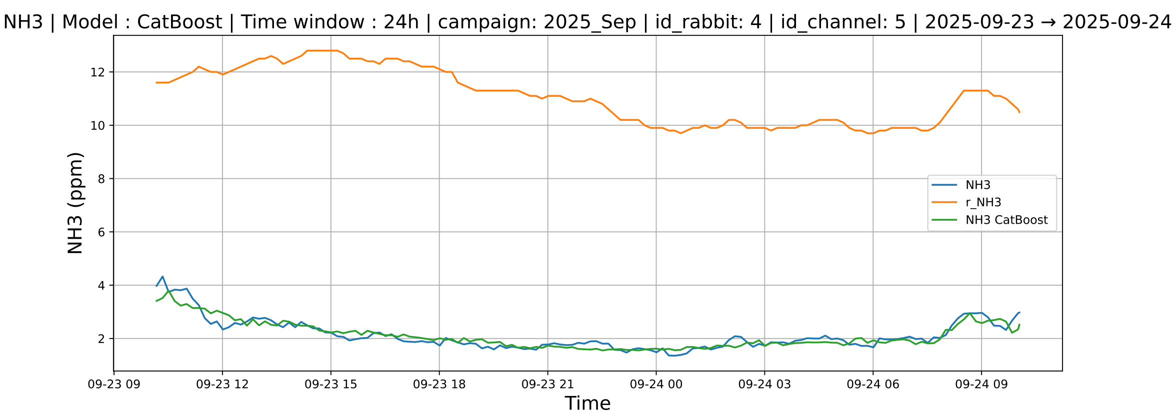


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

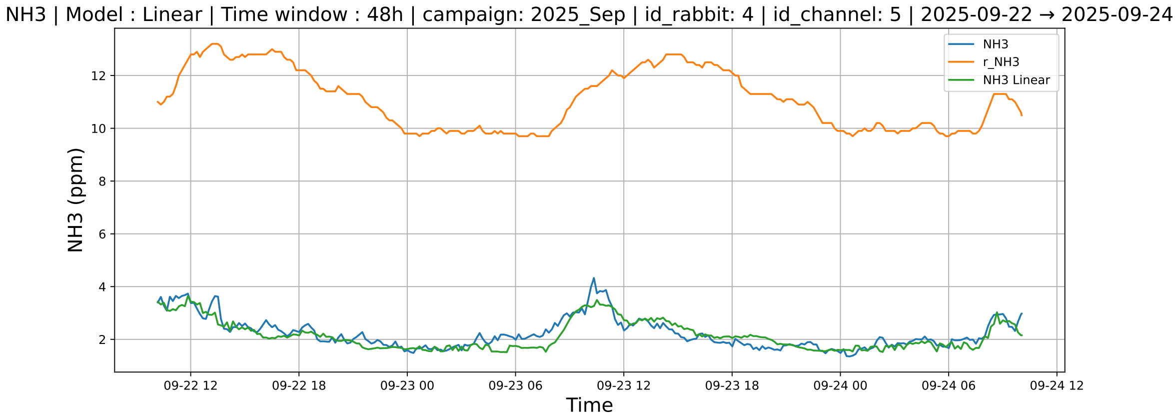
4.1.7.1 Time window : 24

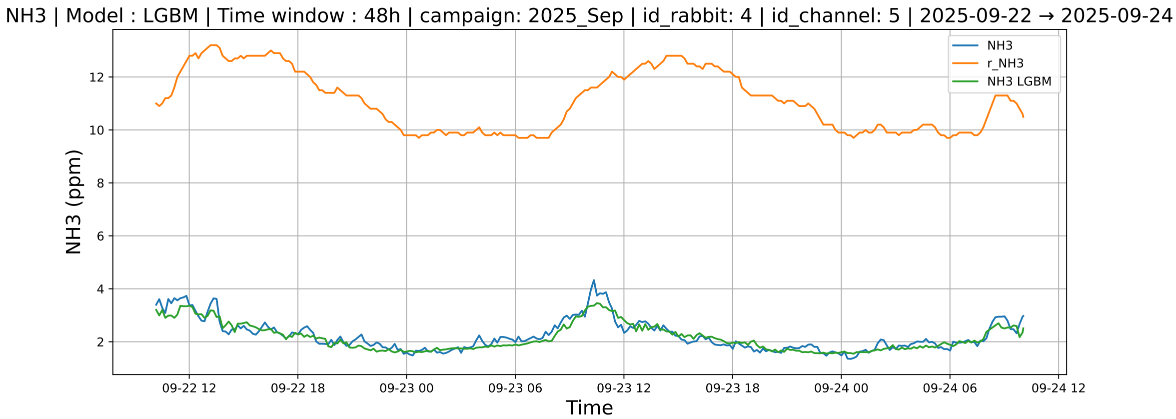


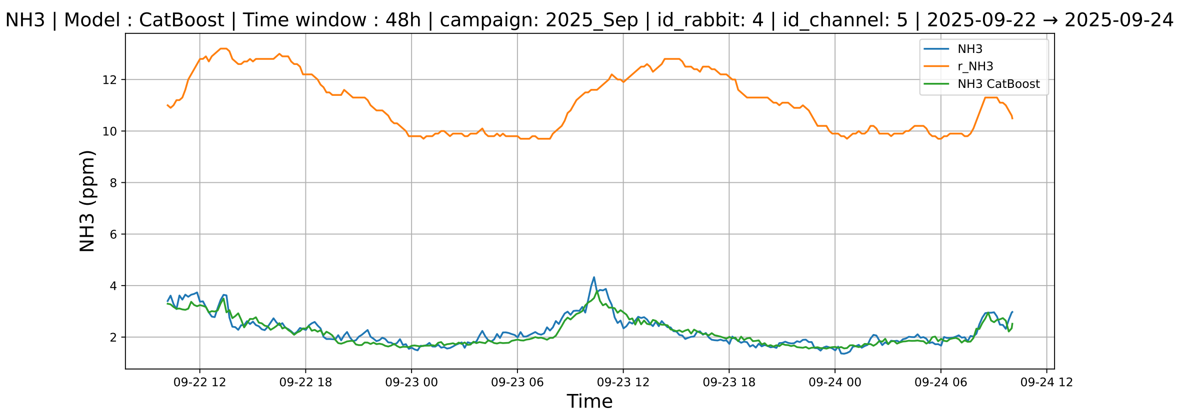




4.1.7.2 Time window : 48

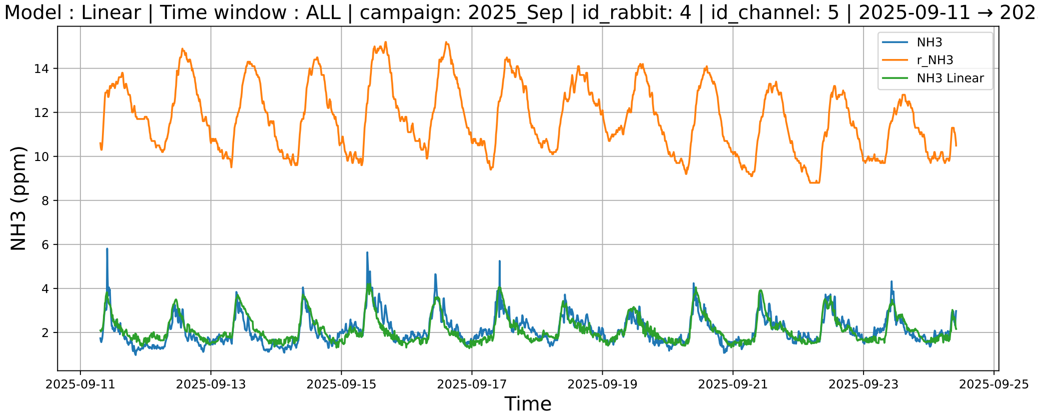




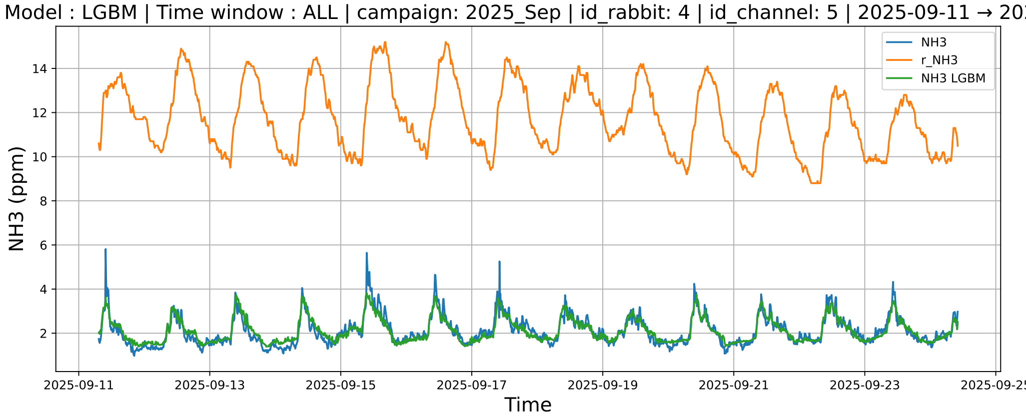


4.1.7.3 Time window : ALL

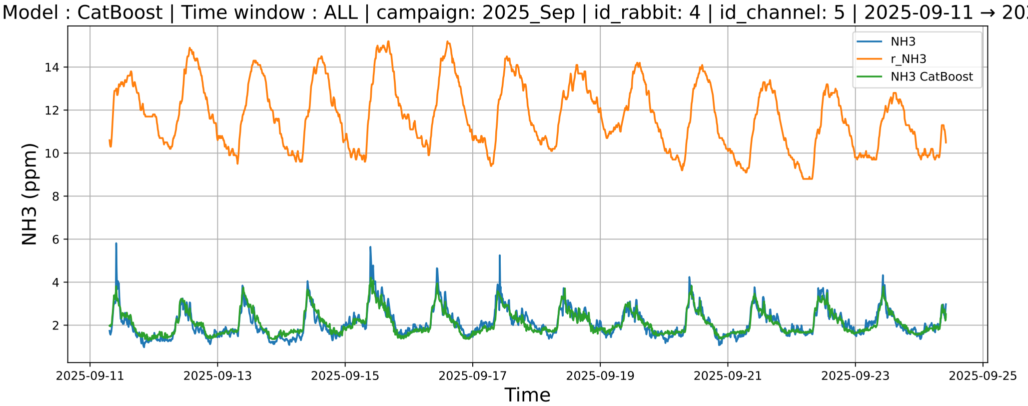
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

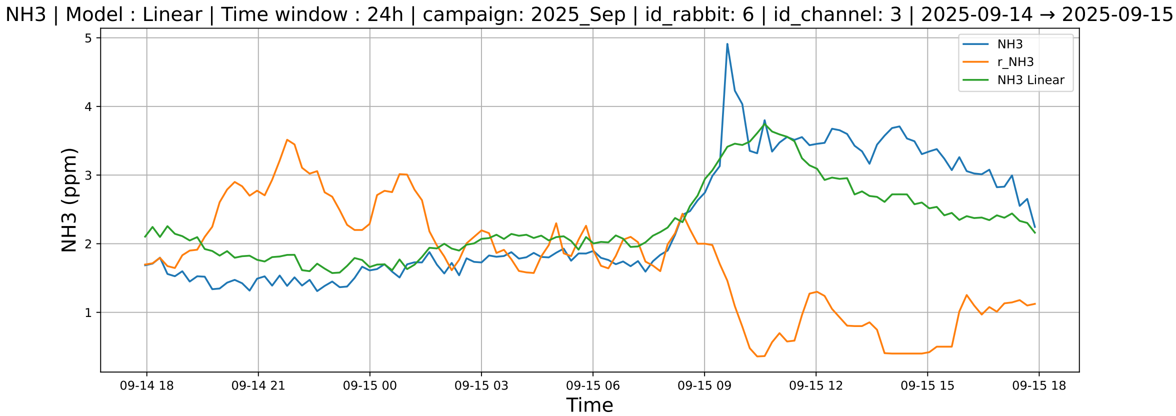


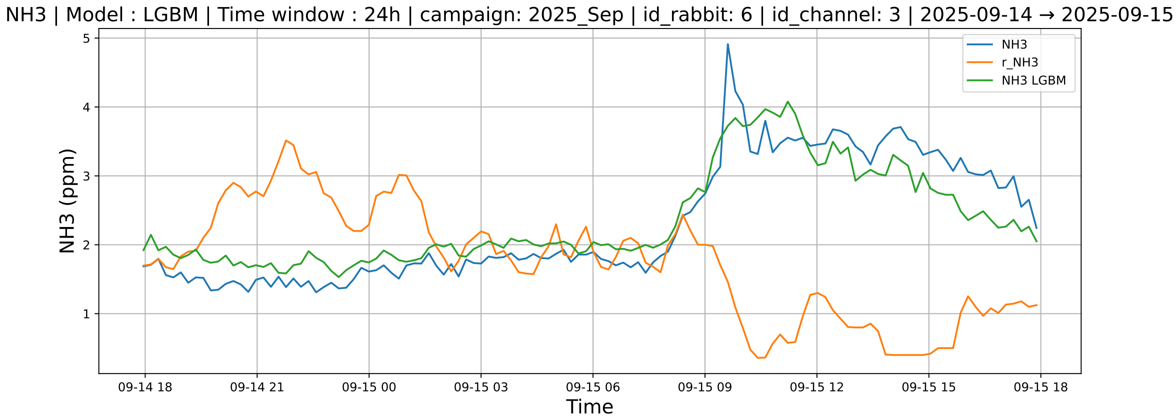
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

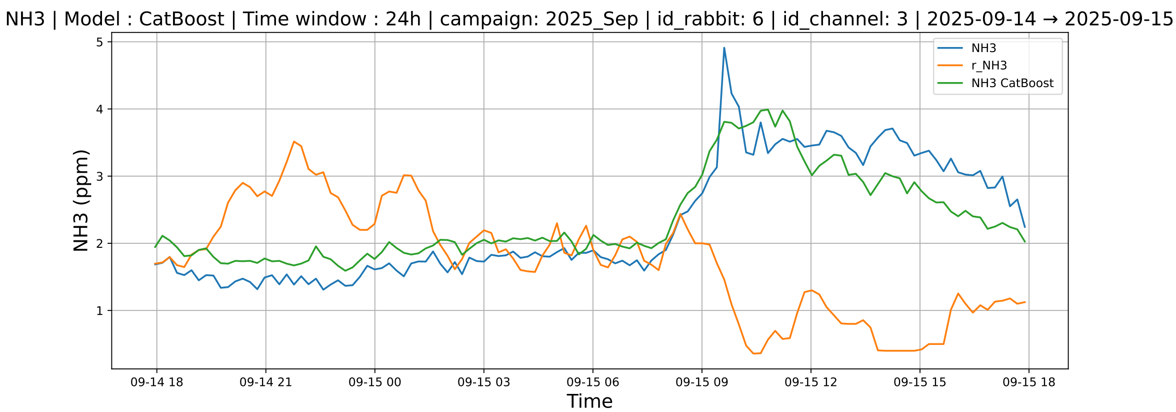


4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

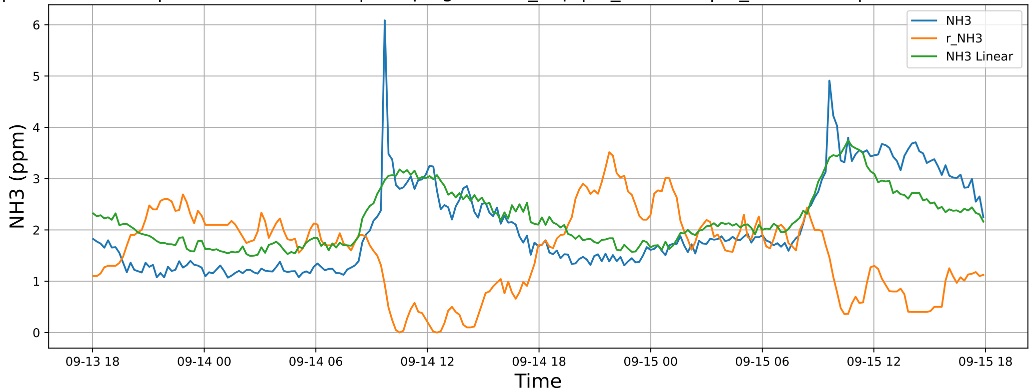




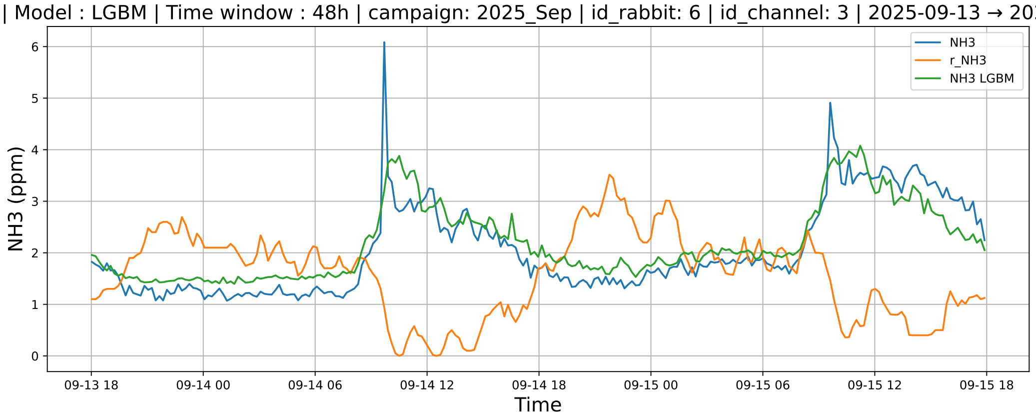


4.1.8.2 Time window : 48

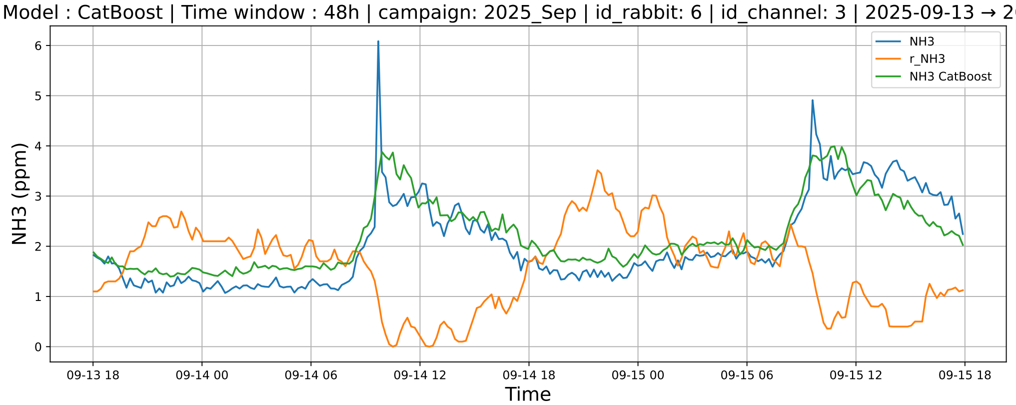
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

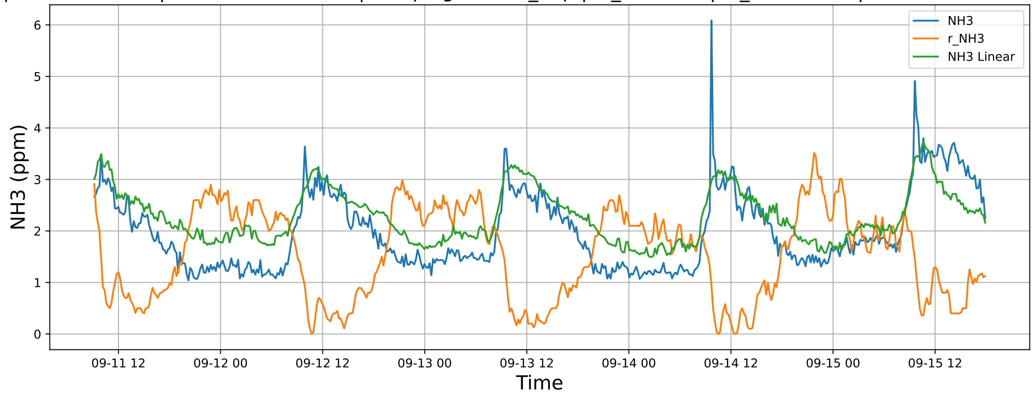


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

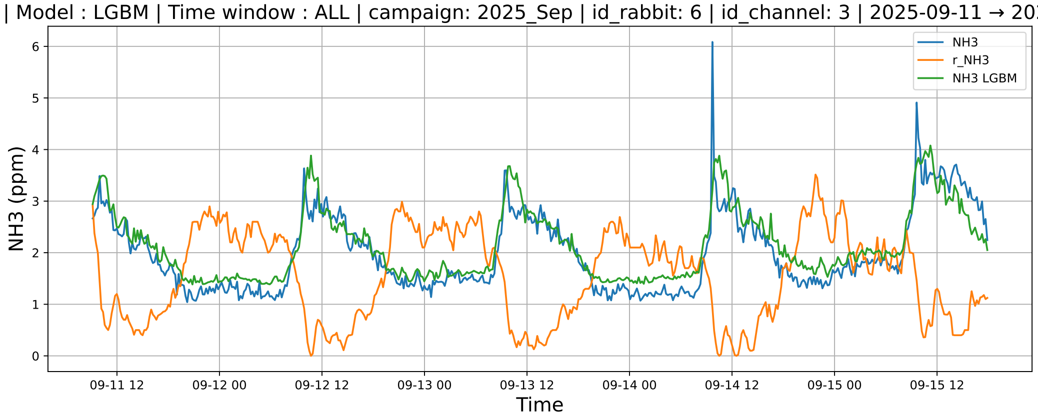


4.1.8.3 Time window : ALL

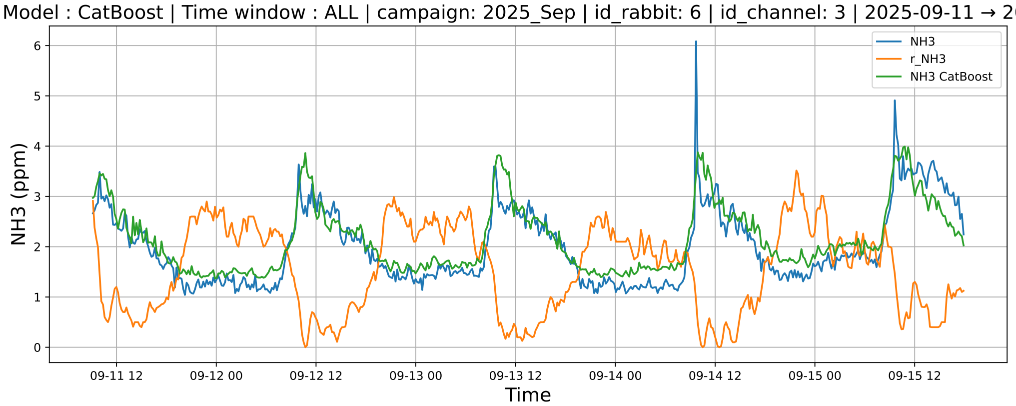
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



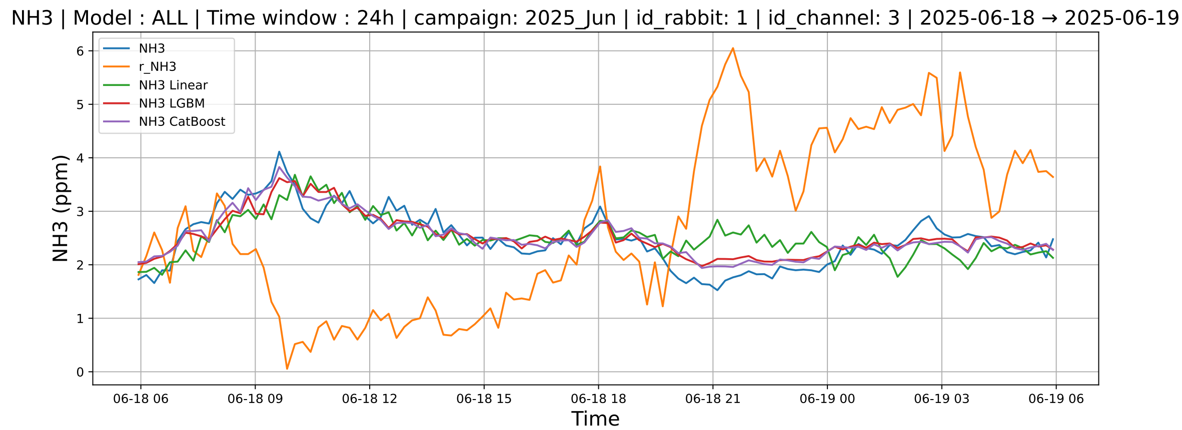
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



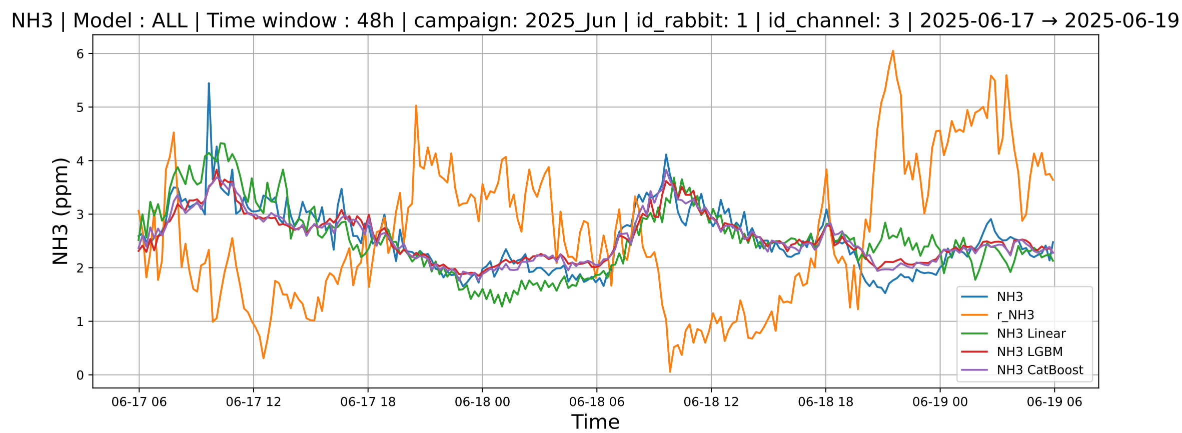
4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

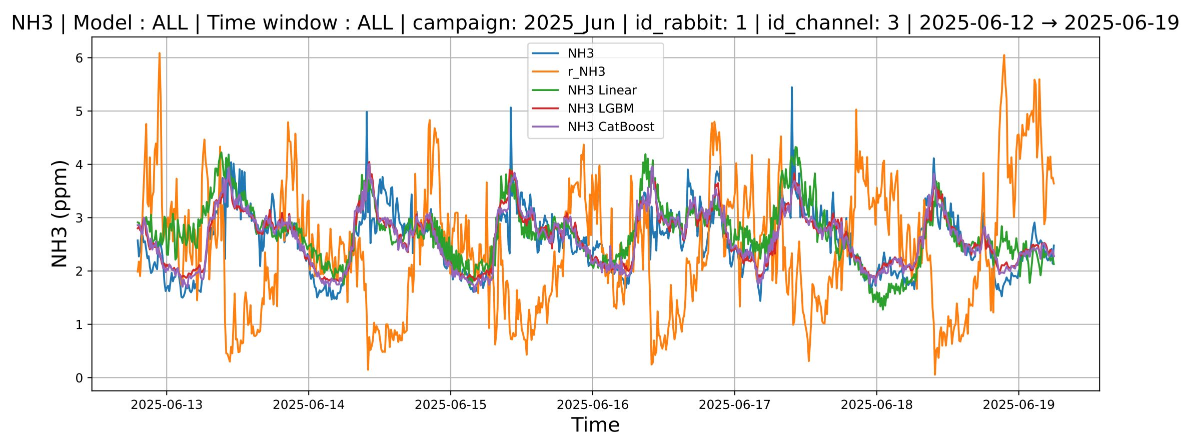
4.2.1.1 Time window : 24



4.2.1.2 Time window : 48

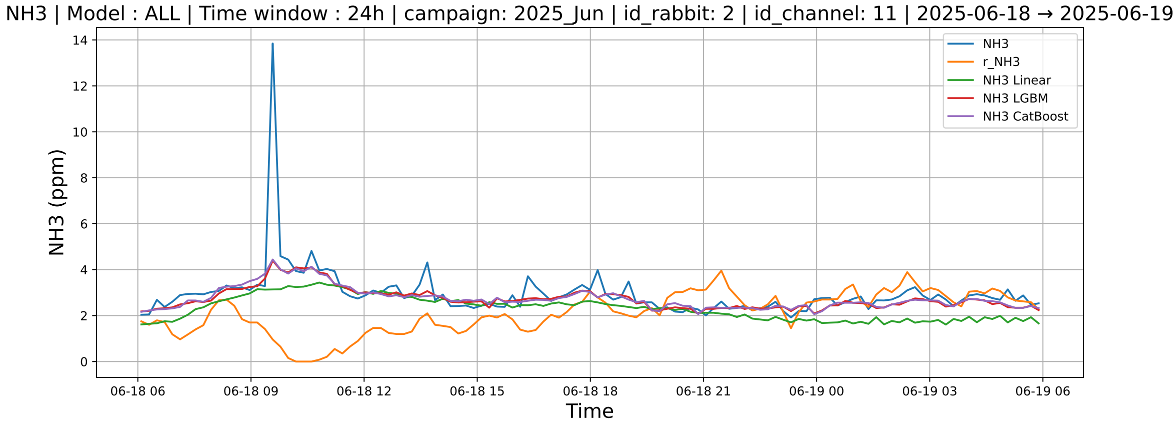


4.2.1.3 Time window : ALL

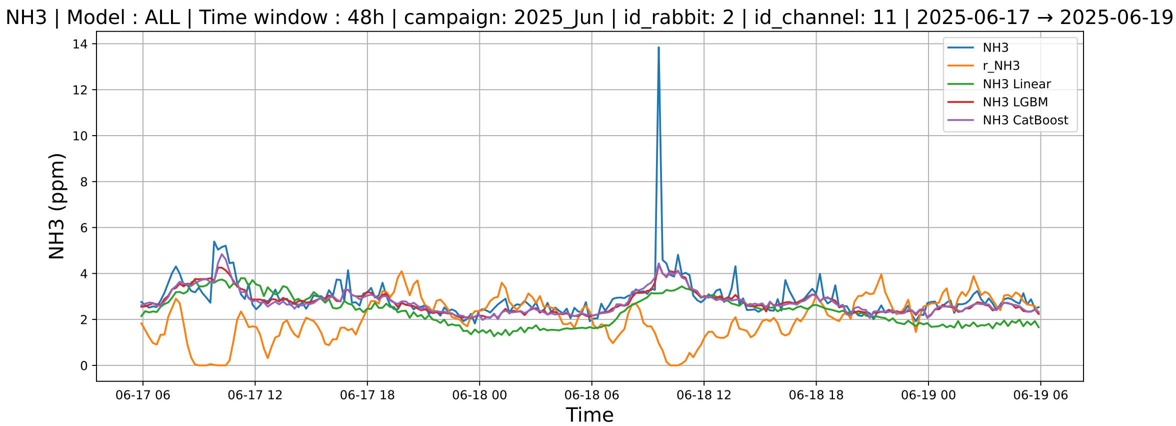


4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

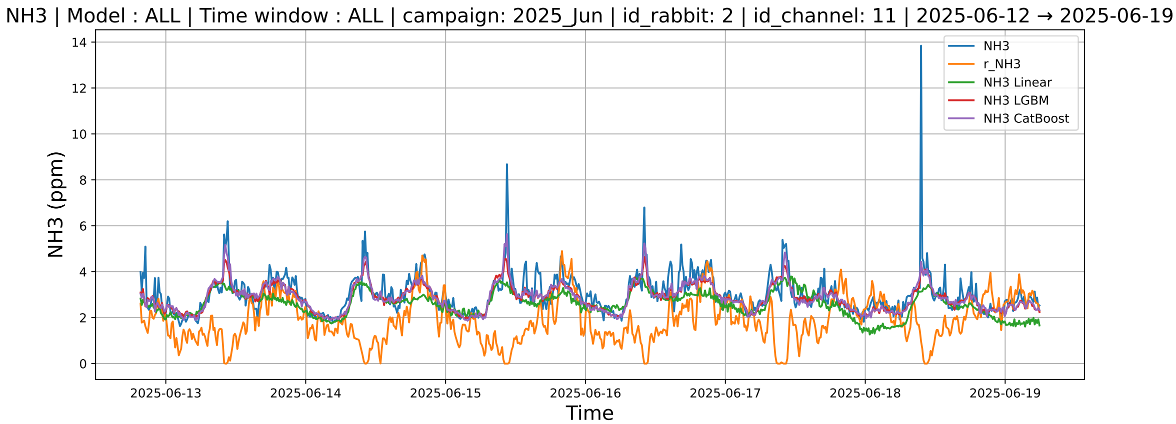
4.2.2.1 Time window : 24



4.2.2.2 Time window : 48

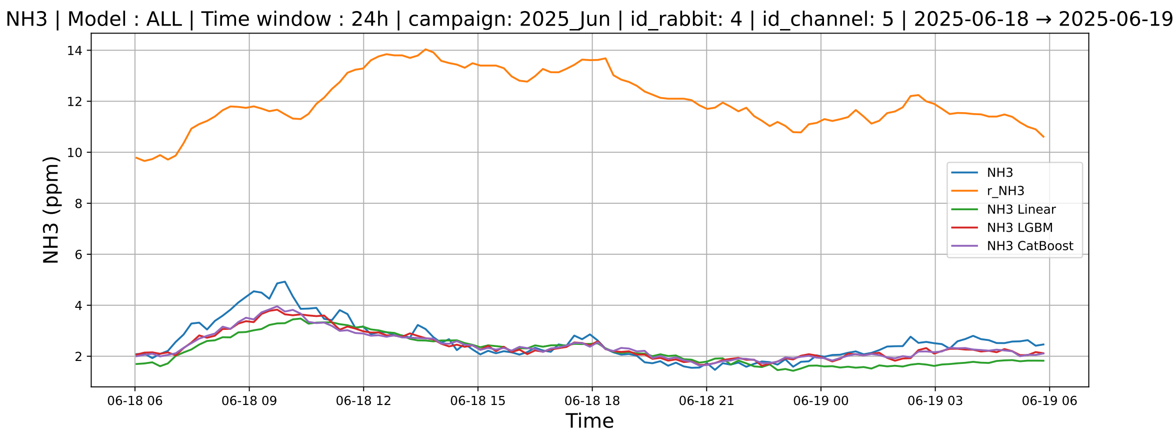


4.2.2.3 Time window : ALL

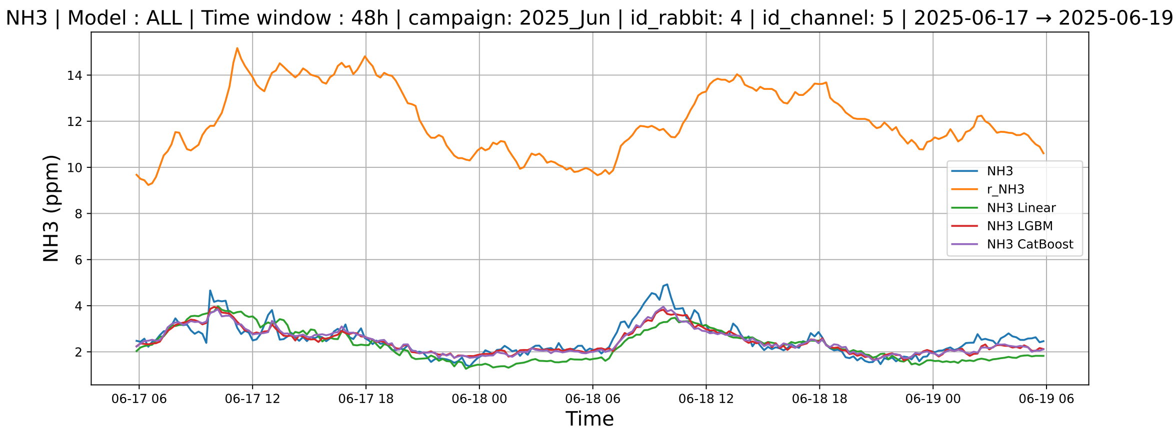


4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

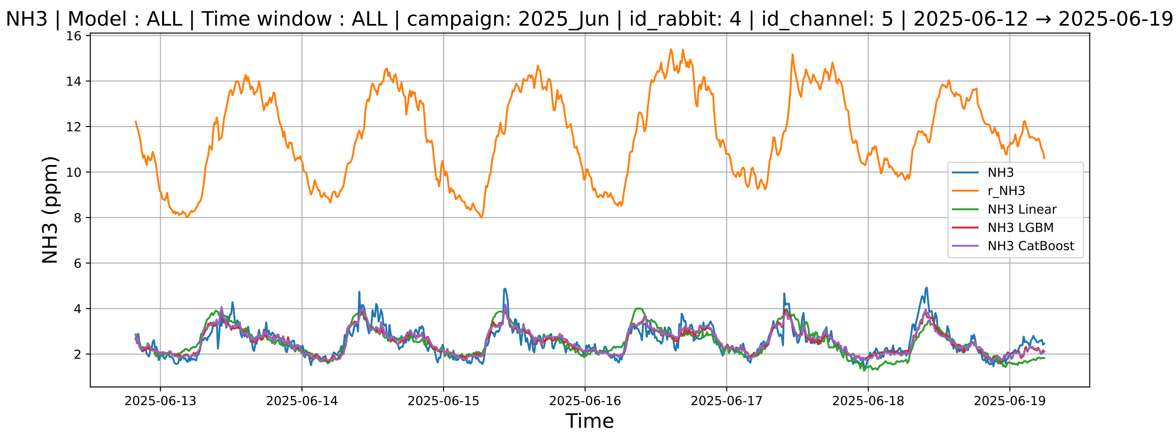
4.2.3.1 Time window : 24



4.2.3.2 Time window : 48

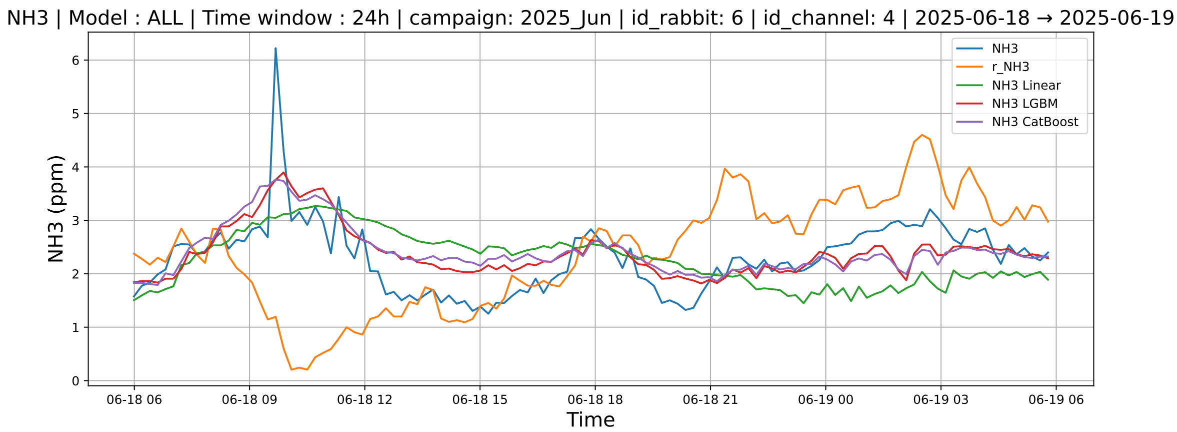


4.2.3.3 Time window : ALL

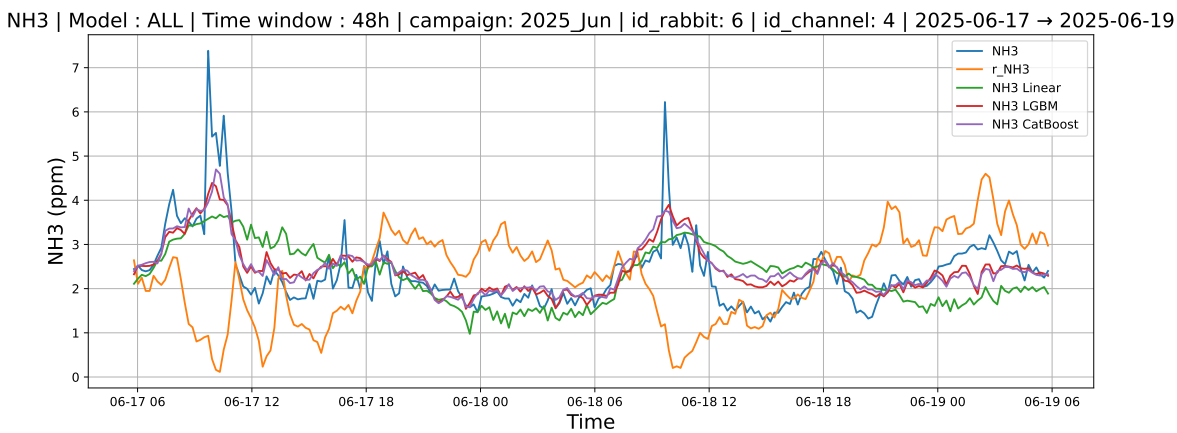


4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

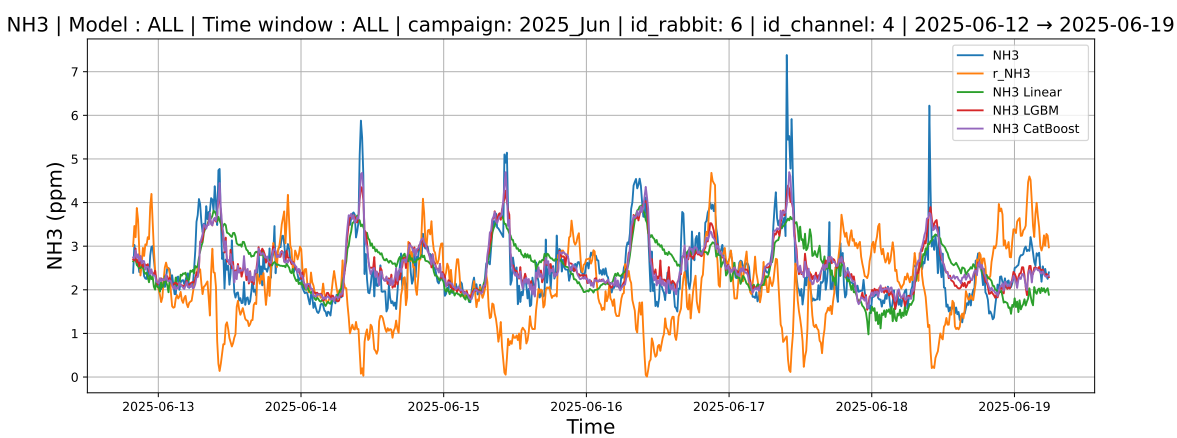
4.2.4.1 Time window : 24



4.2.4.2 Time window : 48

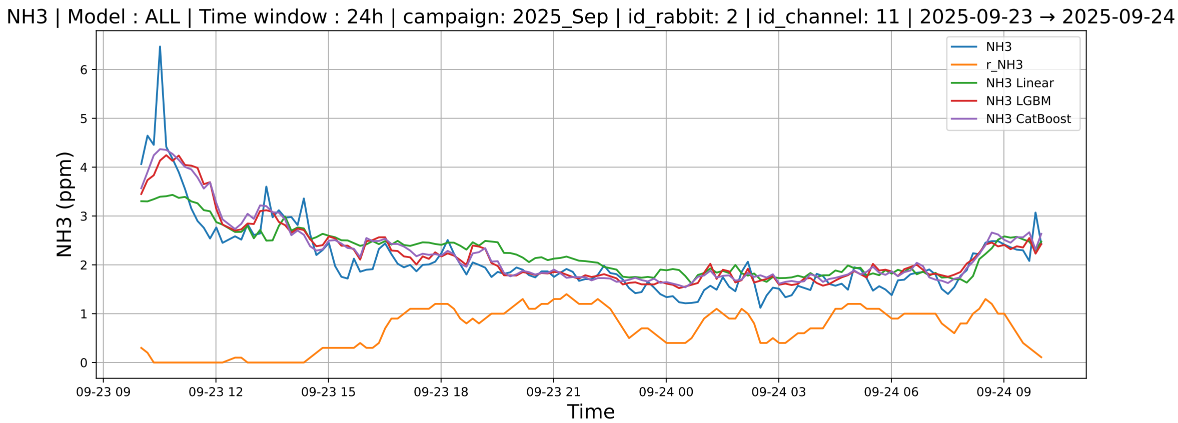


4.2.4.3 Time window : ALL

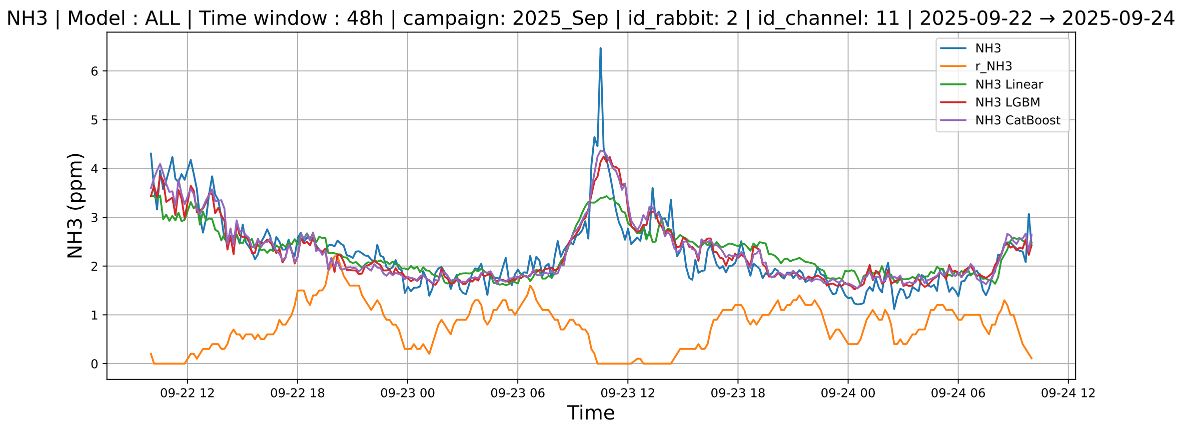


4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

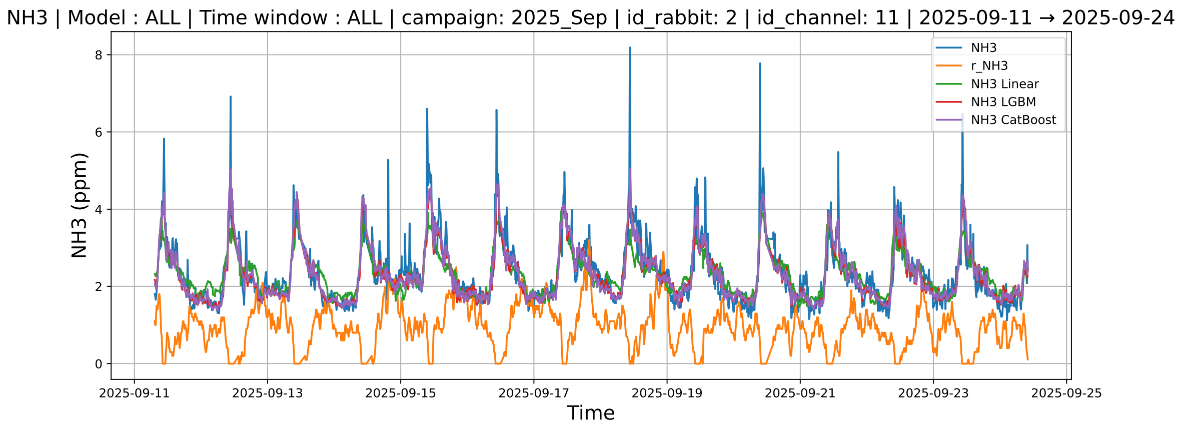
4.2.5.1 Time window : 24



4.2.5.2 Time window : 48

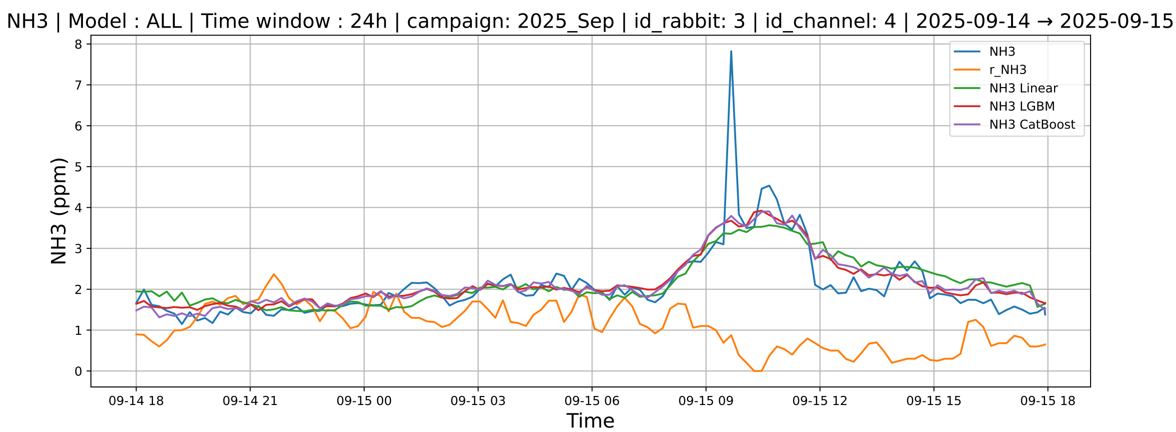


4.2.5.3 Time window : ALL

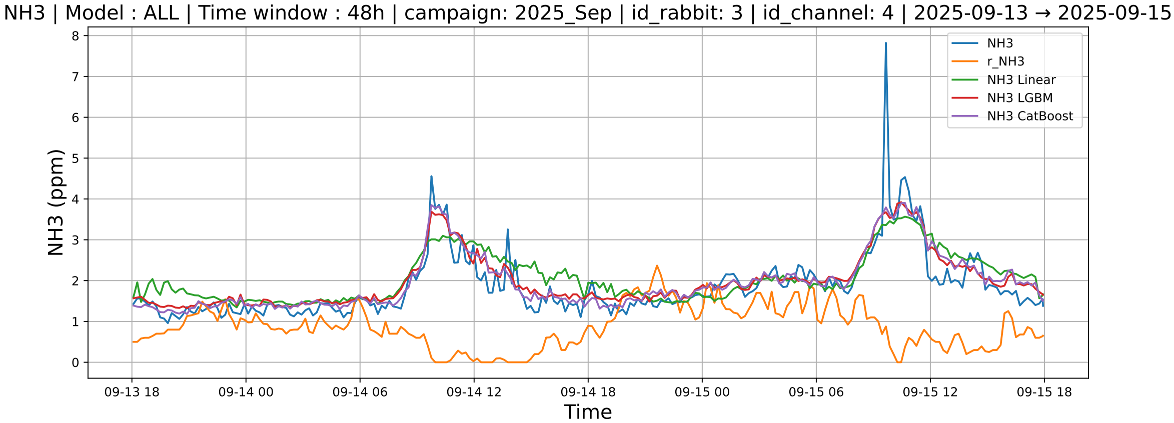


4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

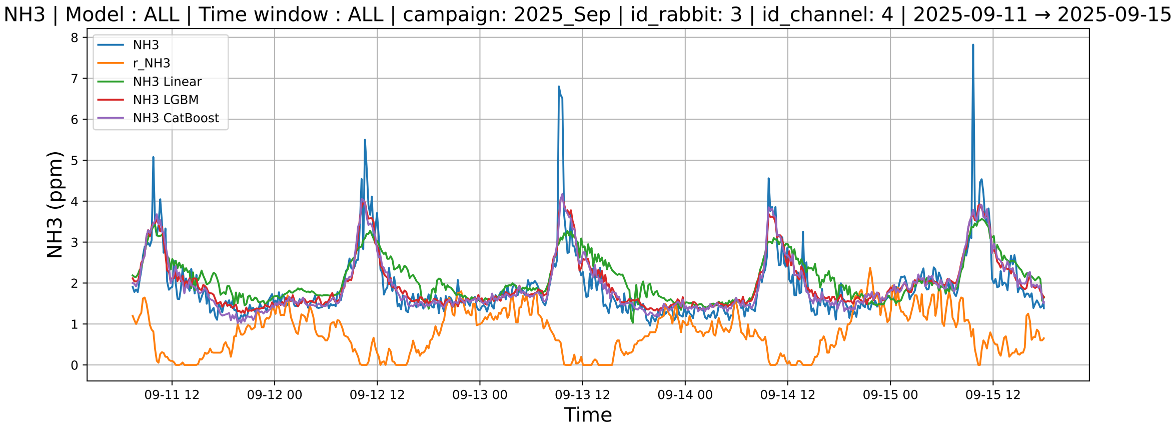
4.2.6.1 Time window : 24



4.2.6.2 Time window : 48

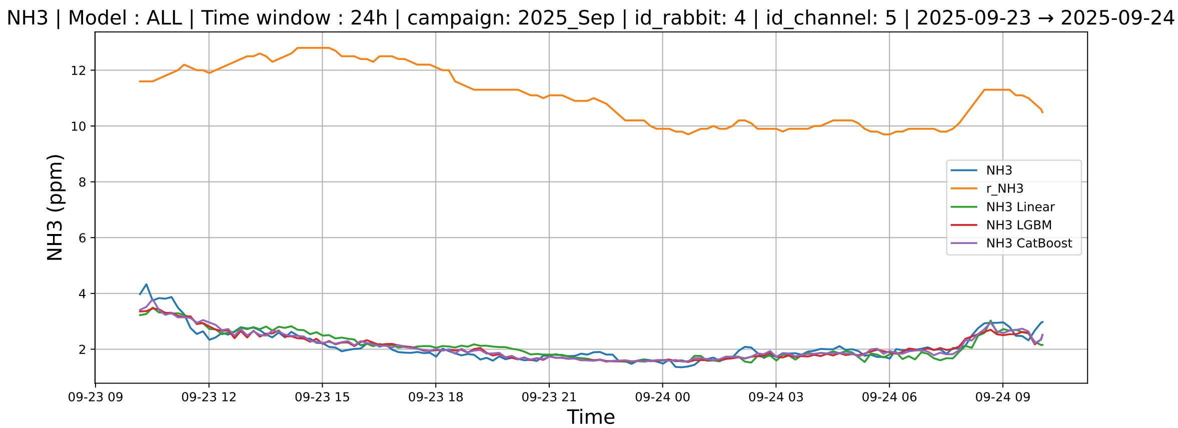


4.2.6.3 Time window : ALL

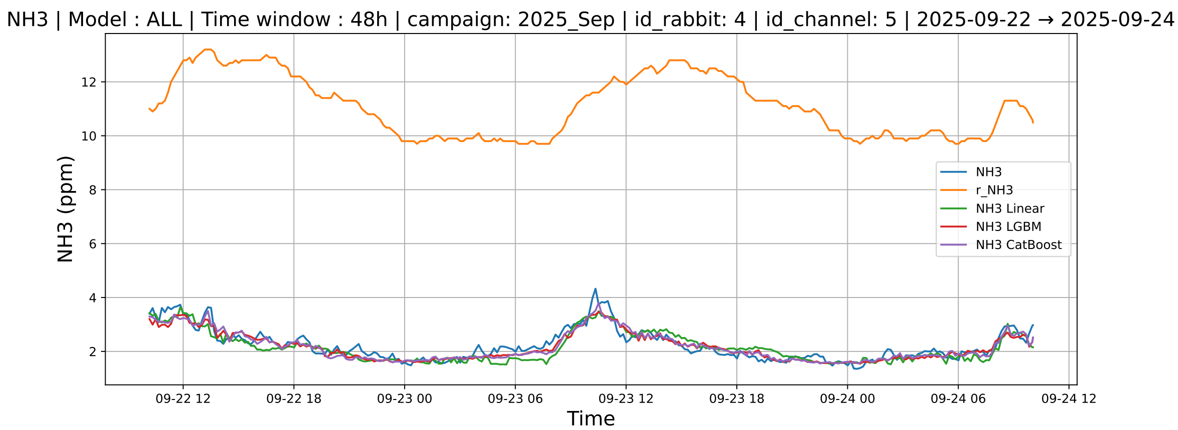


4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

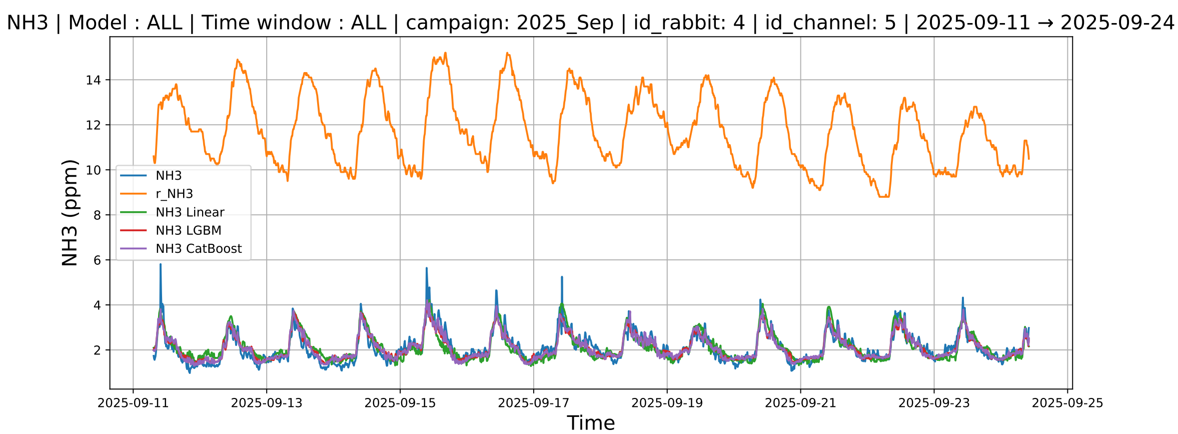
4.2.7.1 Time window : 24



4.2.7.2 Time window : 48

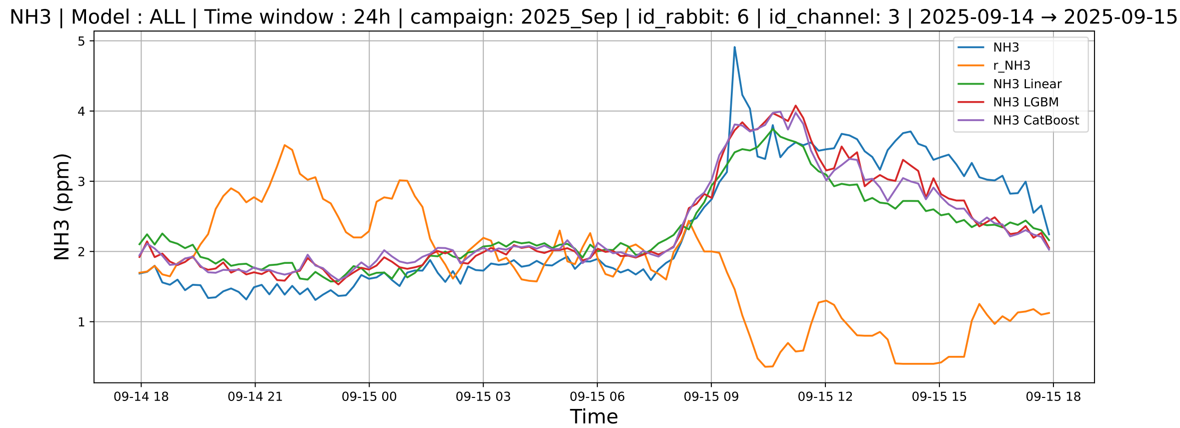


4.2.7.3 Time window : ALL

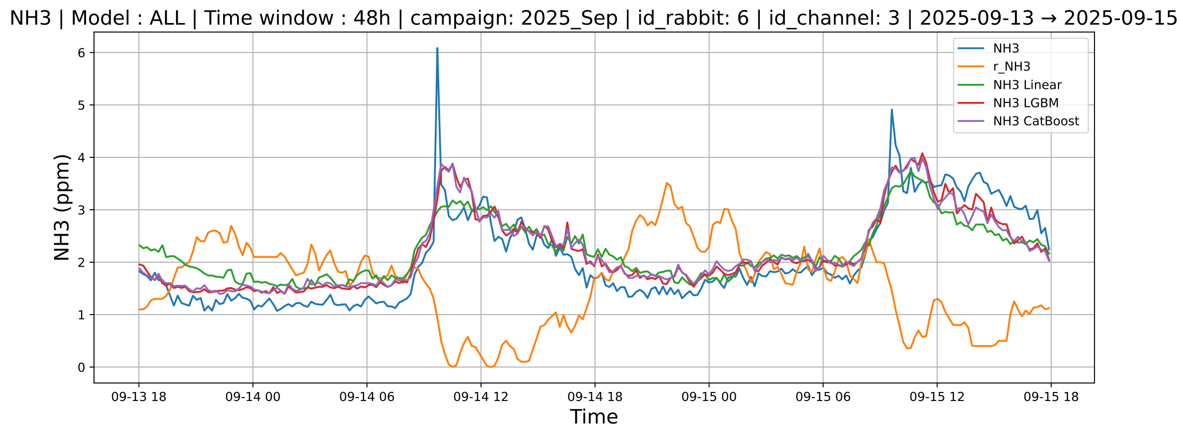


4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

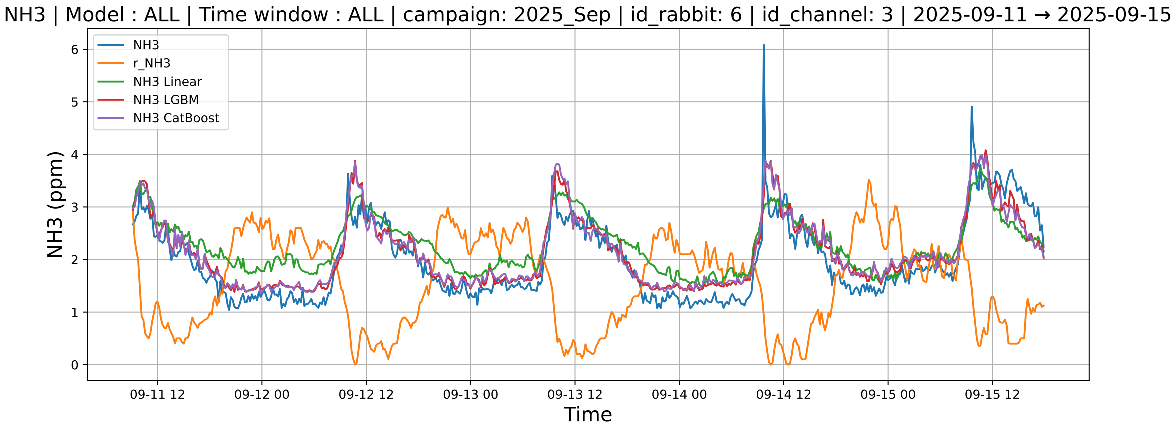
4.2.8.1 Time window : 24



4.2.8.2 Time window : 48



4.2.8.3 Time window : ALL


NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
Time
0
1
2
3
4
5
6
09-11 12
09-12 00
09-12 12
09-13 00
09-13 12
09-14 00
09-14 12
09-15 00
09-15 12